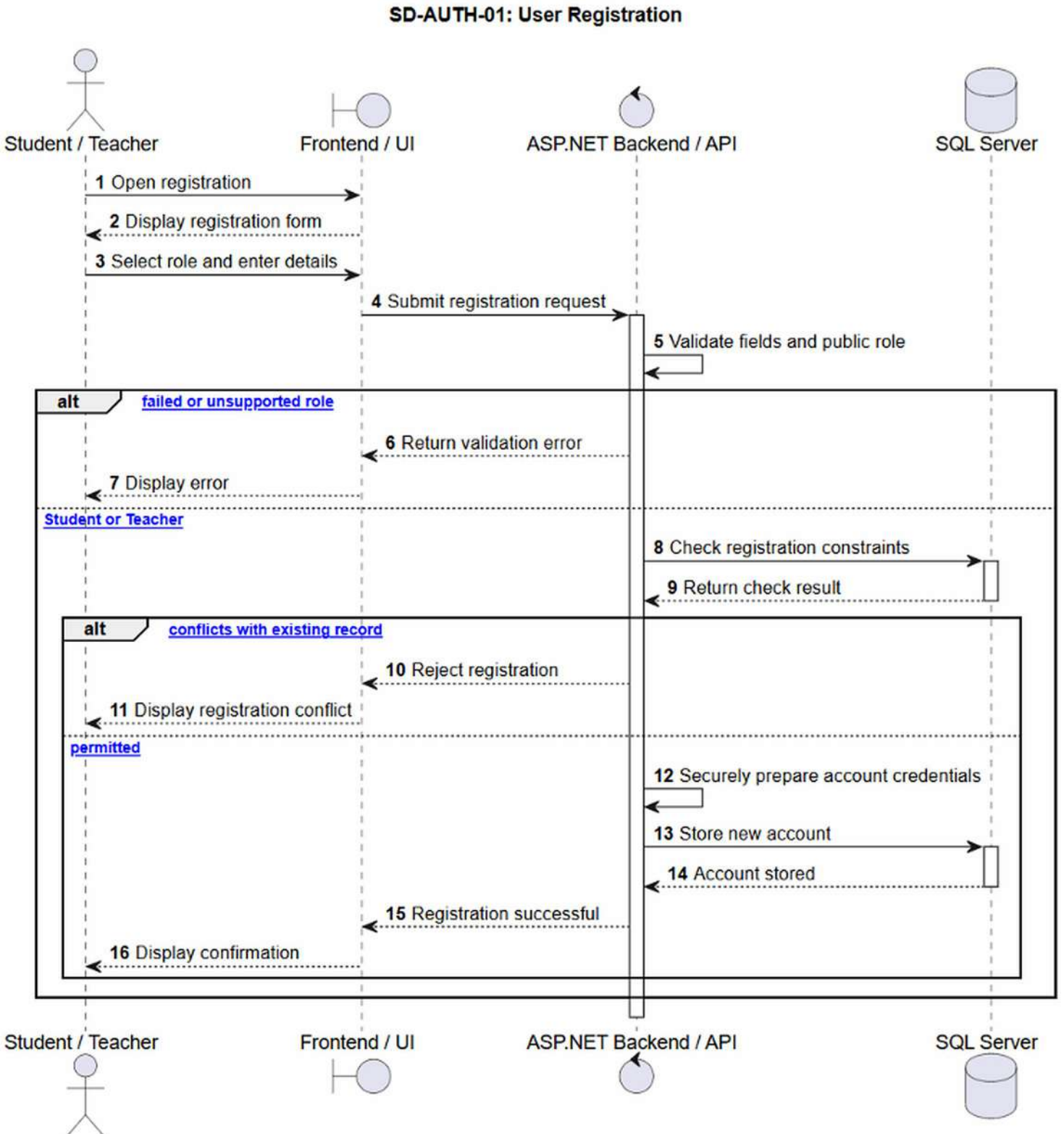
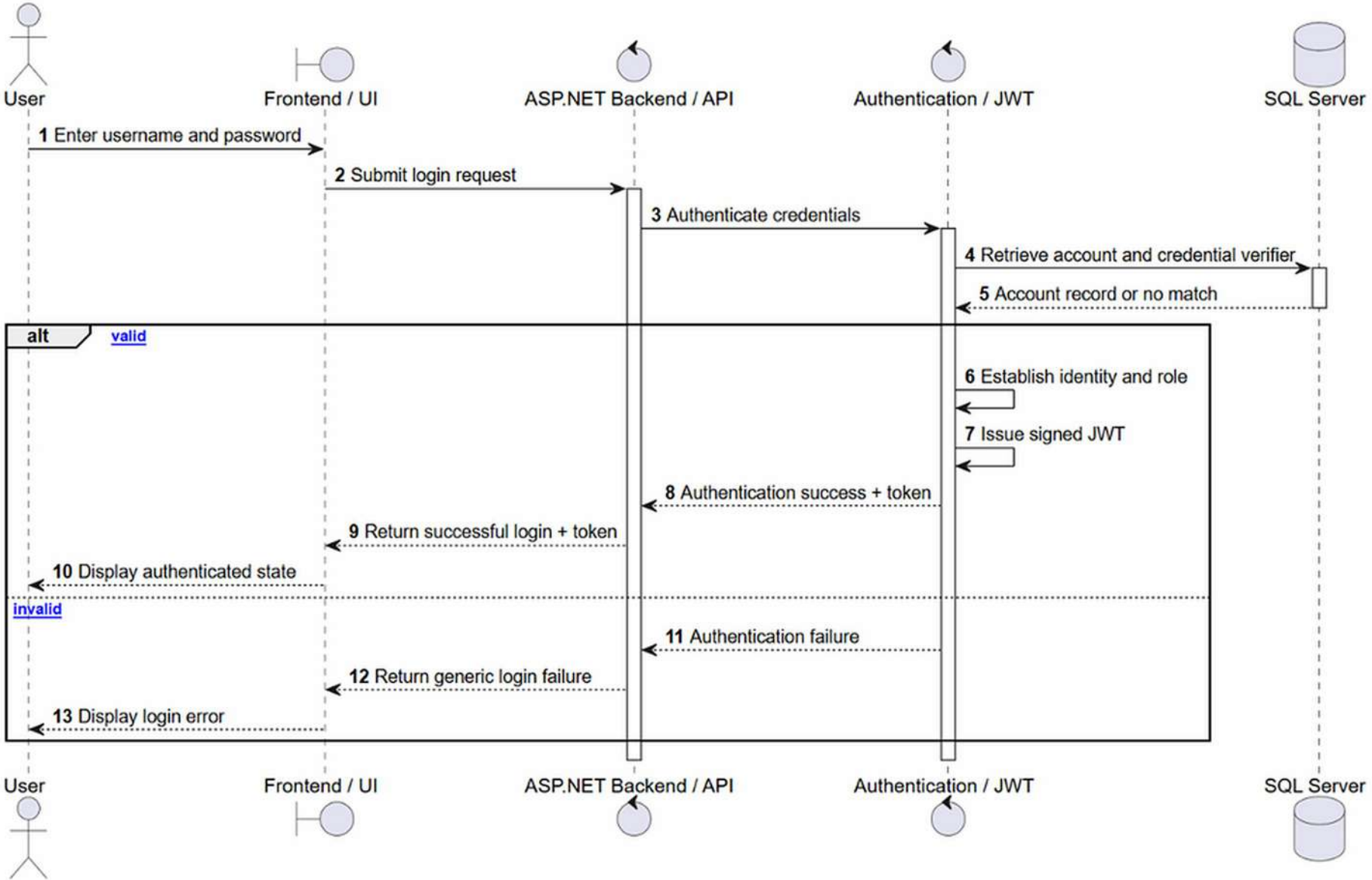
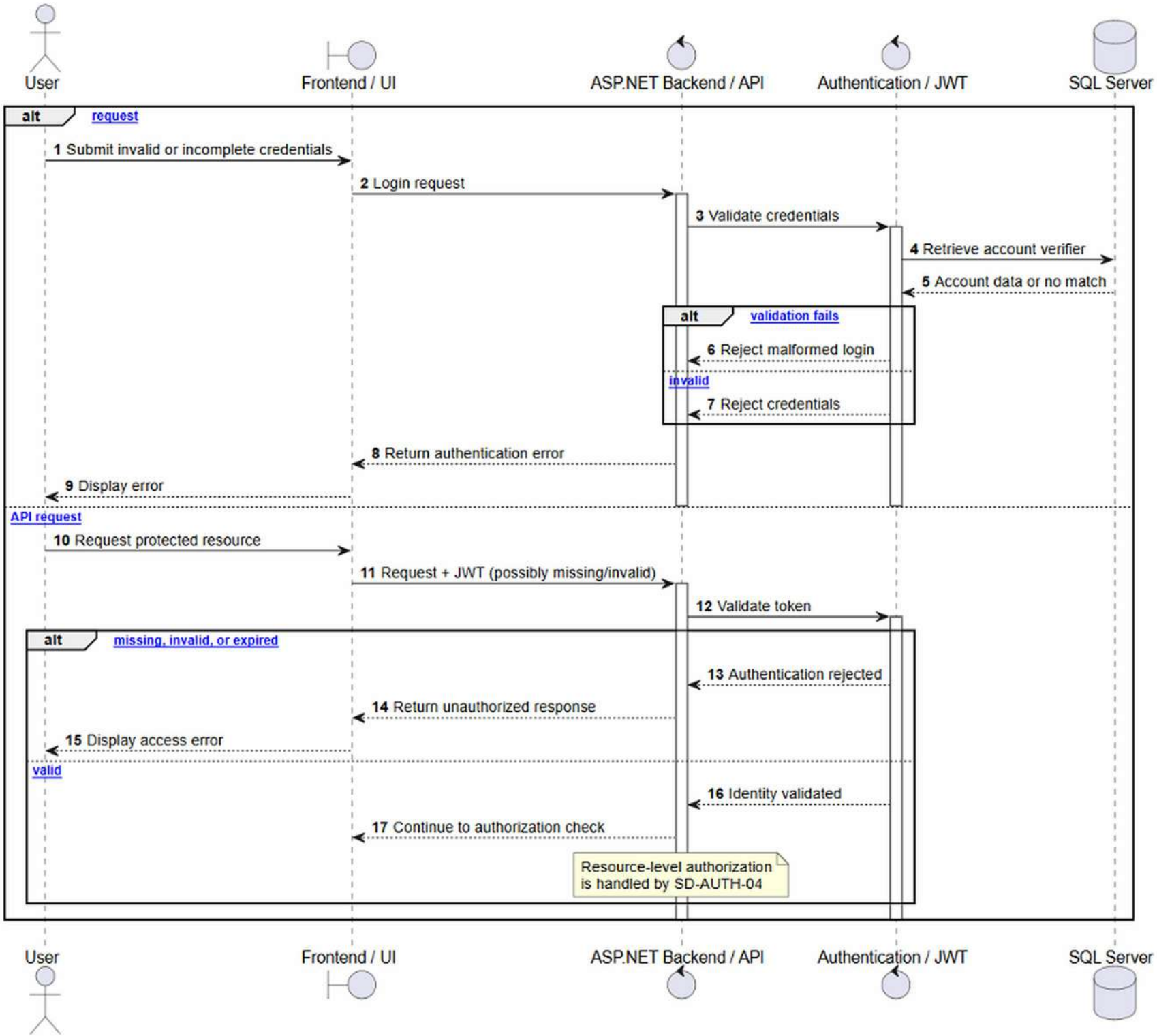

SD-AUTH-01: User Registration
Student / Teacher
Frontend / UI
ASP.NET Backend / API
SQL Server
1 Open registration
2 Display registration form
3 Select role and enter details
4 Submit registration request
5 Validate fields and public role
alt
failed or unsupported role
6 Return validation error
7 Display error
Student or Teacher
8 Check registration constraints
9 Return check result
alt
conflicts with existing record
10 Reject registration
11 Display registration conflict
permitted
12 Securely prepare account credentials
13 Store new account
14 Account stored
15 Registration successful
16 Display confirmation
Student / Teacher
Frontend / UI
ASP.NET Backend / API
SQL Server

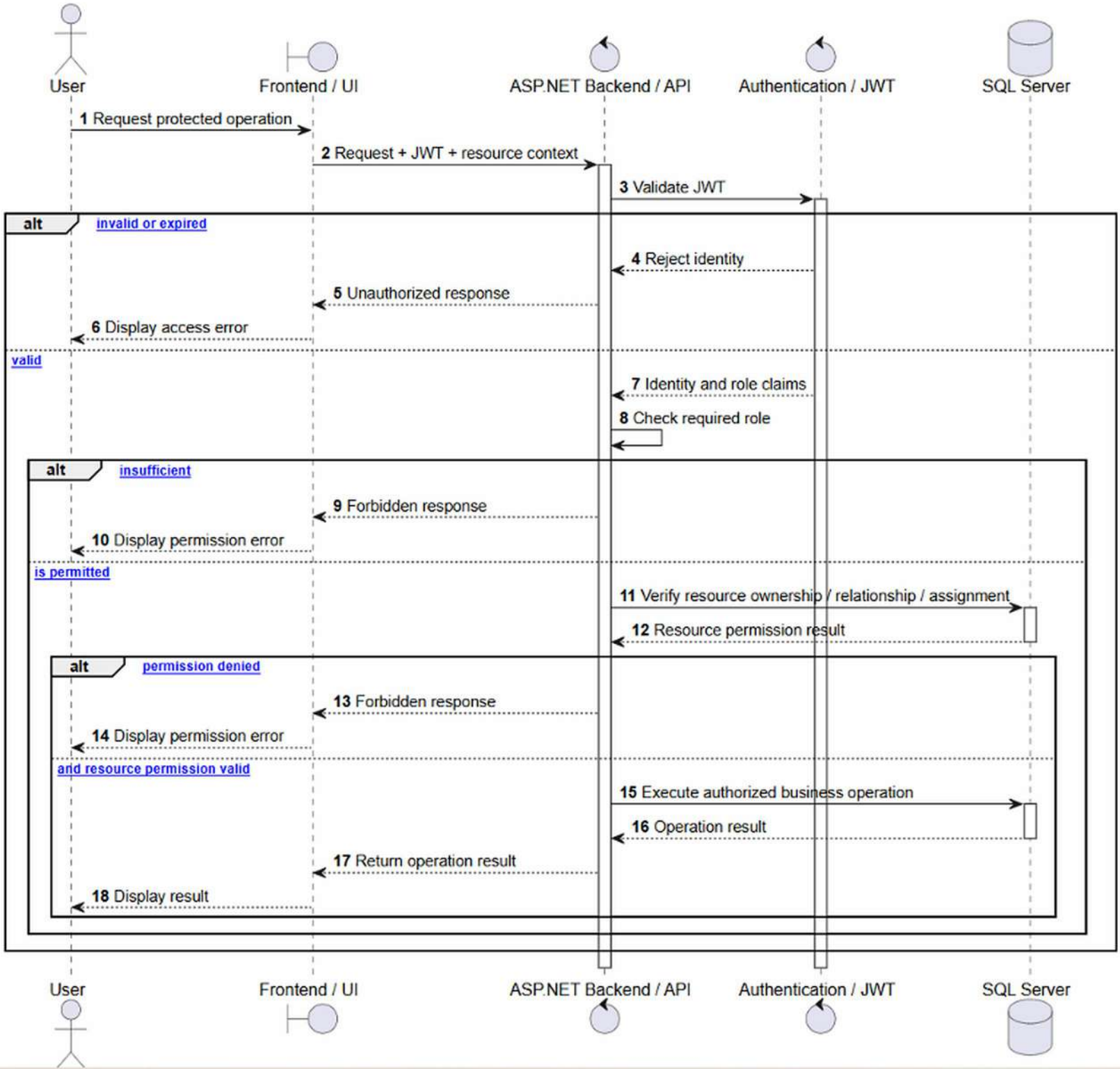
SD-AUTH-02: User Login and JWT Authentication



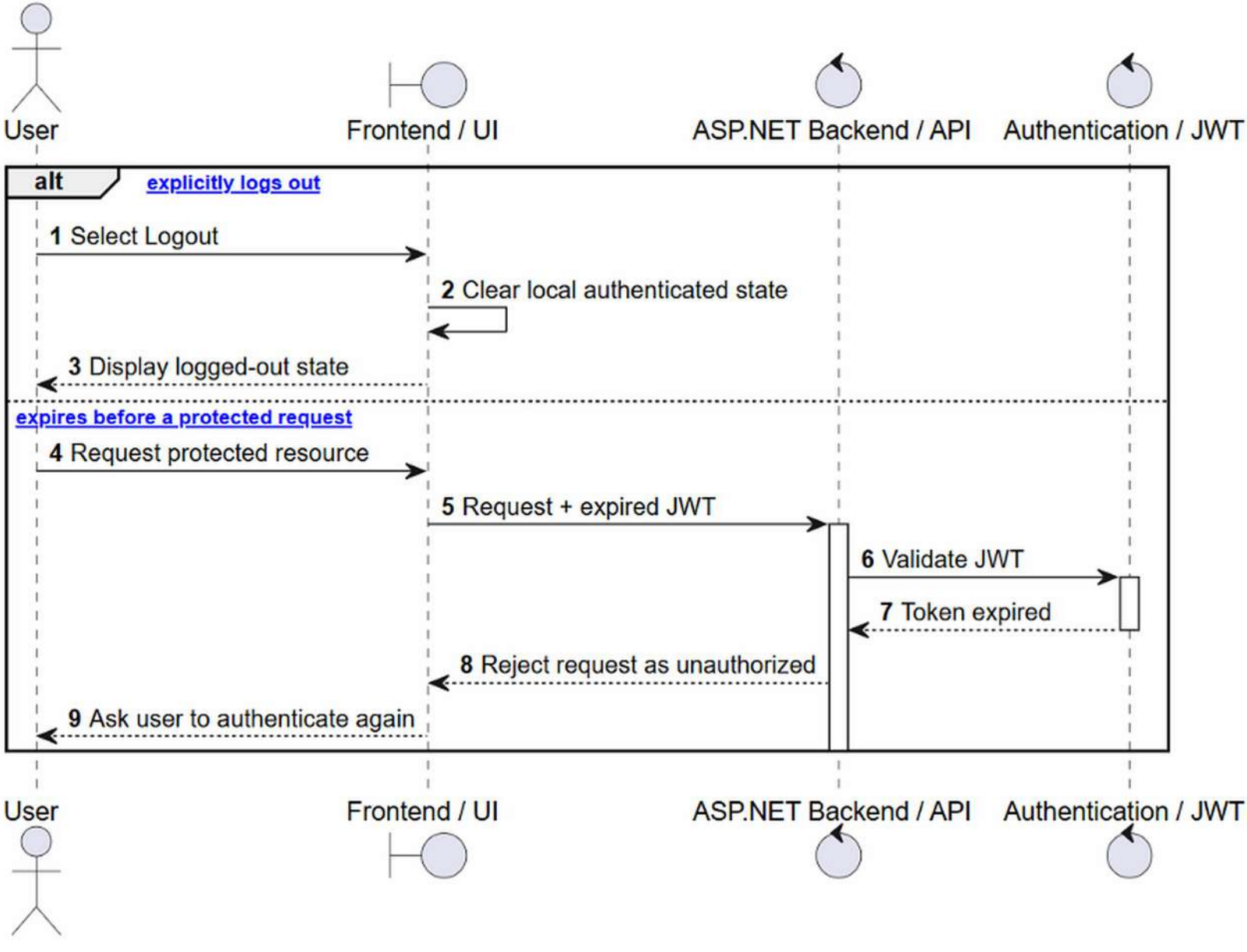
SD-AUTH-03: Authentication Failure

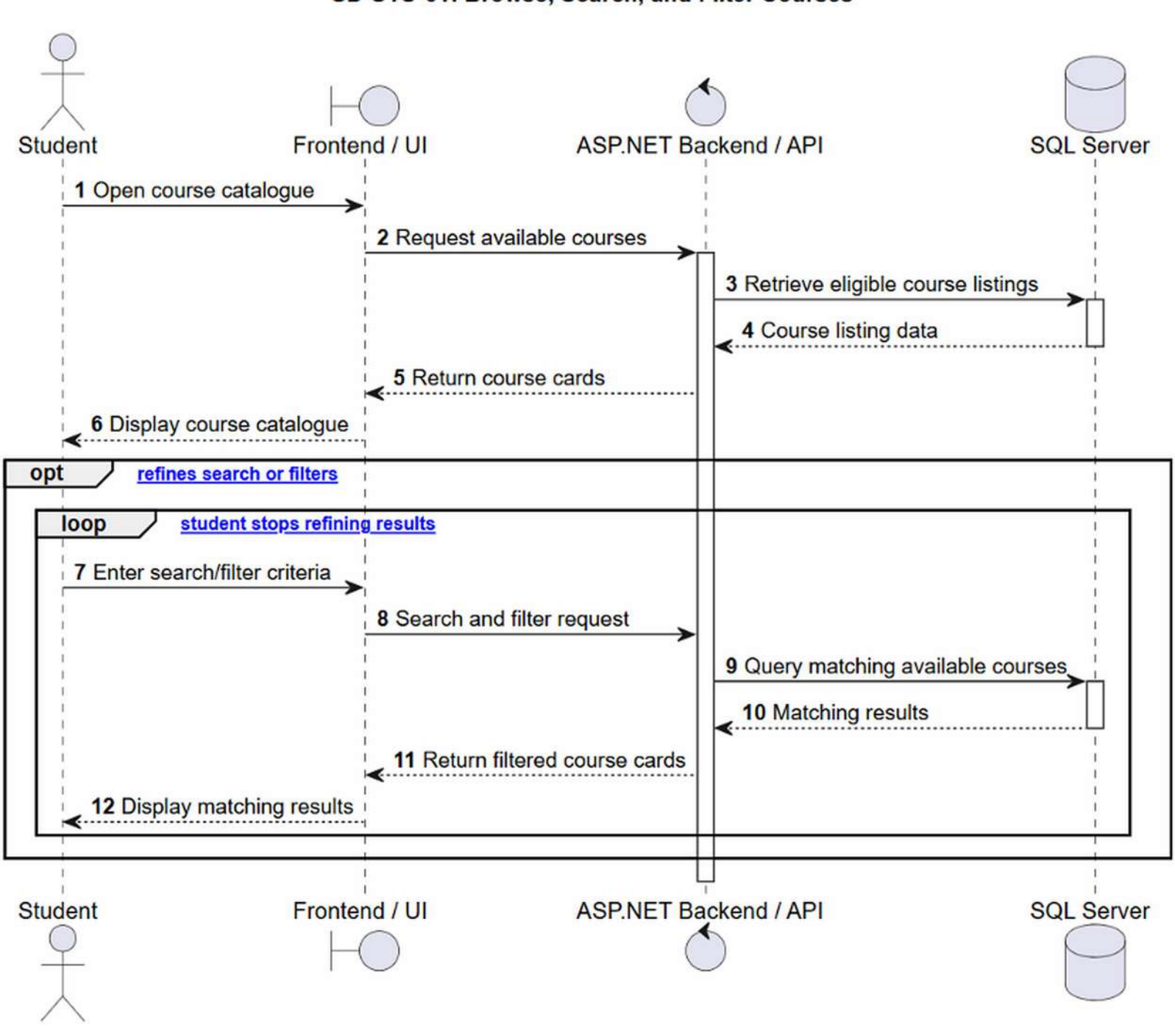


SD-AUTH-04: Role-Based and Resource-Level Authorization

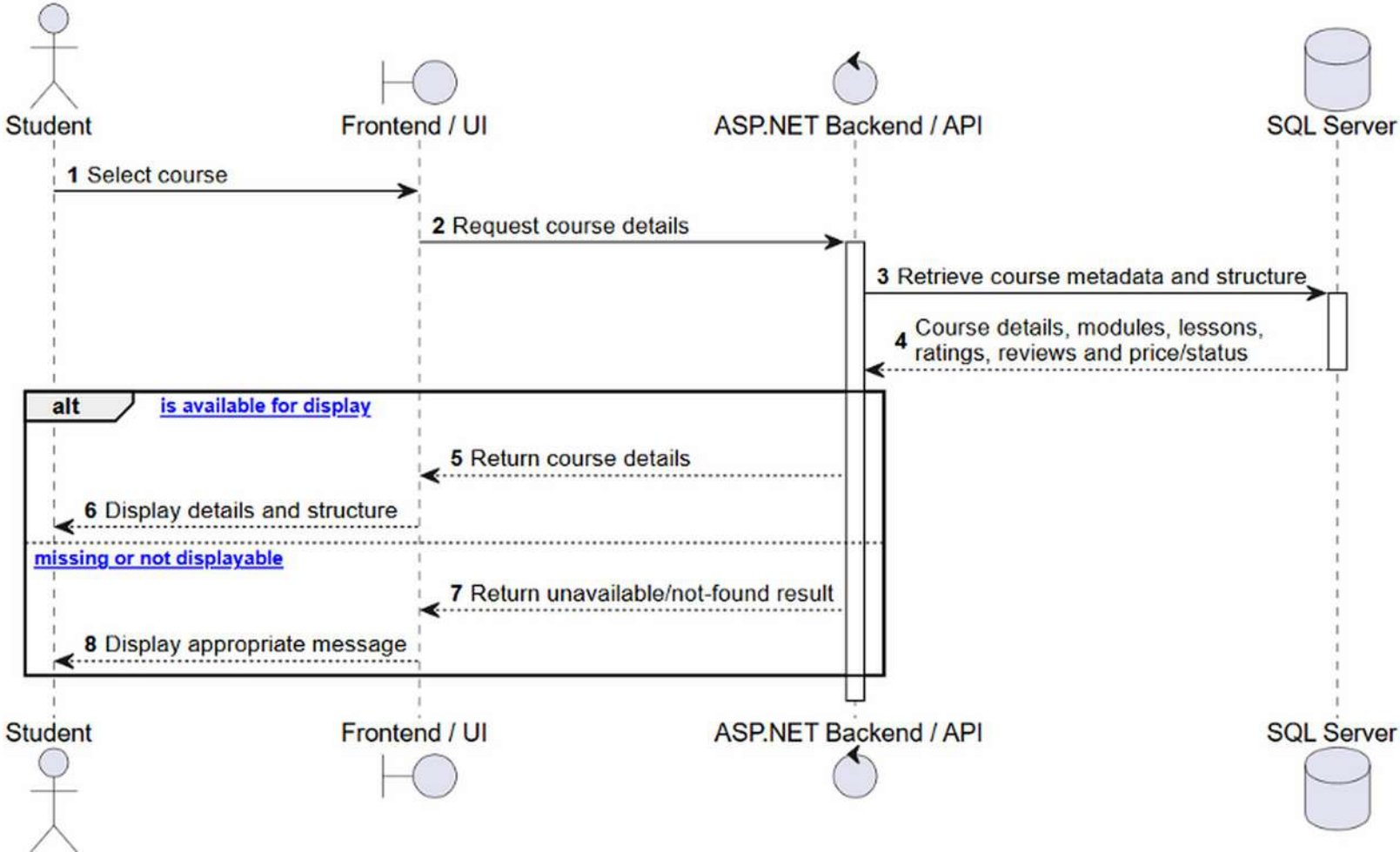


SD-AUTH-05: Logout and Token Expiration

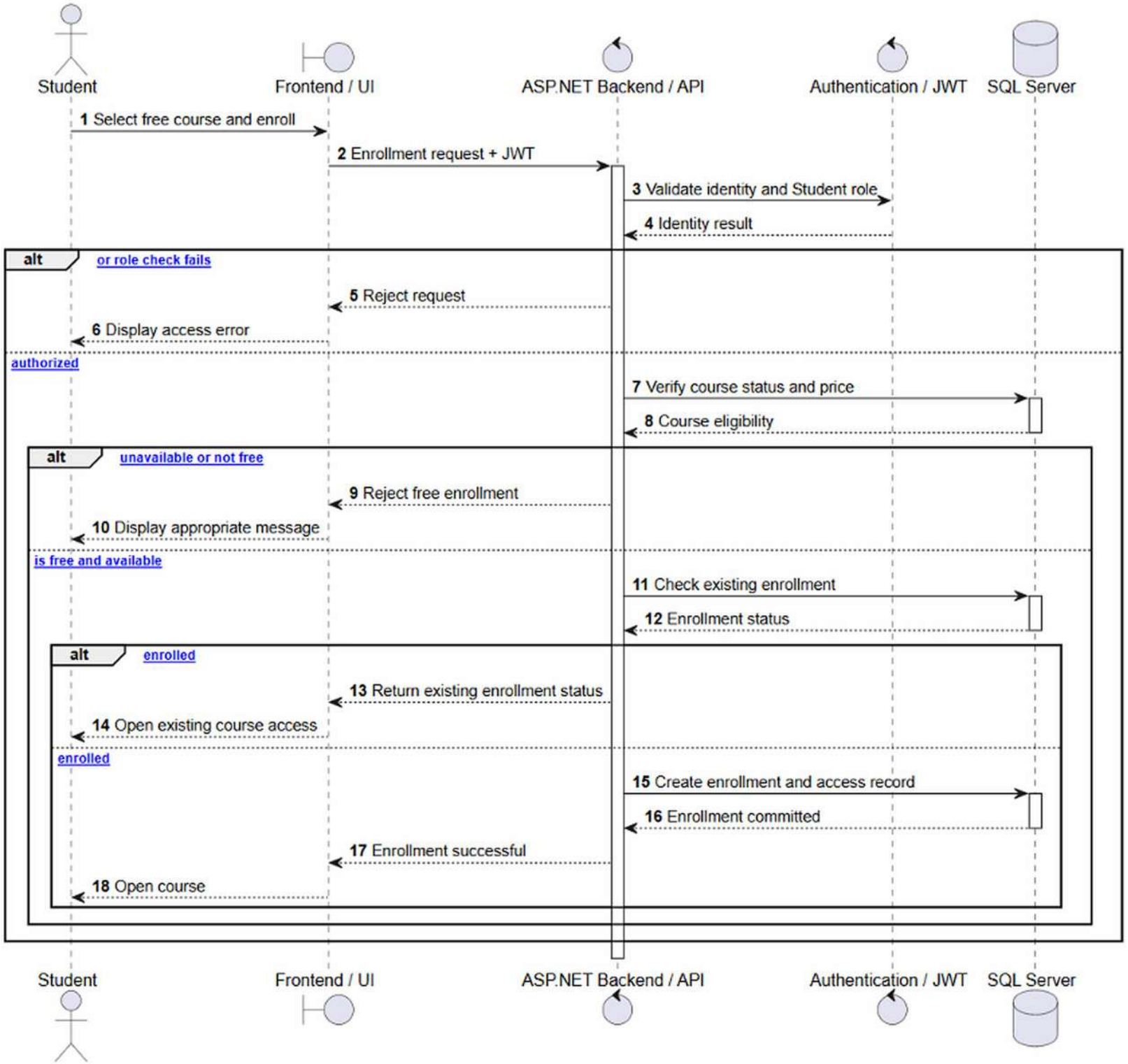



SD-STU-01: Browse, Search, and Filter Courses
Student
Frontend / UI
ASP.NET Backend / API
SQL Server
1 Open course catalogue
2 Request available courses
3 Retrieve eligible course listings
4 Course listing data
5 Return course cards
6 Display course catalogue
opt
refines search or filters
loop
student stops refining results
7 Enter search/filter criteria
8 Search and filter request
9 Query matching available courses
10 Matching results
11 Return filtered course cards
12 Display matching results
Student
Frontend / UI
ASP.NET Backend / API
SQL Server

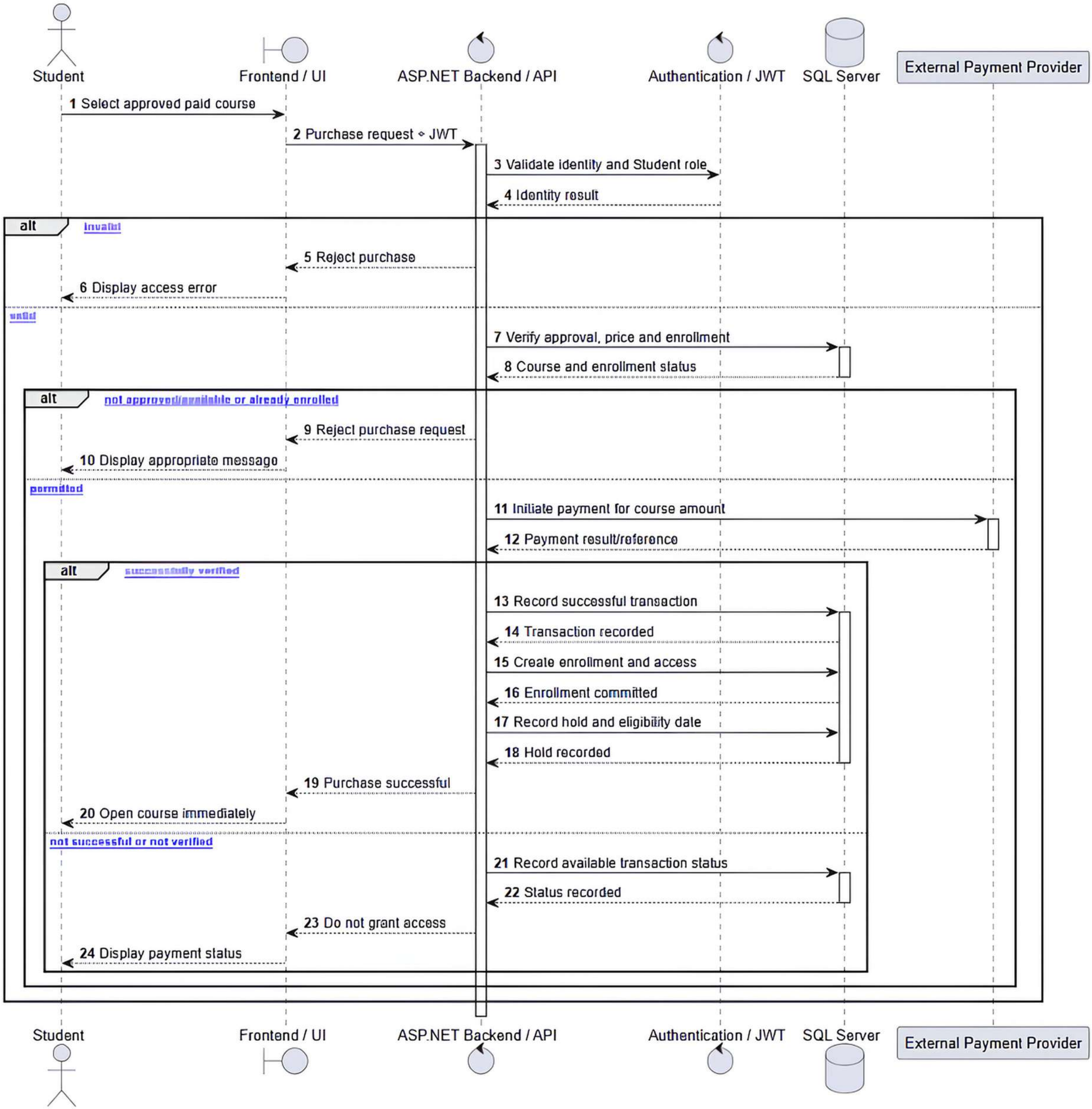
SD-STU-02: View Course Details and Structure



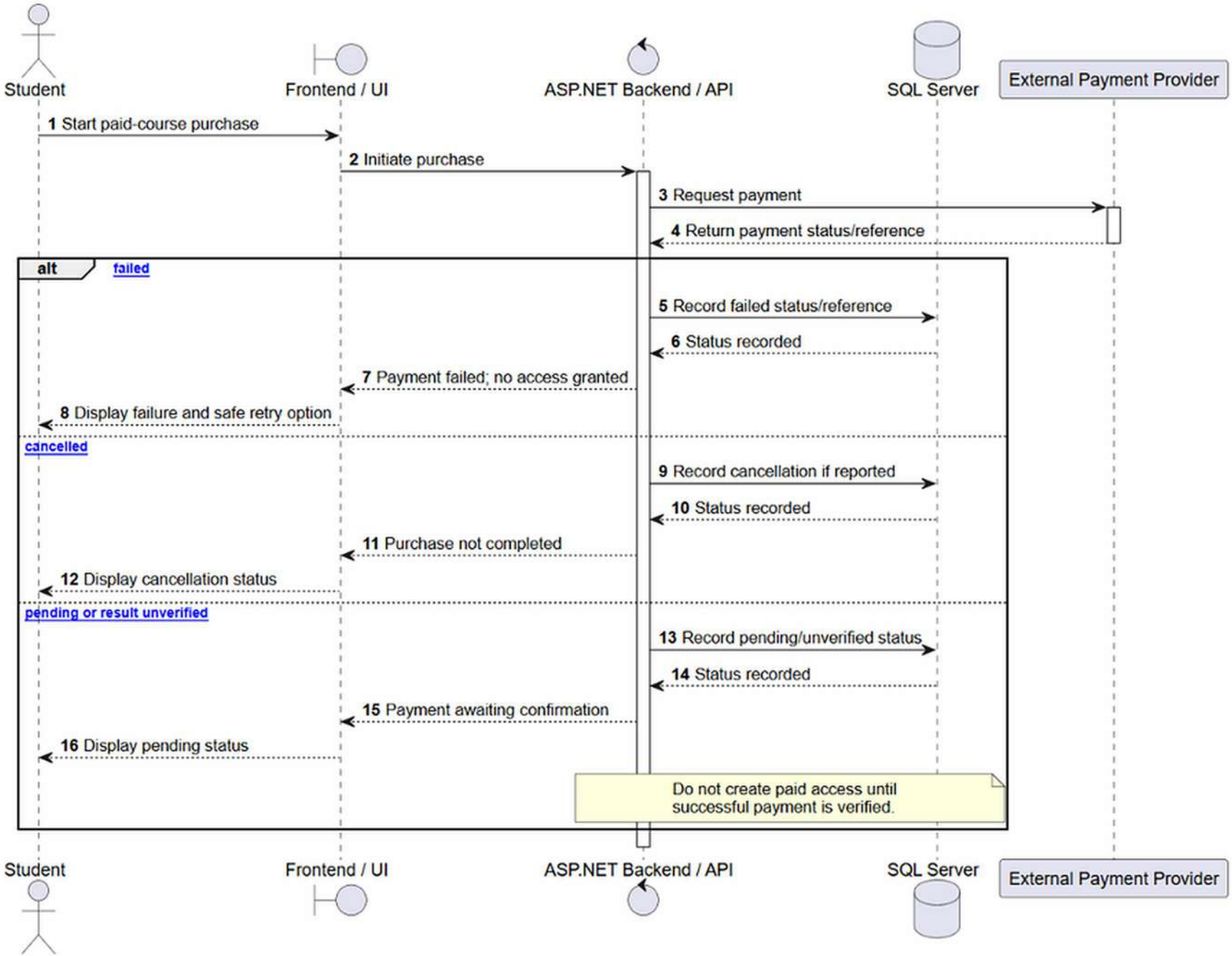
SD-STU-03: Enroll in a Free Course



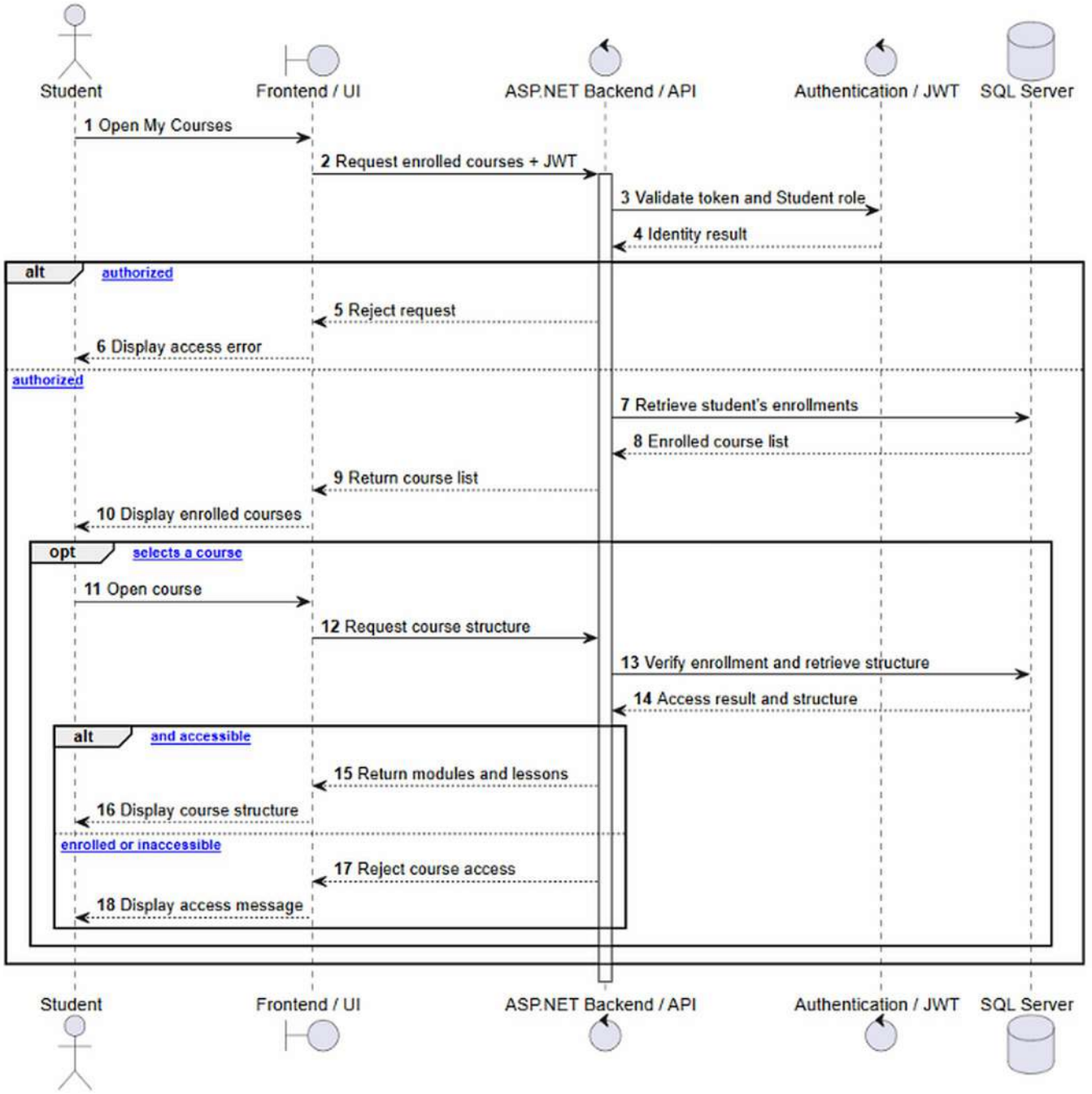
SD-STU-04: Purchase Paid Course and Grant Access



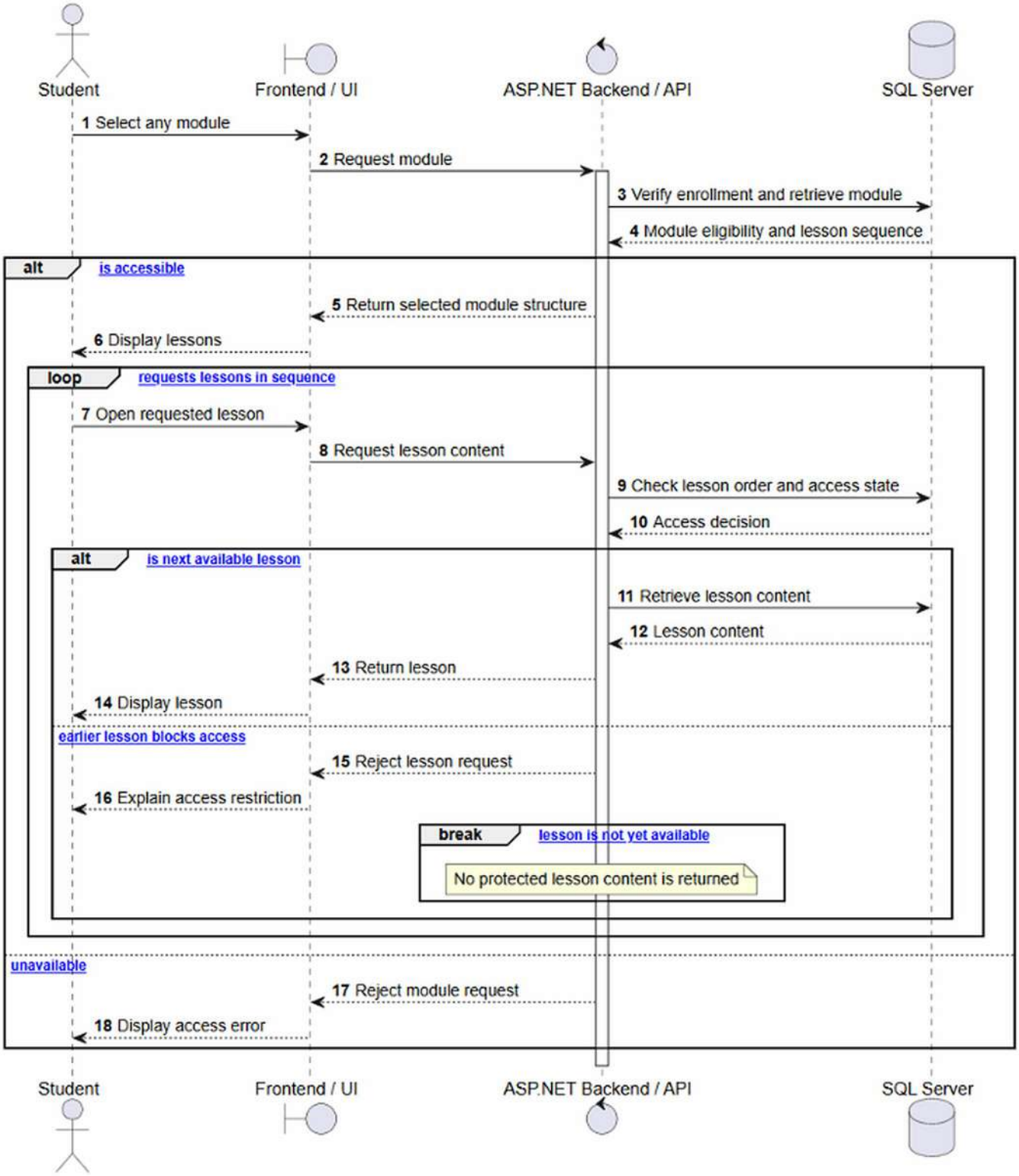
SD-STU-05: Failed, Cancelled, or Pending Payment



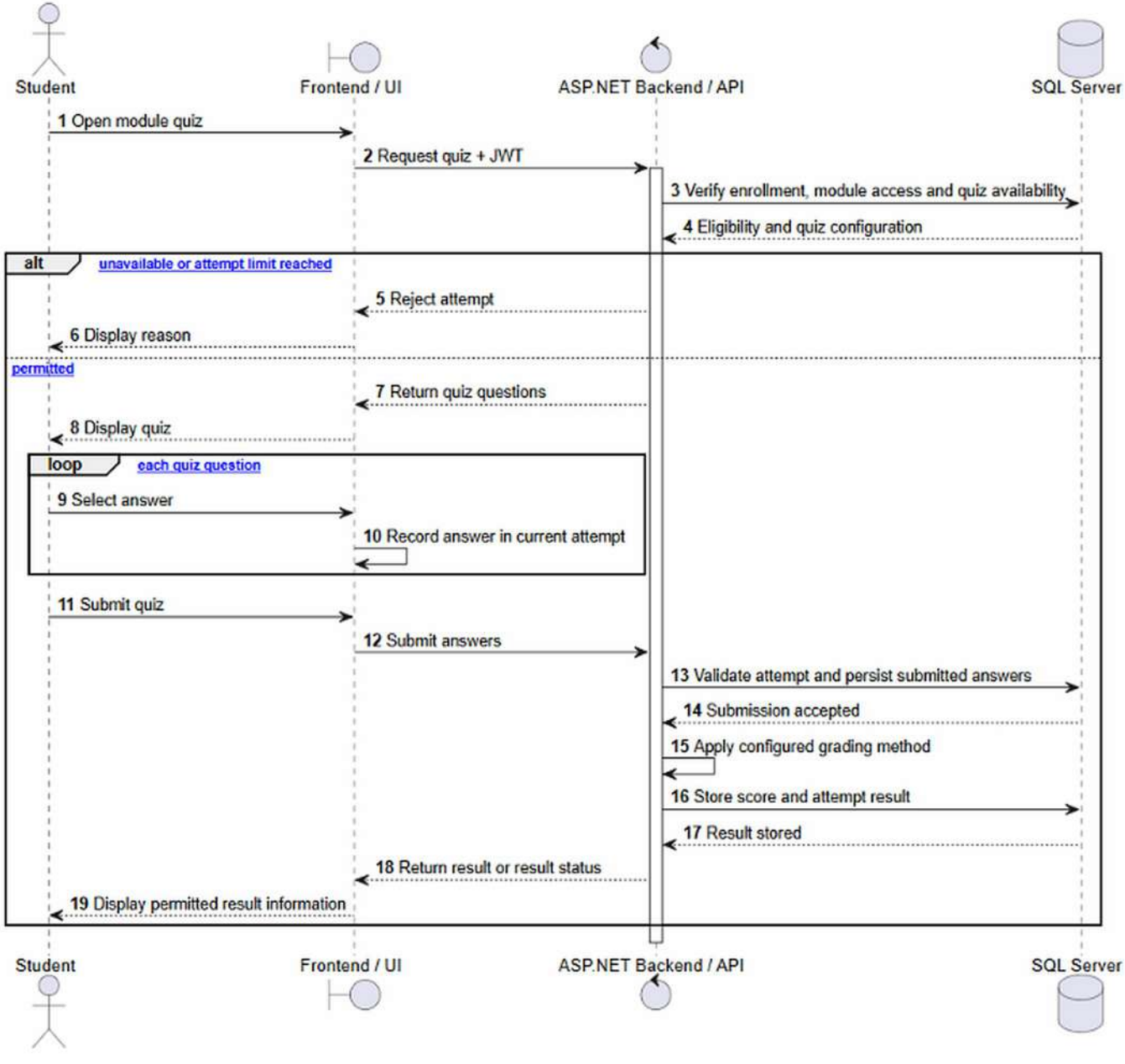
SD-STU-06: Access Enrolled Courses and Course Structure



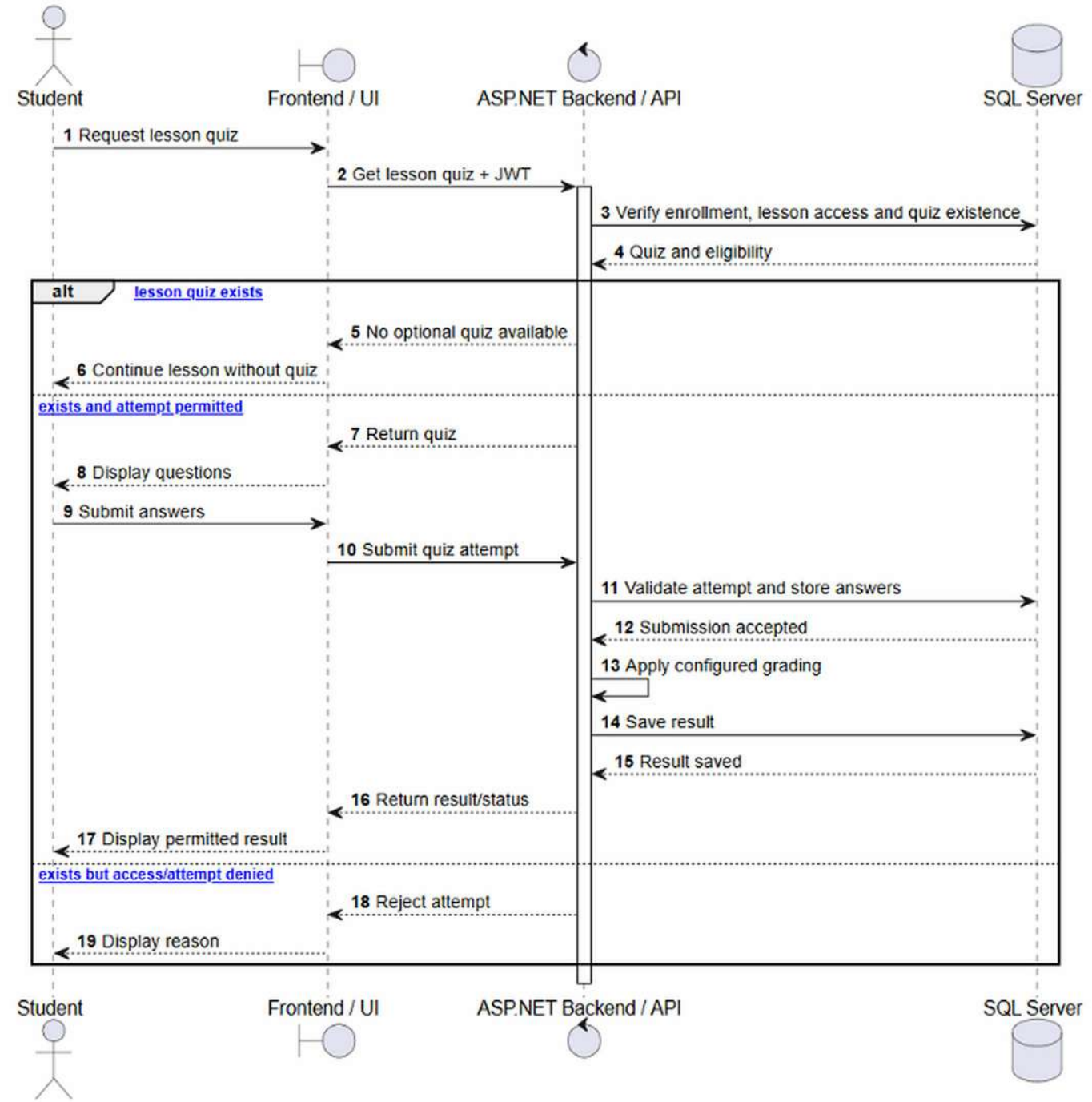
SD-STU-07: Sequential Lesson Access

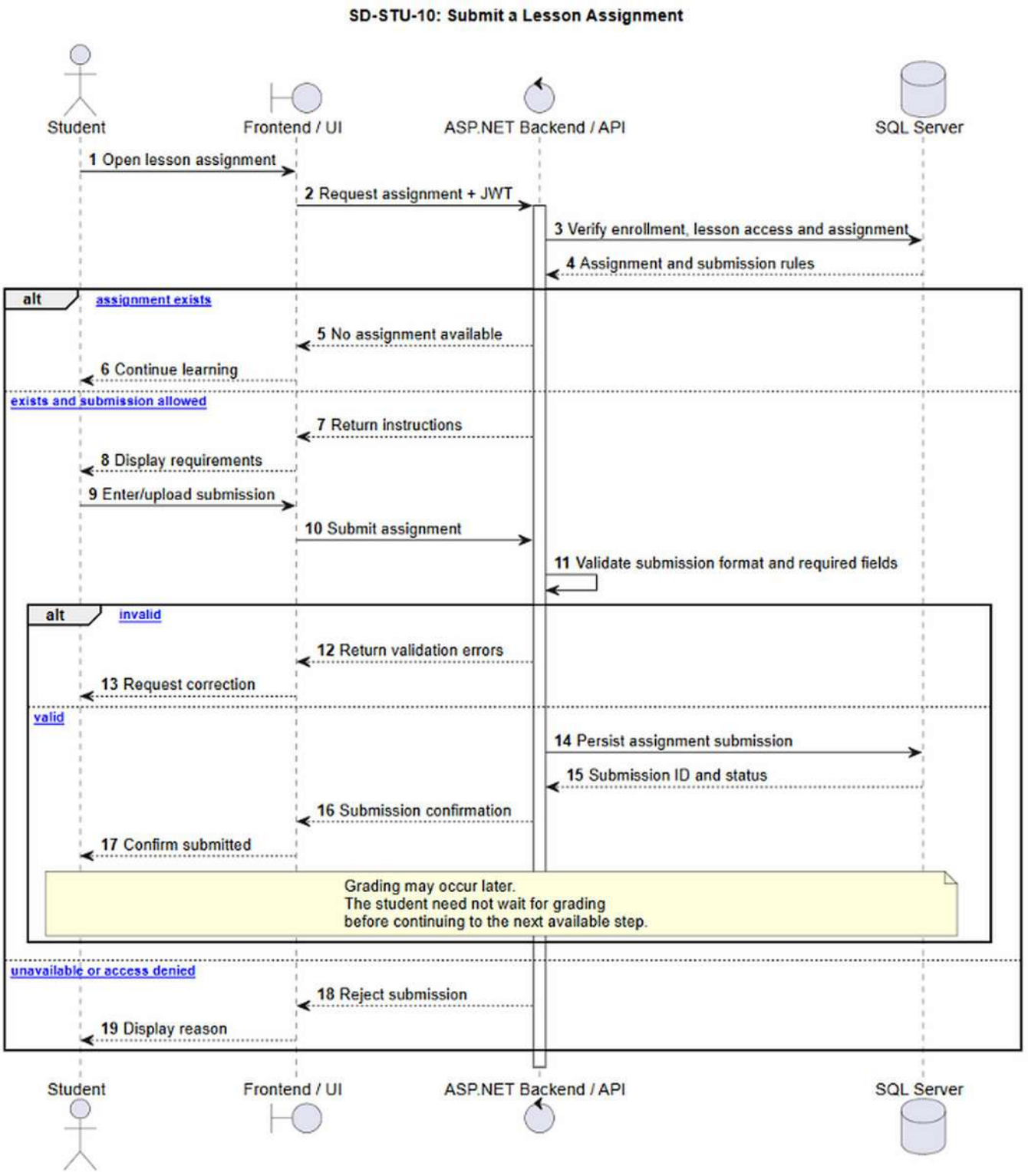


SD-STU-08: Attempt Mandatory Module Quiz

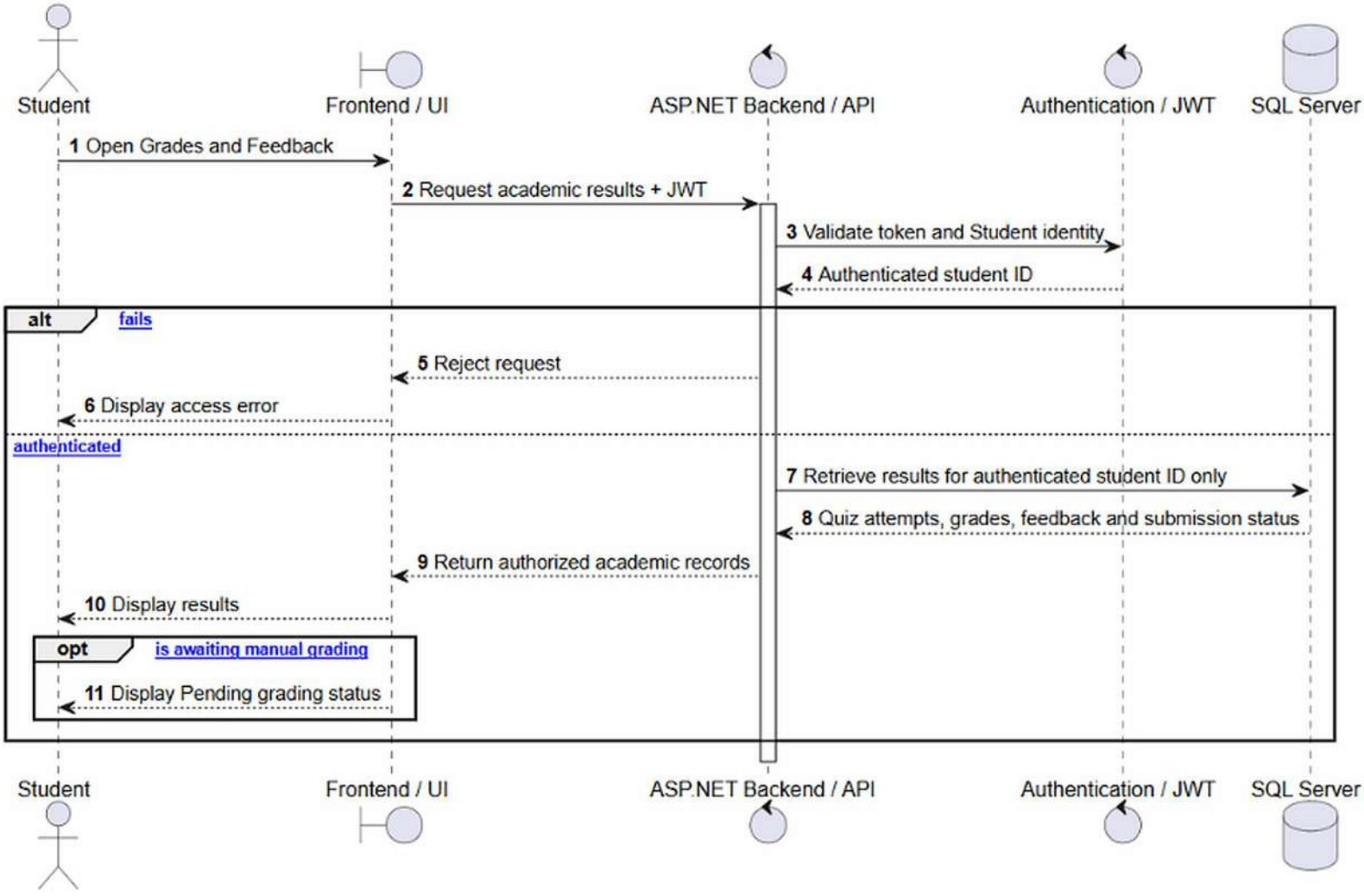


SD-STU-09: Attempt Optional Lesson Quiz

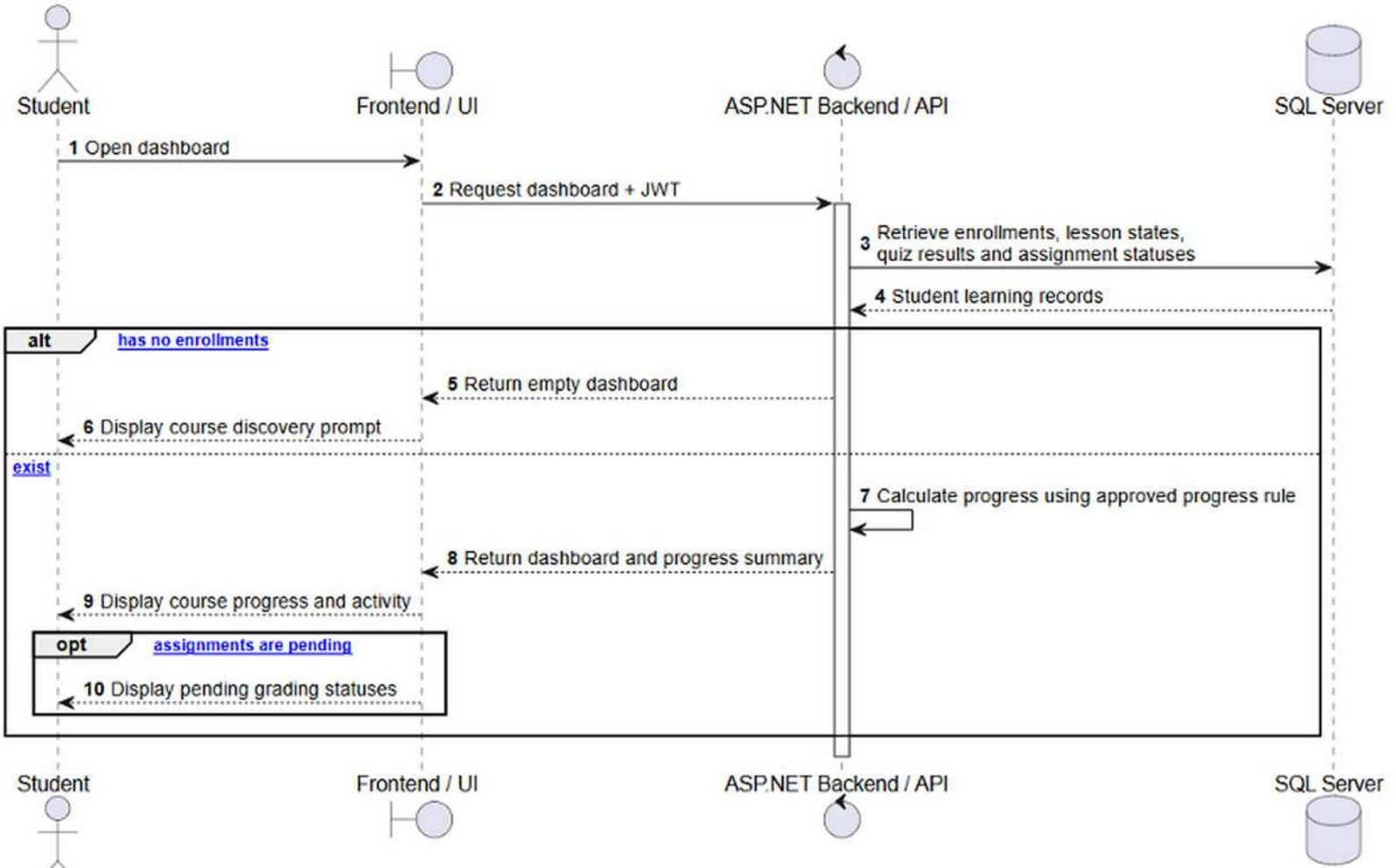



SD-STU-10: Submit a Lesson Assignment
Student
Frontend / UI
ASP.NET Backend / API
SQL Server
1 Open lesson assignment
2 Request assignment + JWT
3 Verify enrollment, lesson access and assignment
4 Assignment and submission rules
alt
assignment exists
5 No assignment available
6 Continue learning
exists and submission allowed
7 Return instructions
8 Display requirements
9 Enter/upload submission
10 Submit assignment
11 Validate submission format and required fields
alt
invalid
12 Return validation errors
13 Request correction
valid
14 Persist assignment submission
15 Submission ID and status
16 Submission confirmation
17 Confirm submitted
Grading may occur later.
The student need not wait for grading
before continuing to the next available step.
unavailable or access denied
18 Reject submission
19 Display reason
Student
Frontend / UI
ASP.NET Backend / API
SQL Server

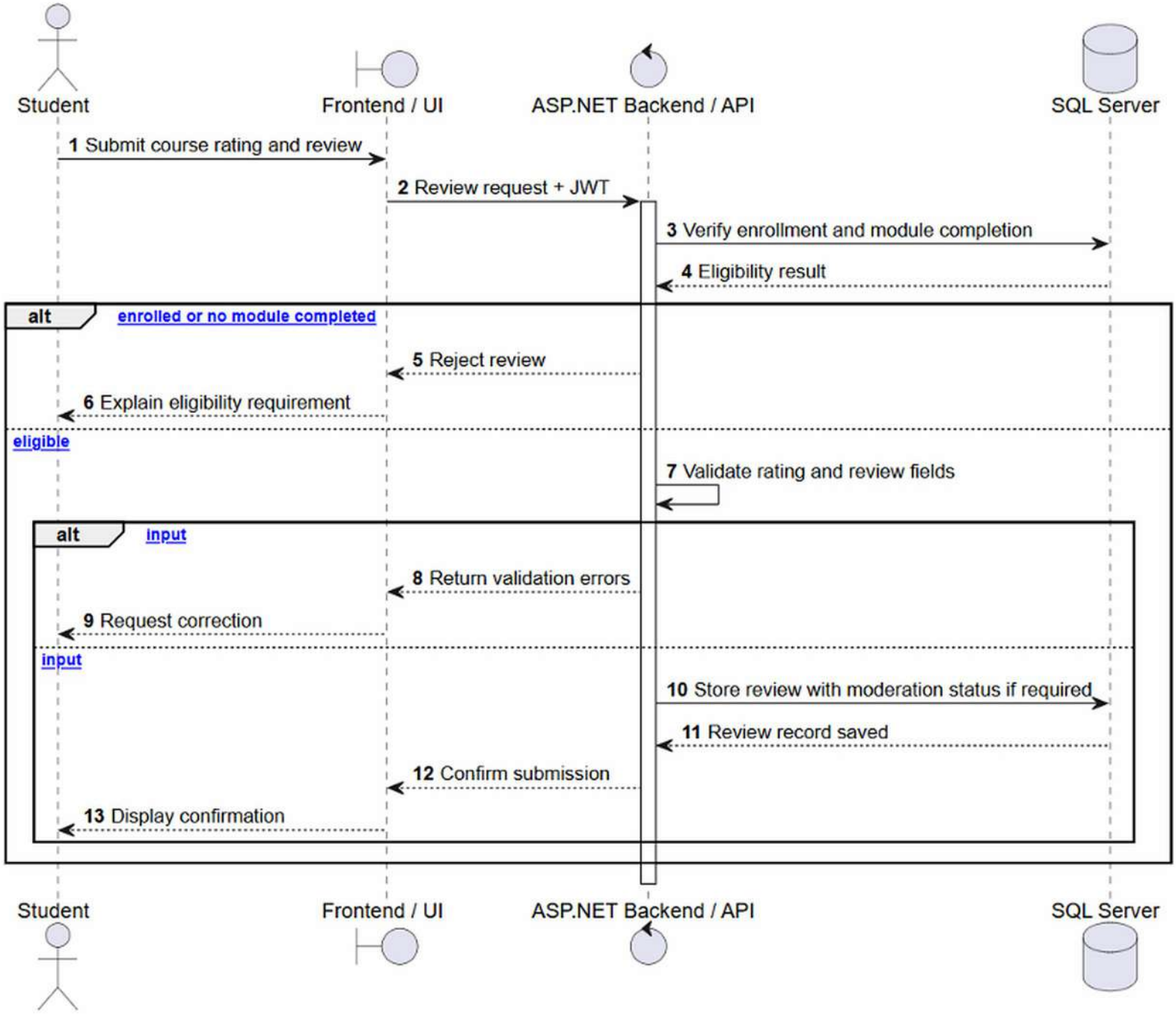
SD-STU-11: View Grades and Feedback



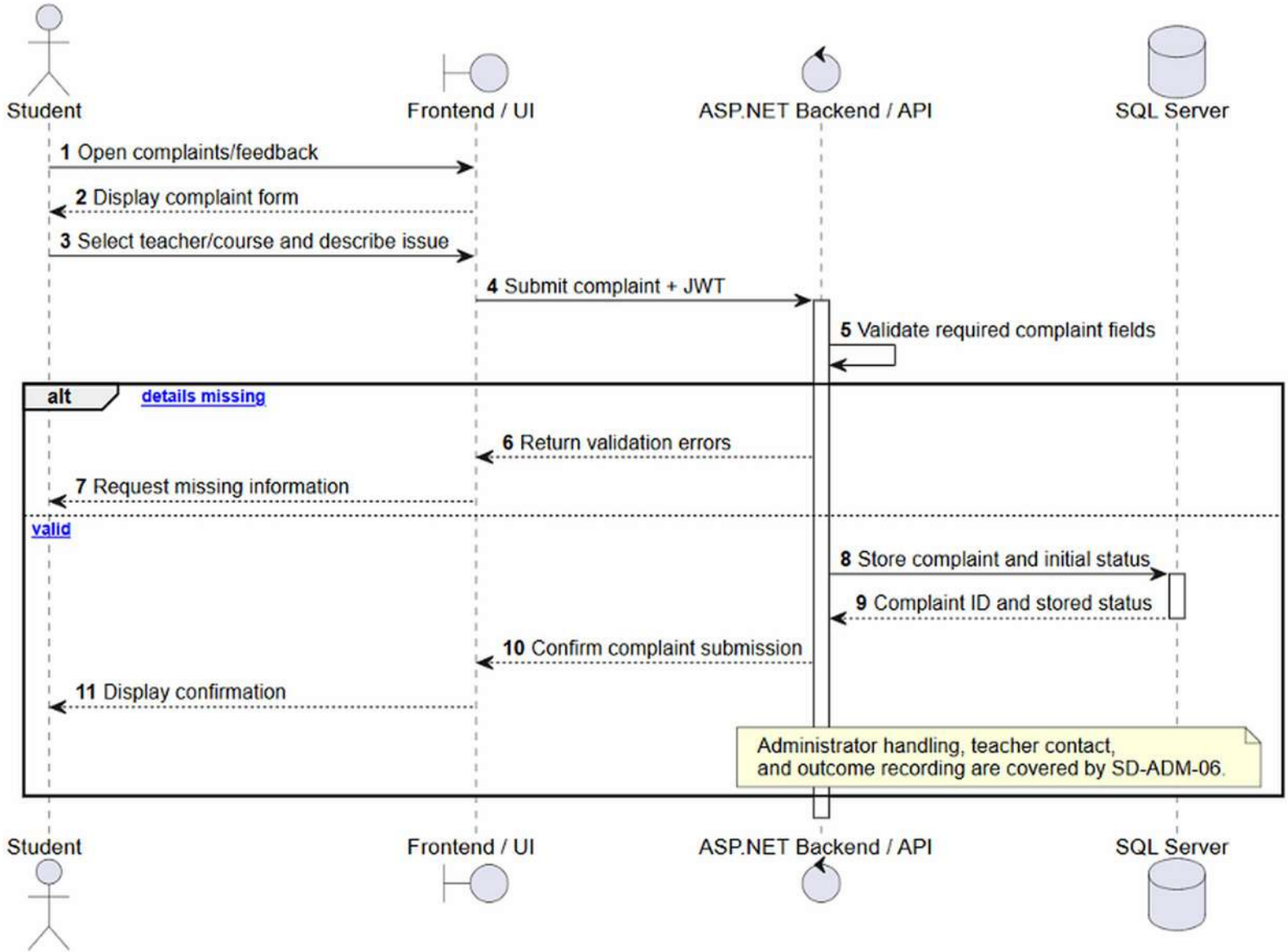
SD-STU-12: Student Dashboard and Learning Progress



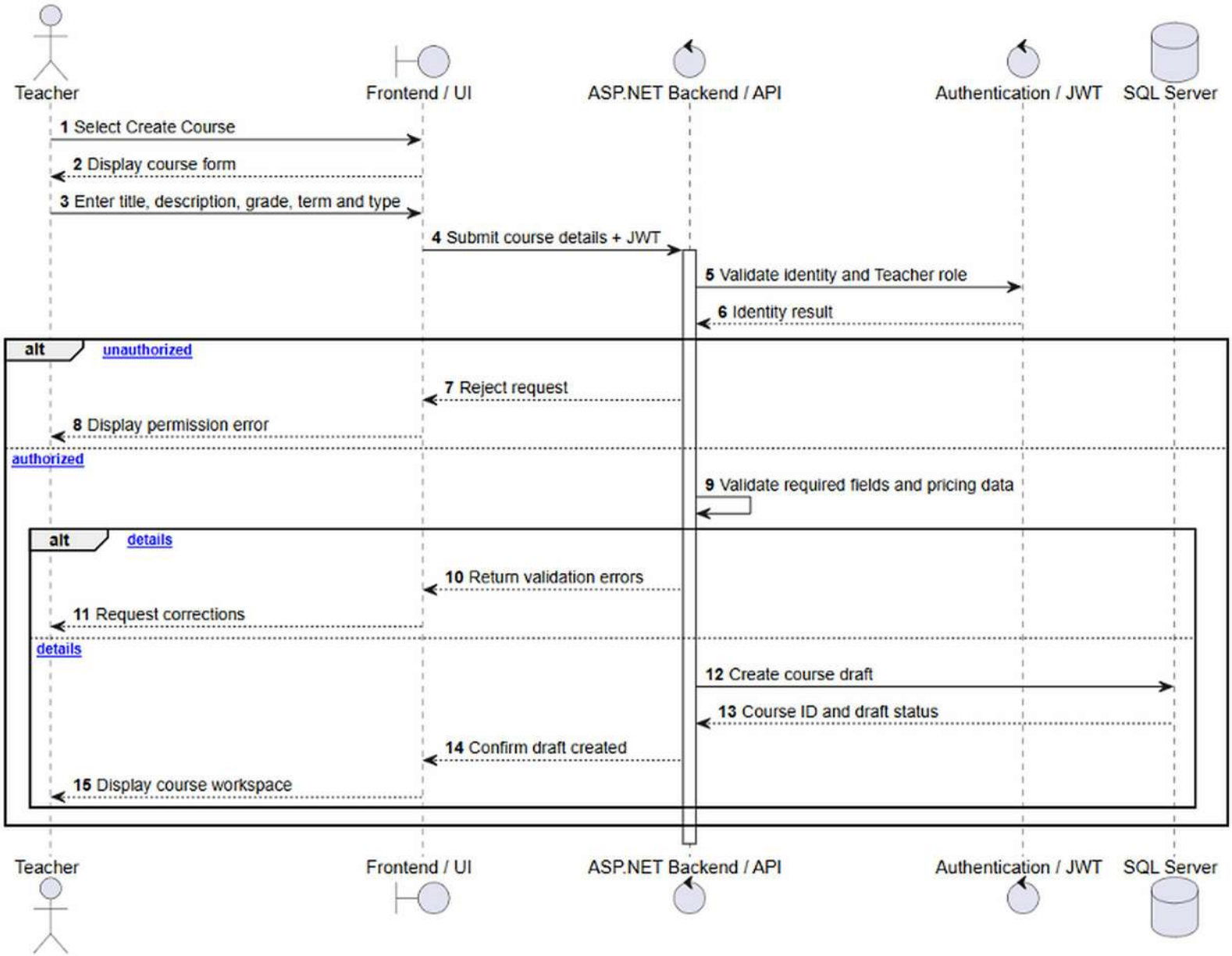
SD-STU-13: Submit Course Review and Rating



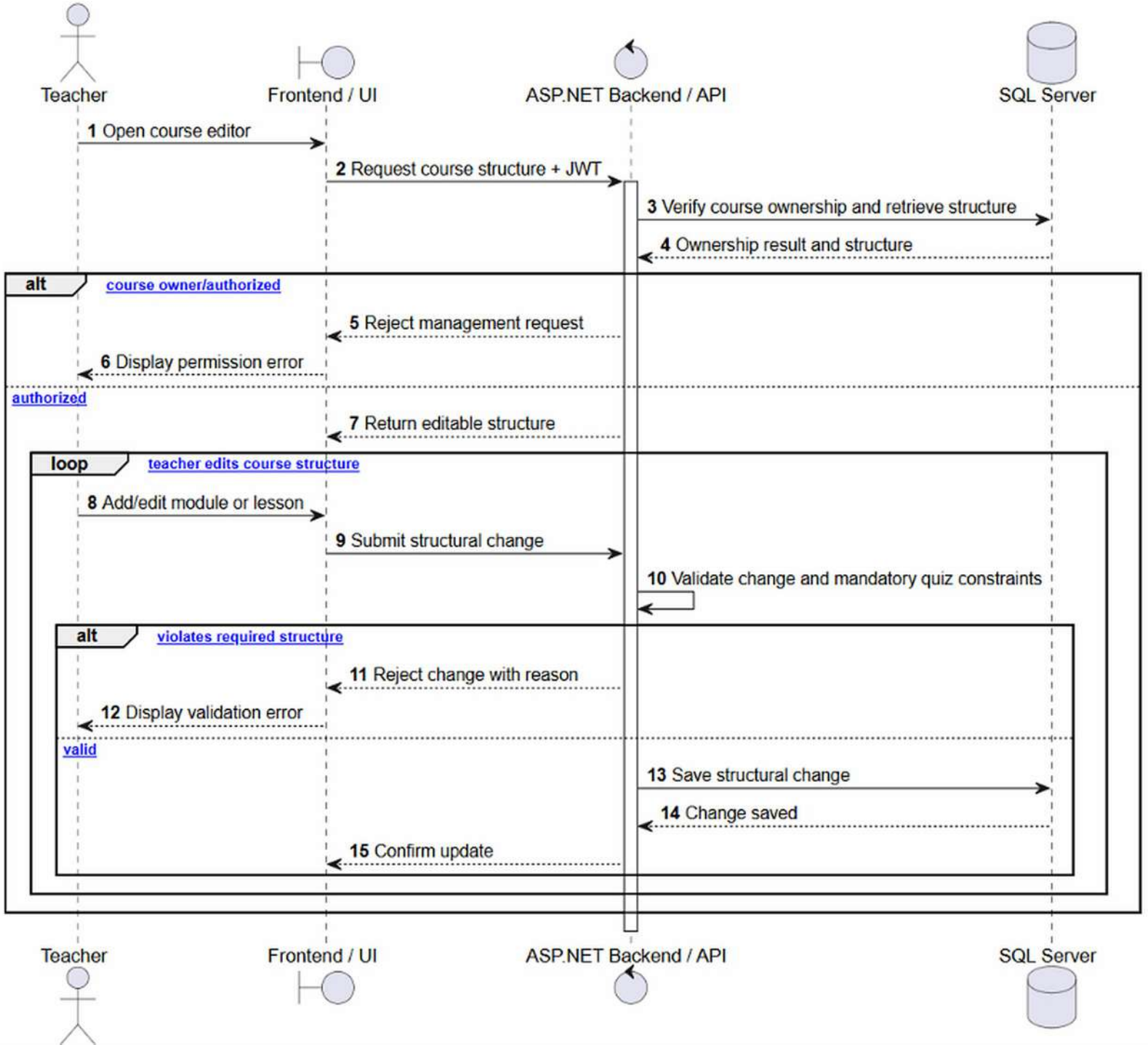
SD-STU-14: Submit Complaint or Feedback



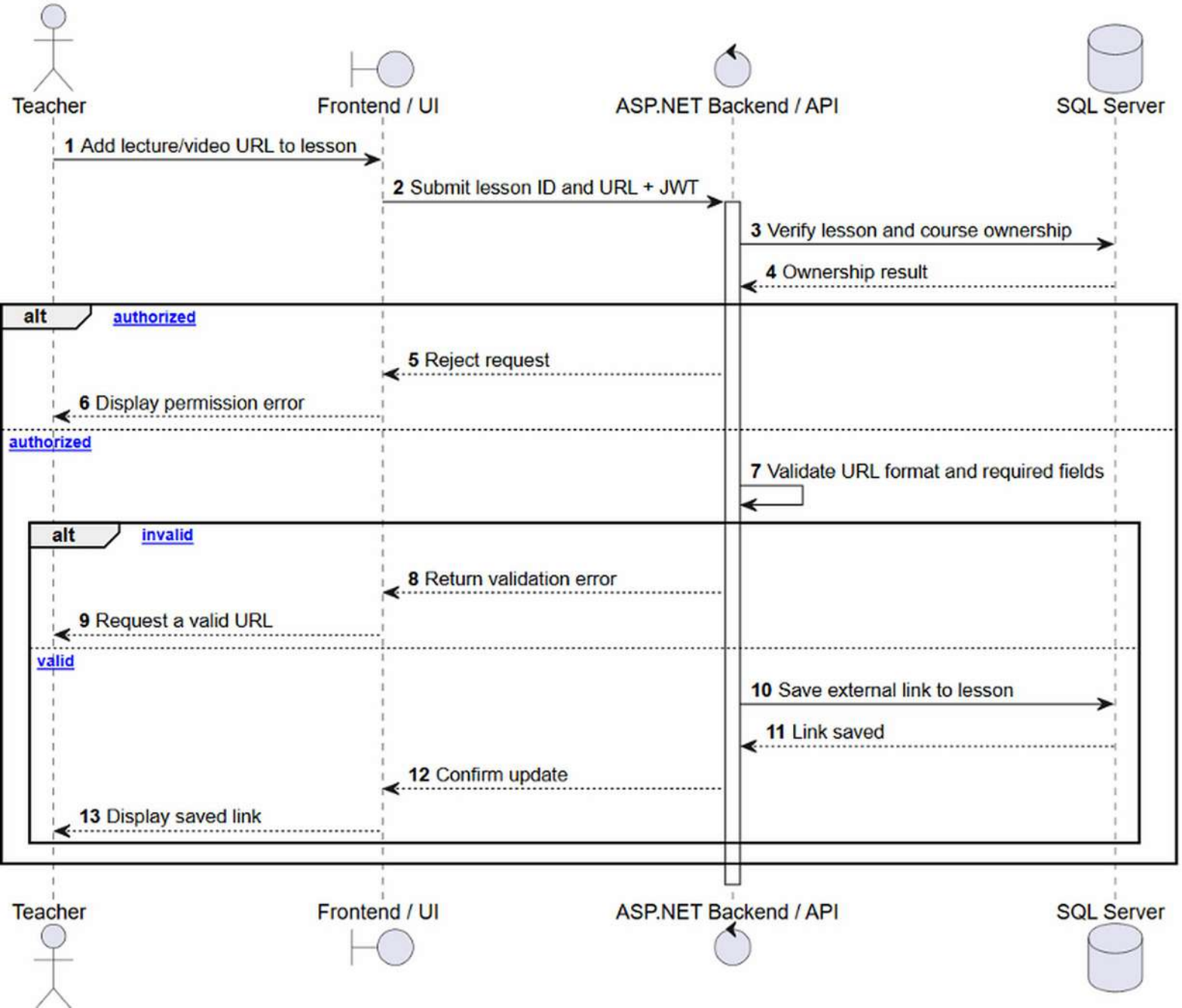
SD-TEA-01: Create a Course



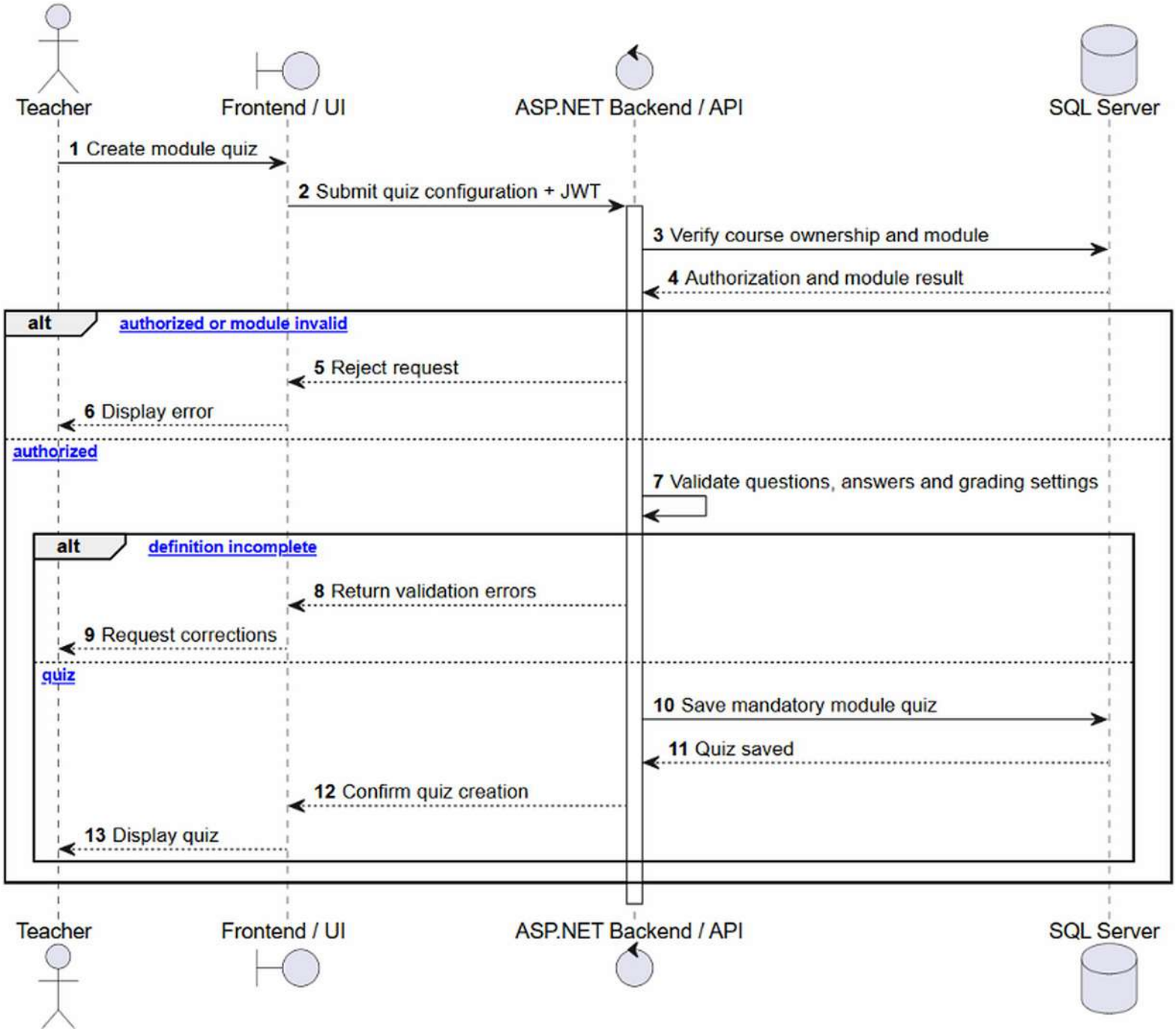
SD-TEA-02: Manage Modules and Lessons



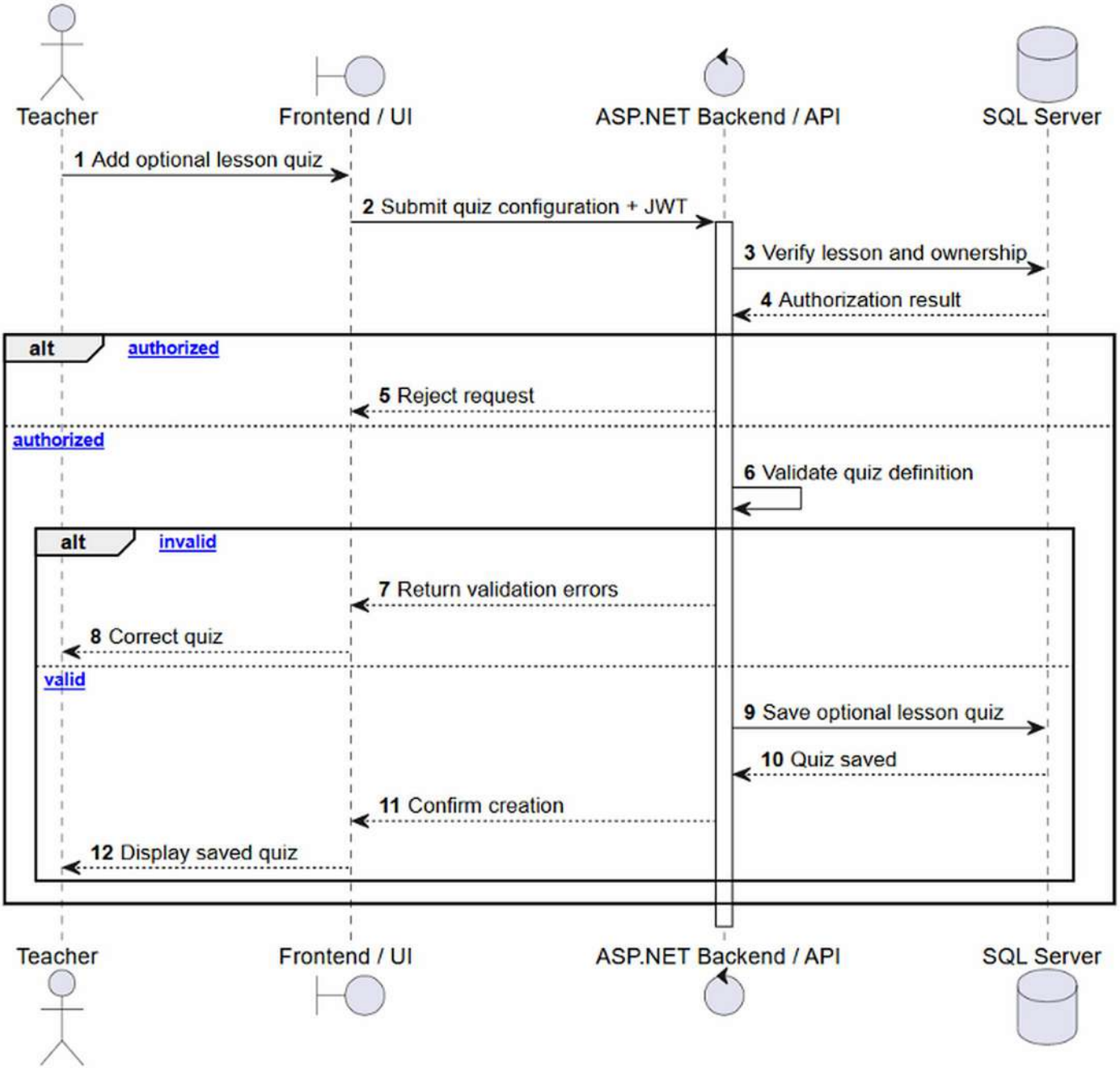
SD-TEA-03: Add External Lecture or Video Link



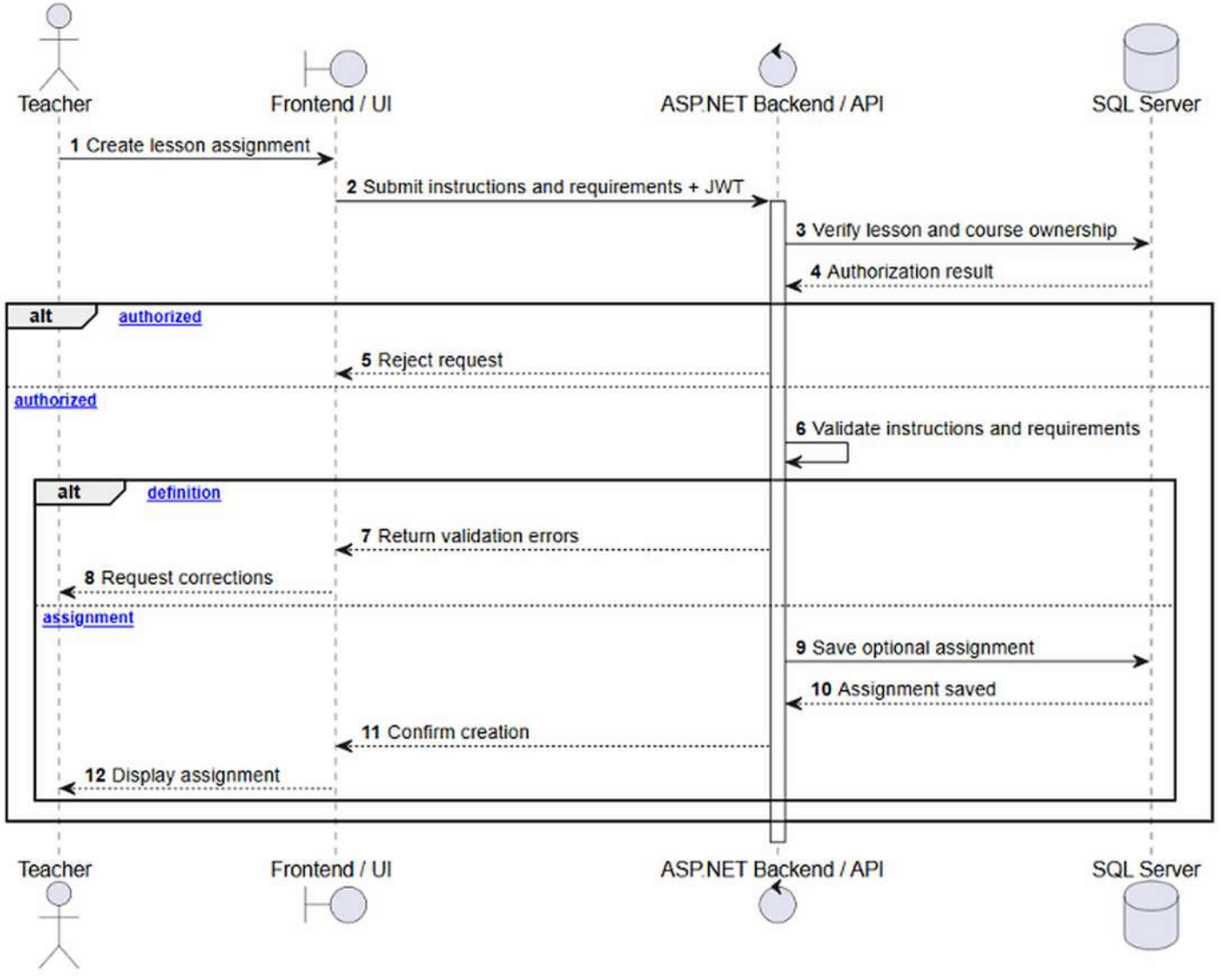
SD-TEA-04: Create Mandatory Module Quiz



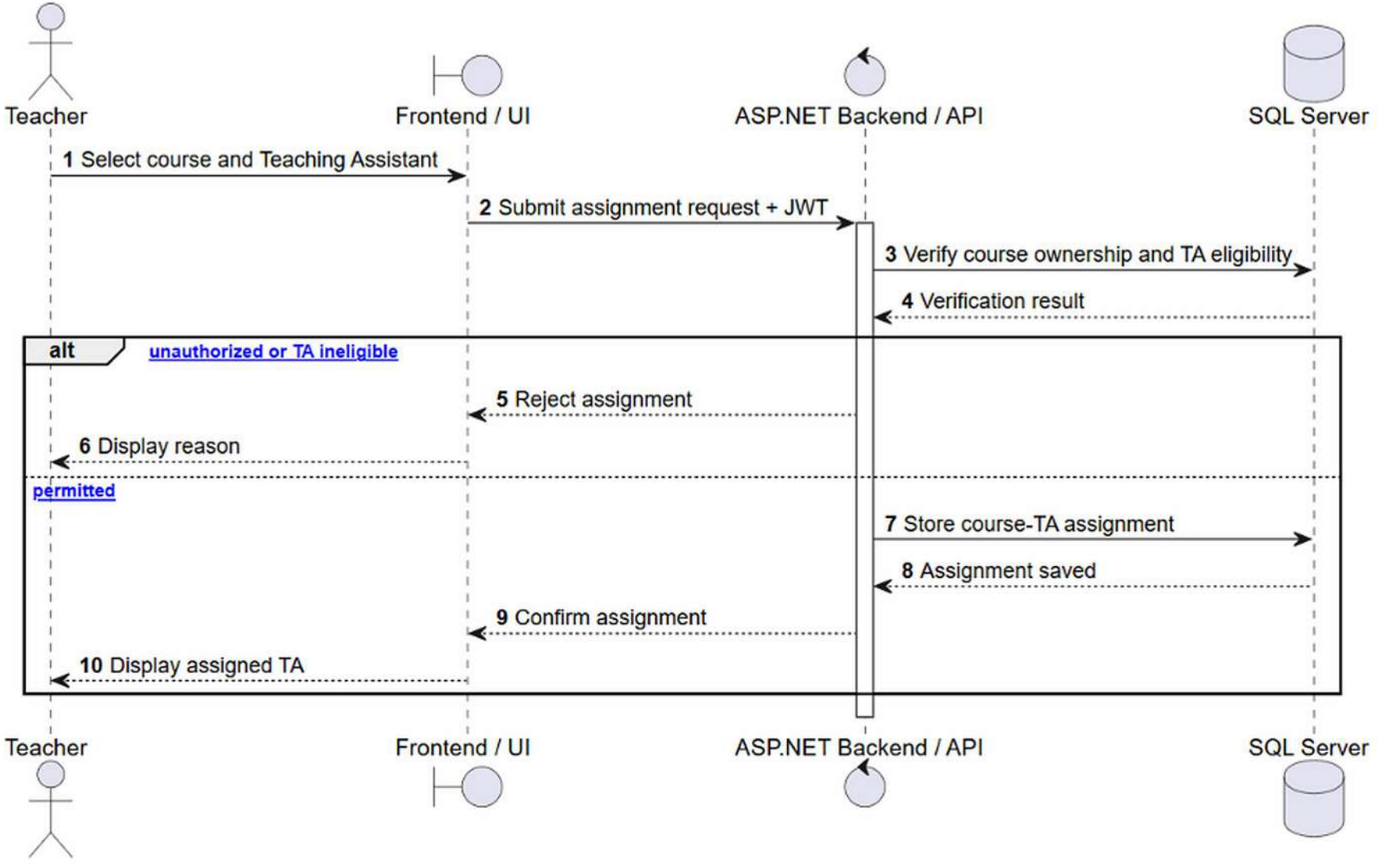
SD-TEA-05: Create Optional Lesson Quiz



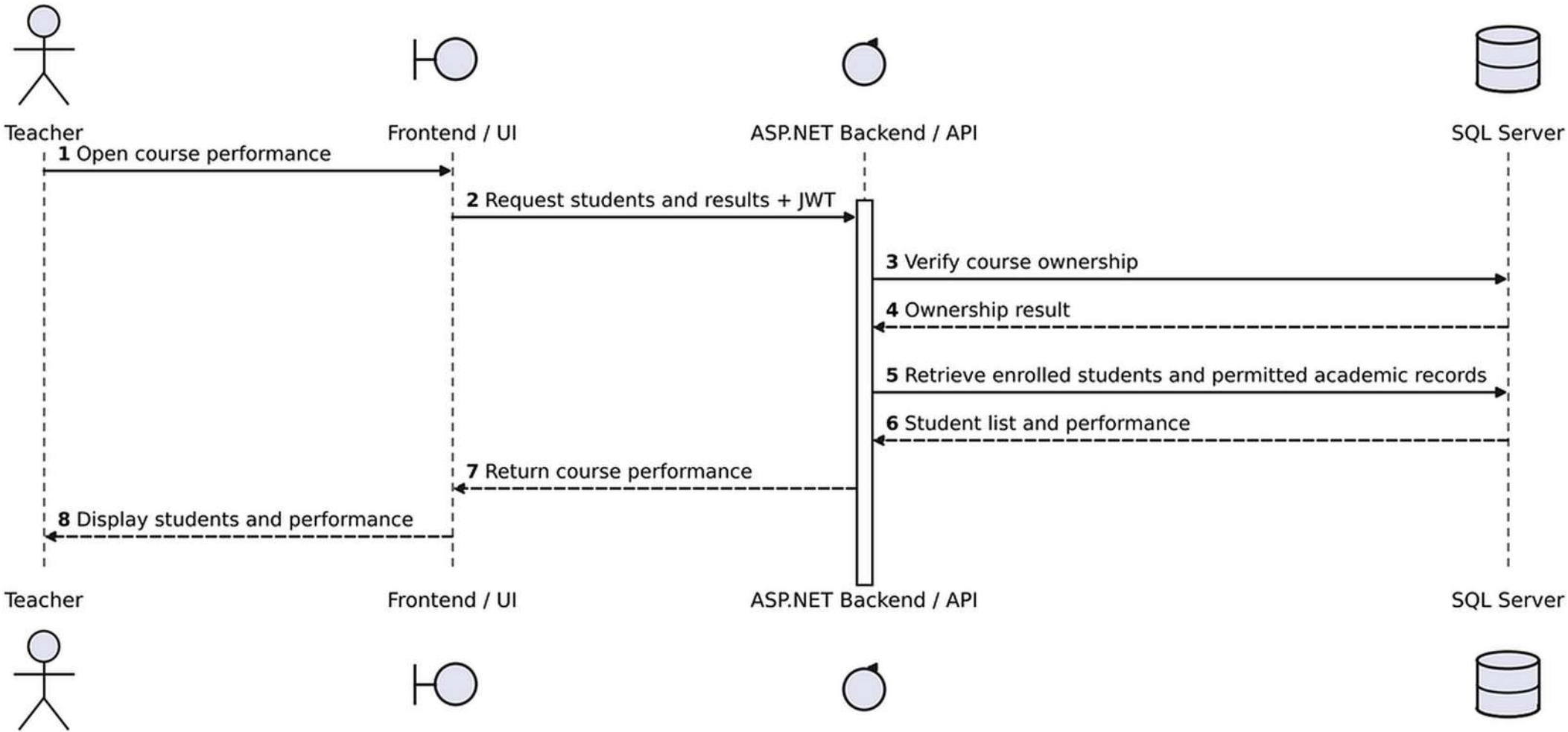
SD-TEA-06: Create Optional Lesson Assignment



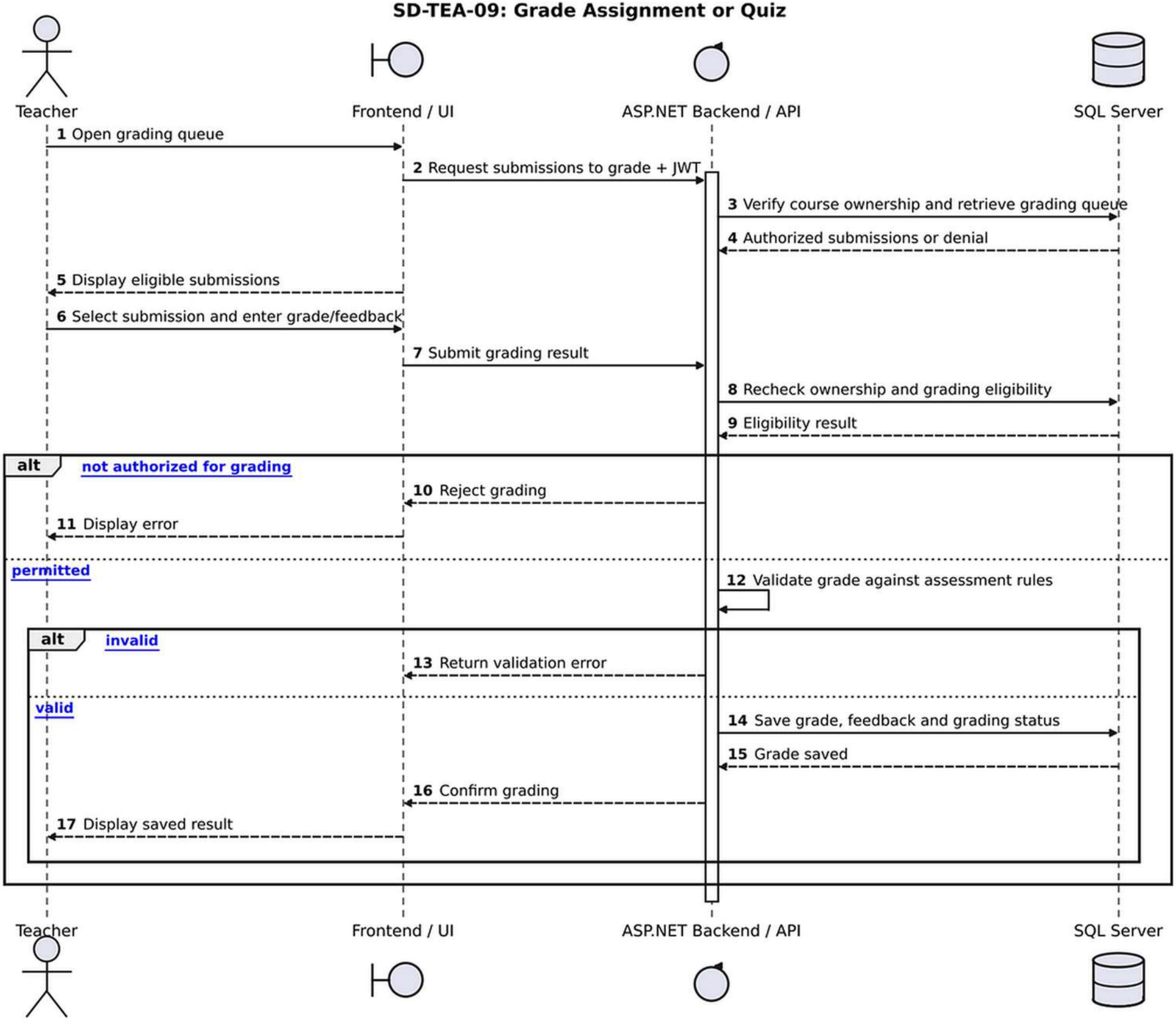
SD-TEA-07: Assign Teaching Assistant



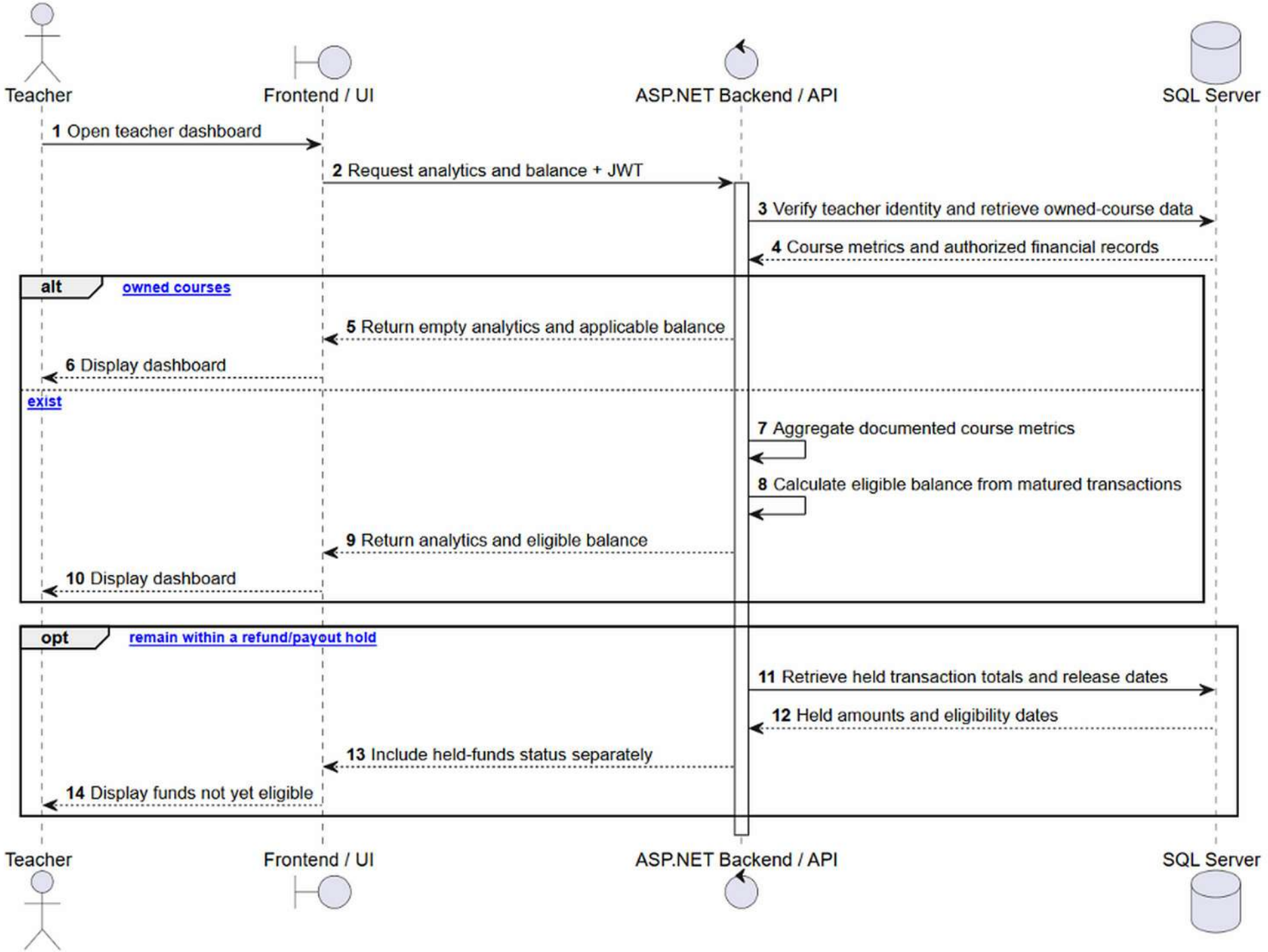
SD-TEA-08: View Students and Performance



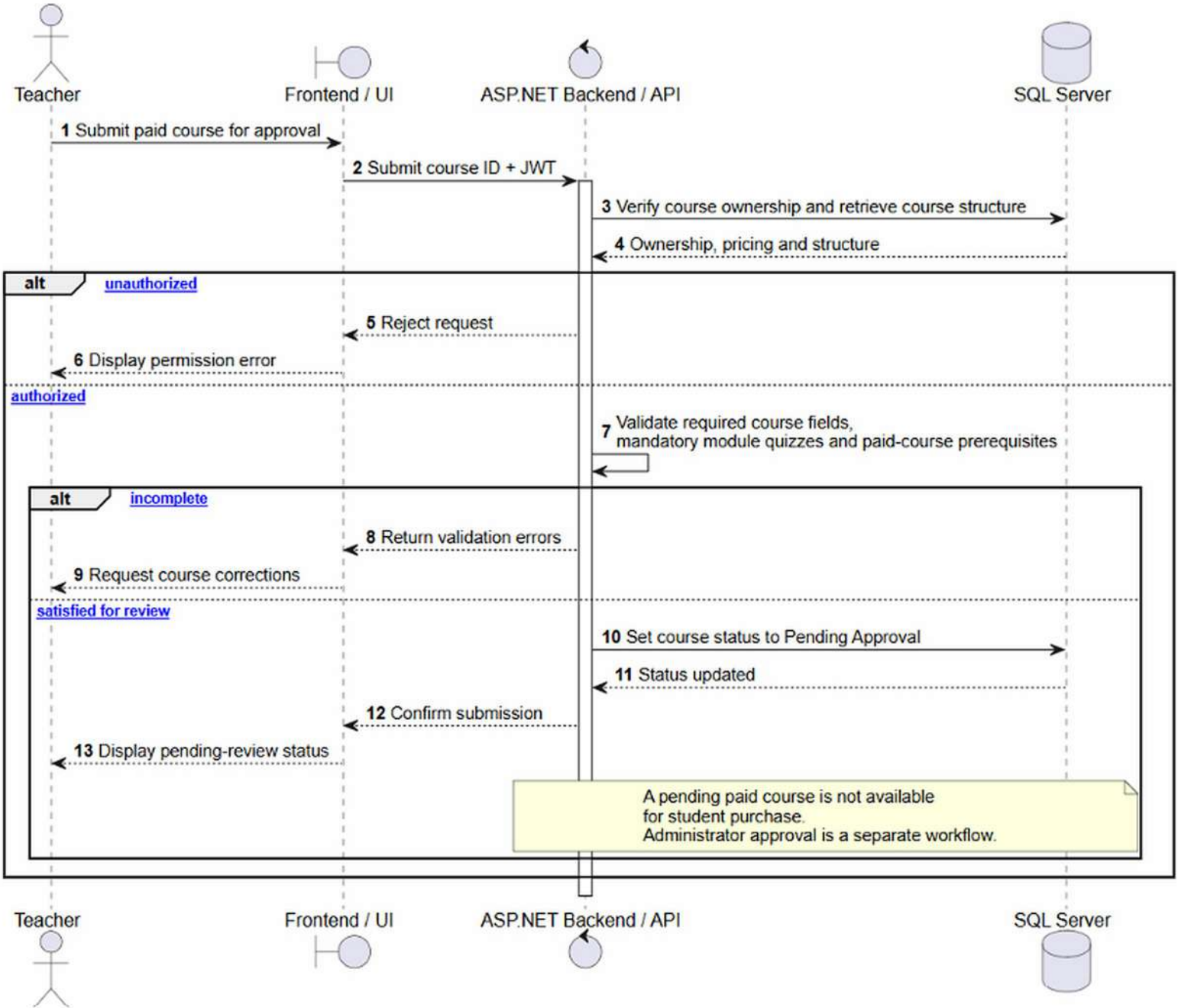
SD-TEA-09: Grade Assignment or Quiz



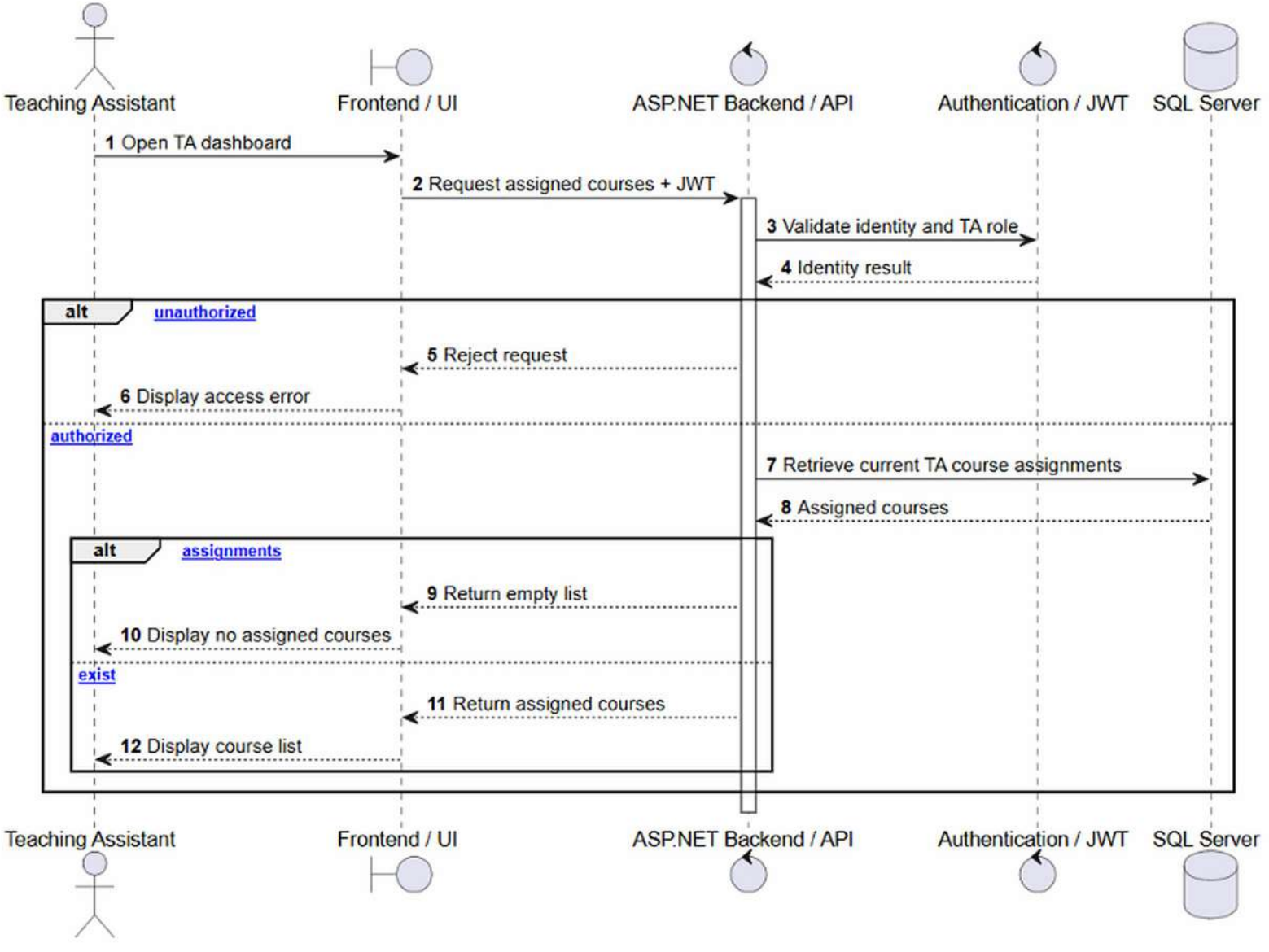
SD-TEA-10: Course Analytics and Eligible Balance



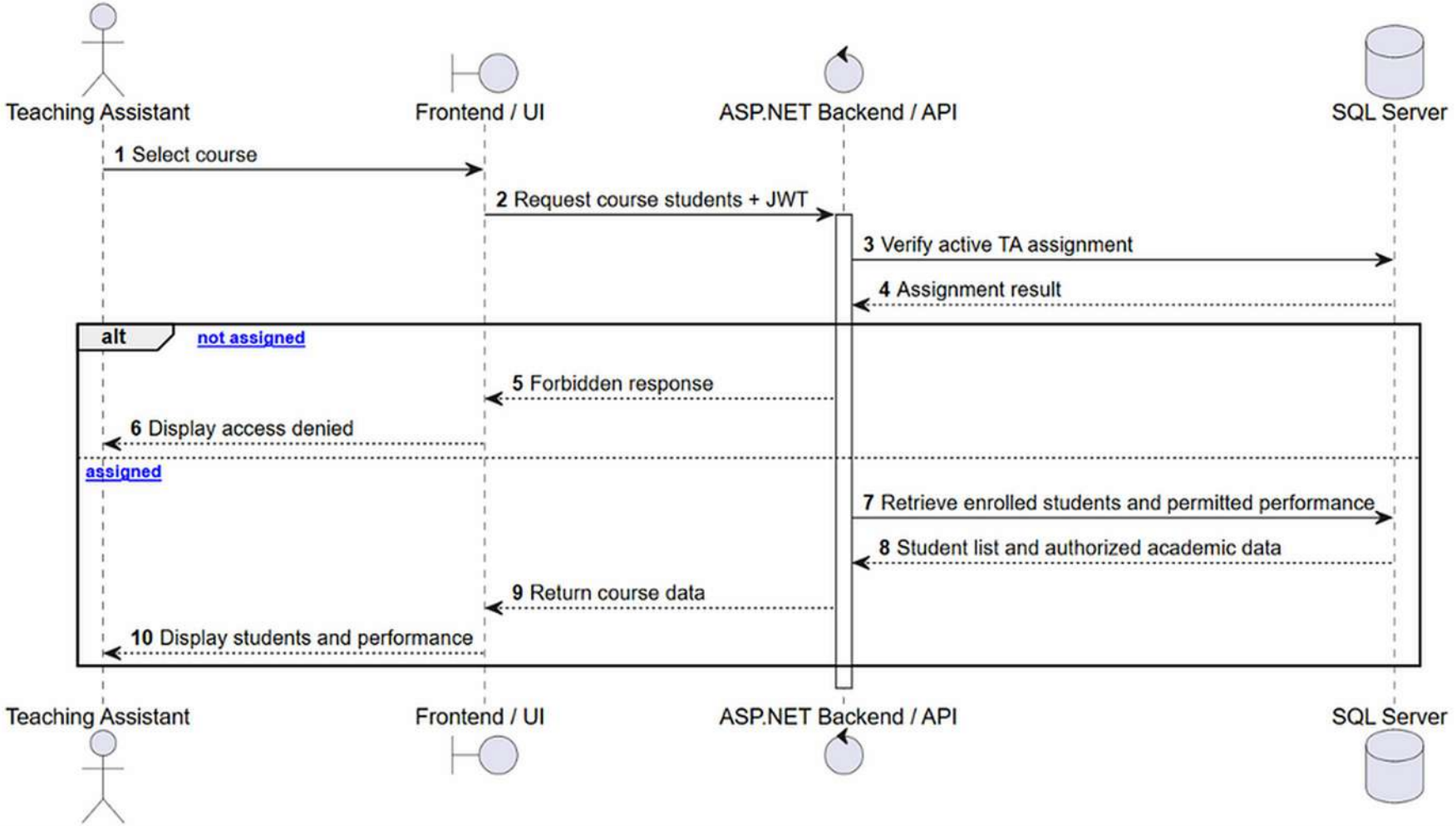
SD-TEA-11: Submit Paid Course for Approval



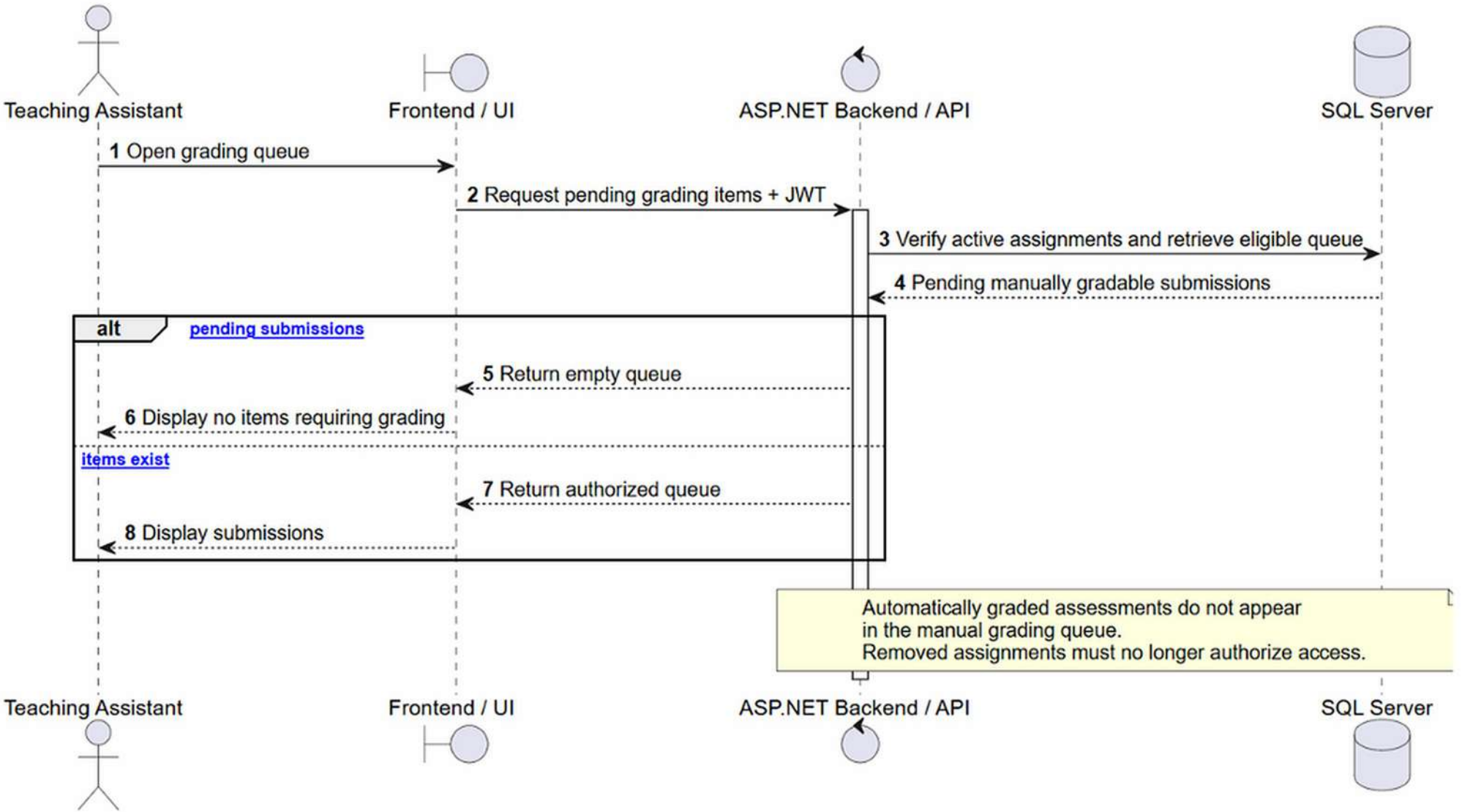
SD-TA-01: View Assigned Courses



SD-TA-02: View Students in Assigned Course



SD-TA-03: View Pending Grading Queue



SD-TA-04: Grade Authorized Assignment

Teaching Assistant
Frontend / UI
ASP.NET Backend / API
SQL Server

1 Select assignment submission
2 Request submission + JWT
3 Verify active assignment and grading eligibility
4 Authorization result and submission

alt or assignment unavailable
5 Reject request
6 Display access error

authorized
7 Return submission for grading
8 Display submission
9 Enter grade and feedback
10 Submit grading result

alt revoked or submission unavailable
11 Reject grading
12 Display error

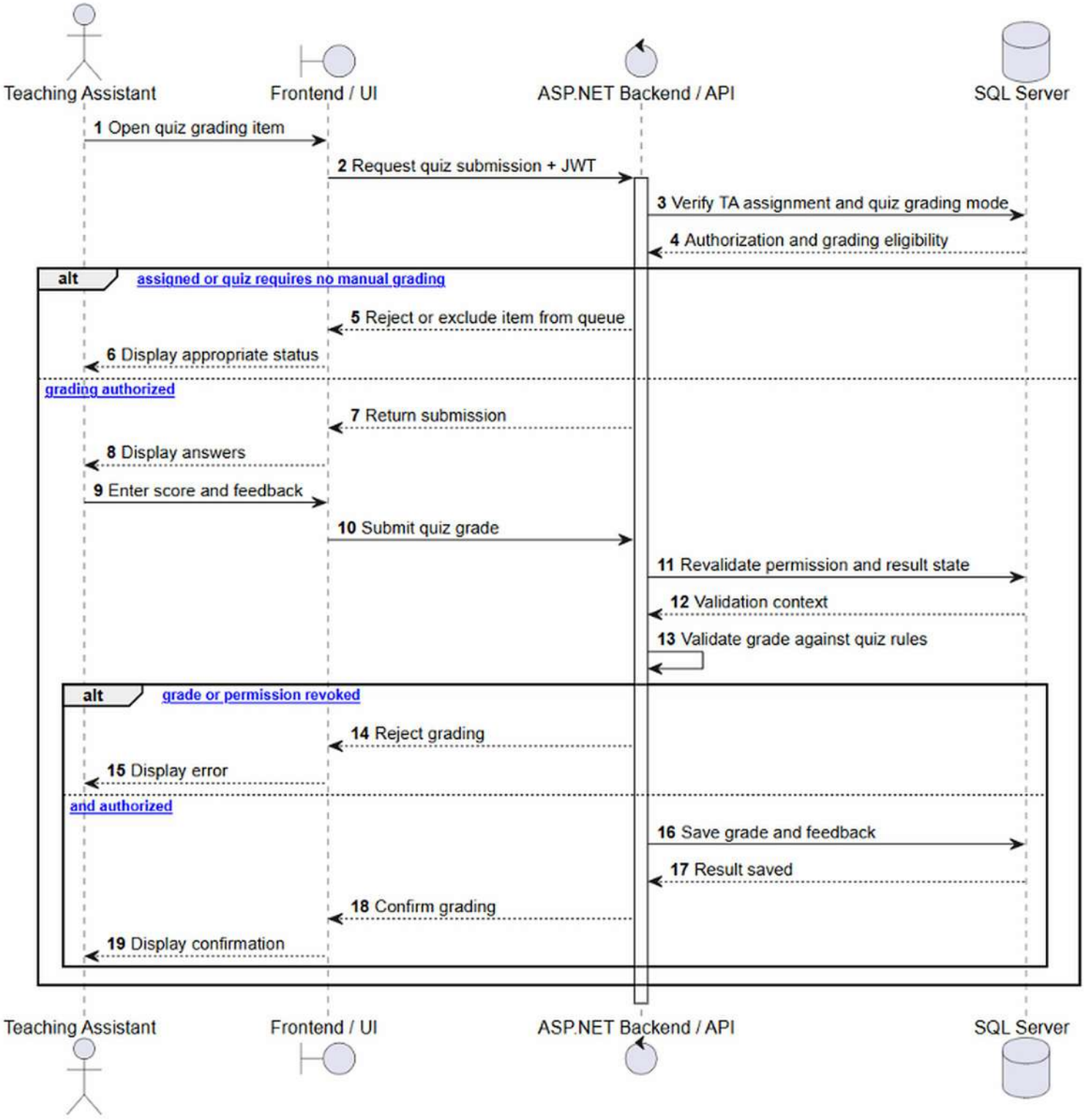
permitted
13 Validate grade

alt invalid
14 Return validation error
15 Display error

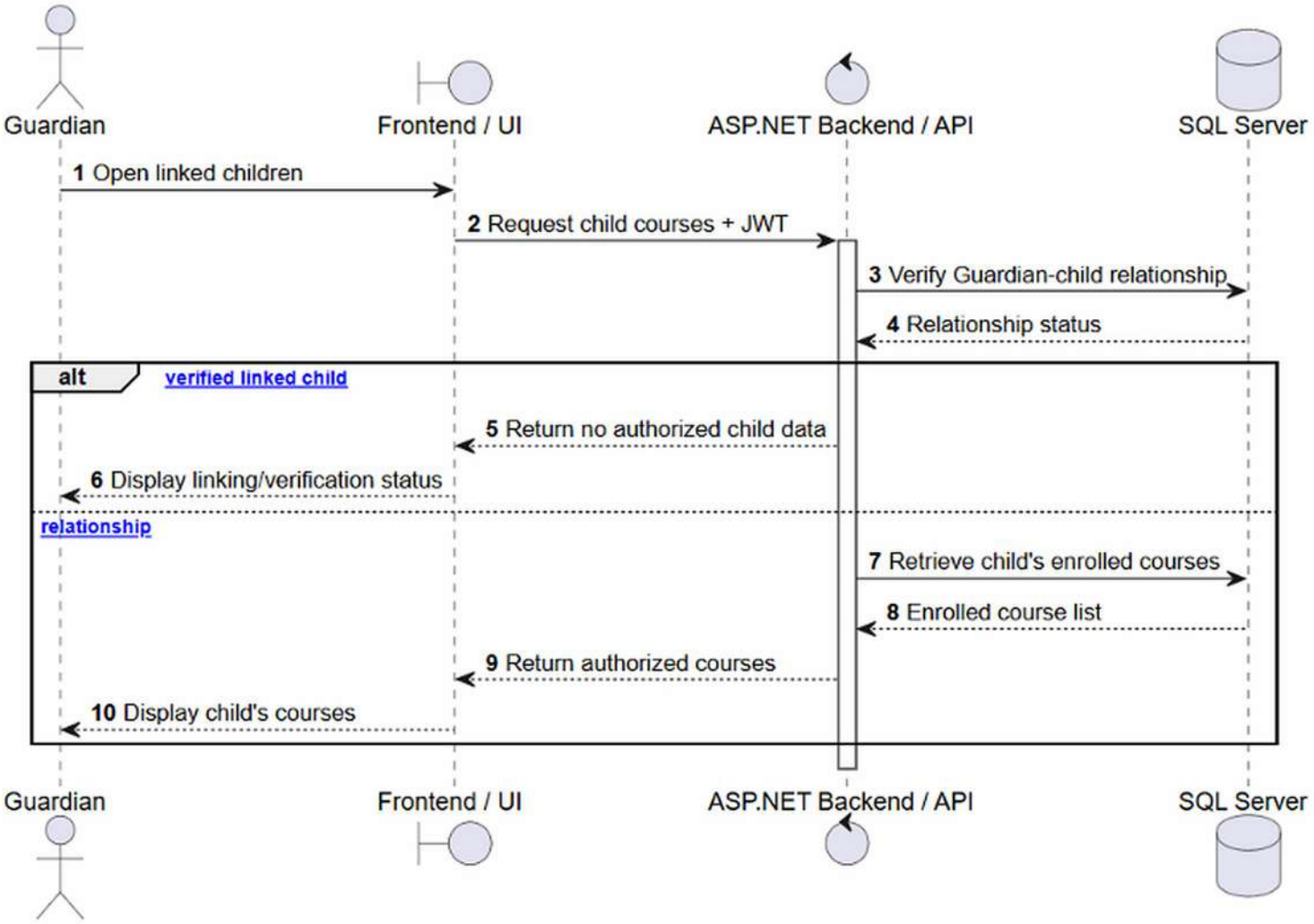
valid
16 Save grade, feedback and grading status
17 Grade saved
18 Confirm grading
19 Display confirmation

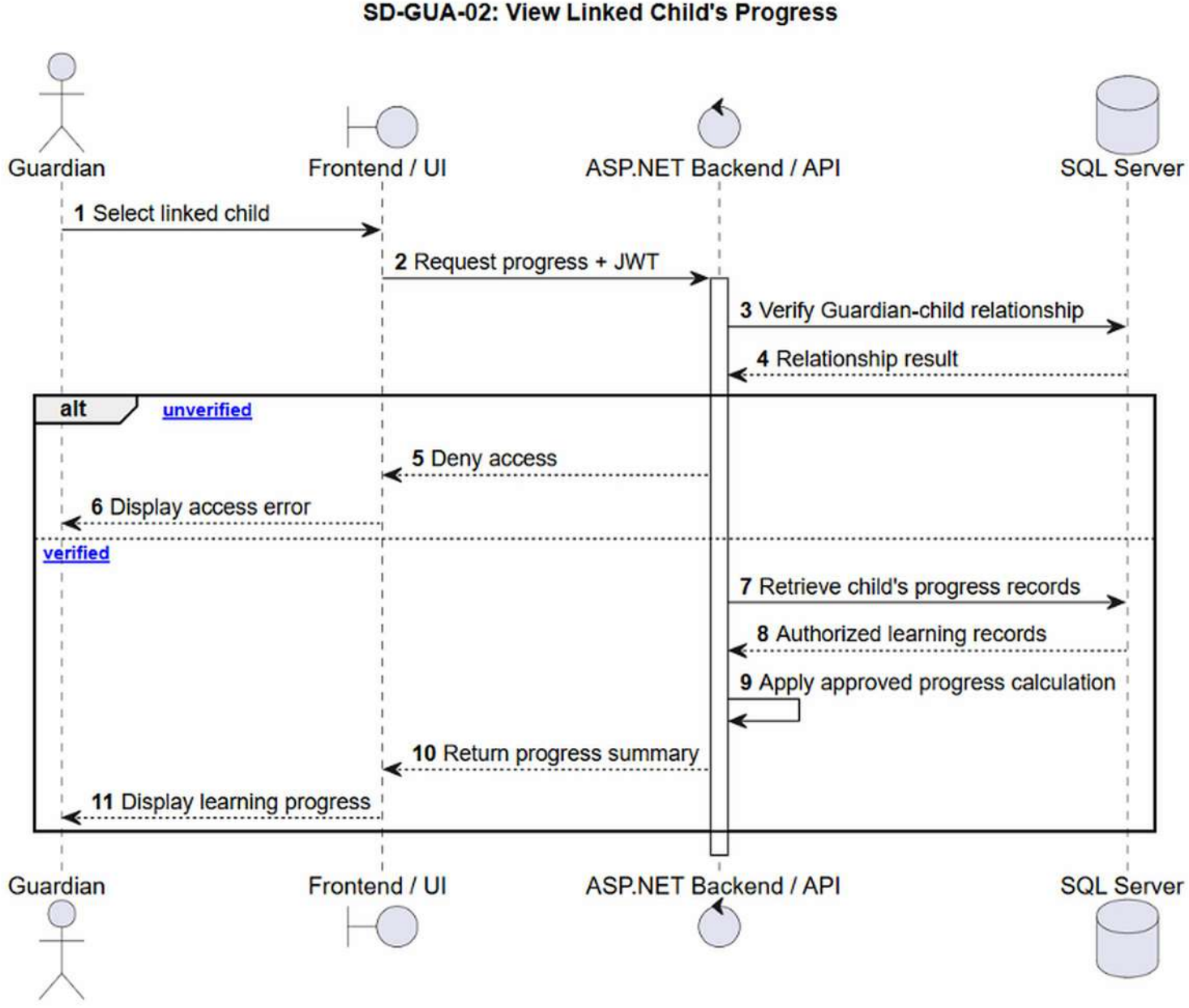
Teaching Assistant
Frontend / UI
ASP.NET Backend / API
SQL Server

SD-TA-05: Grade Authorized Quiz

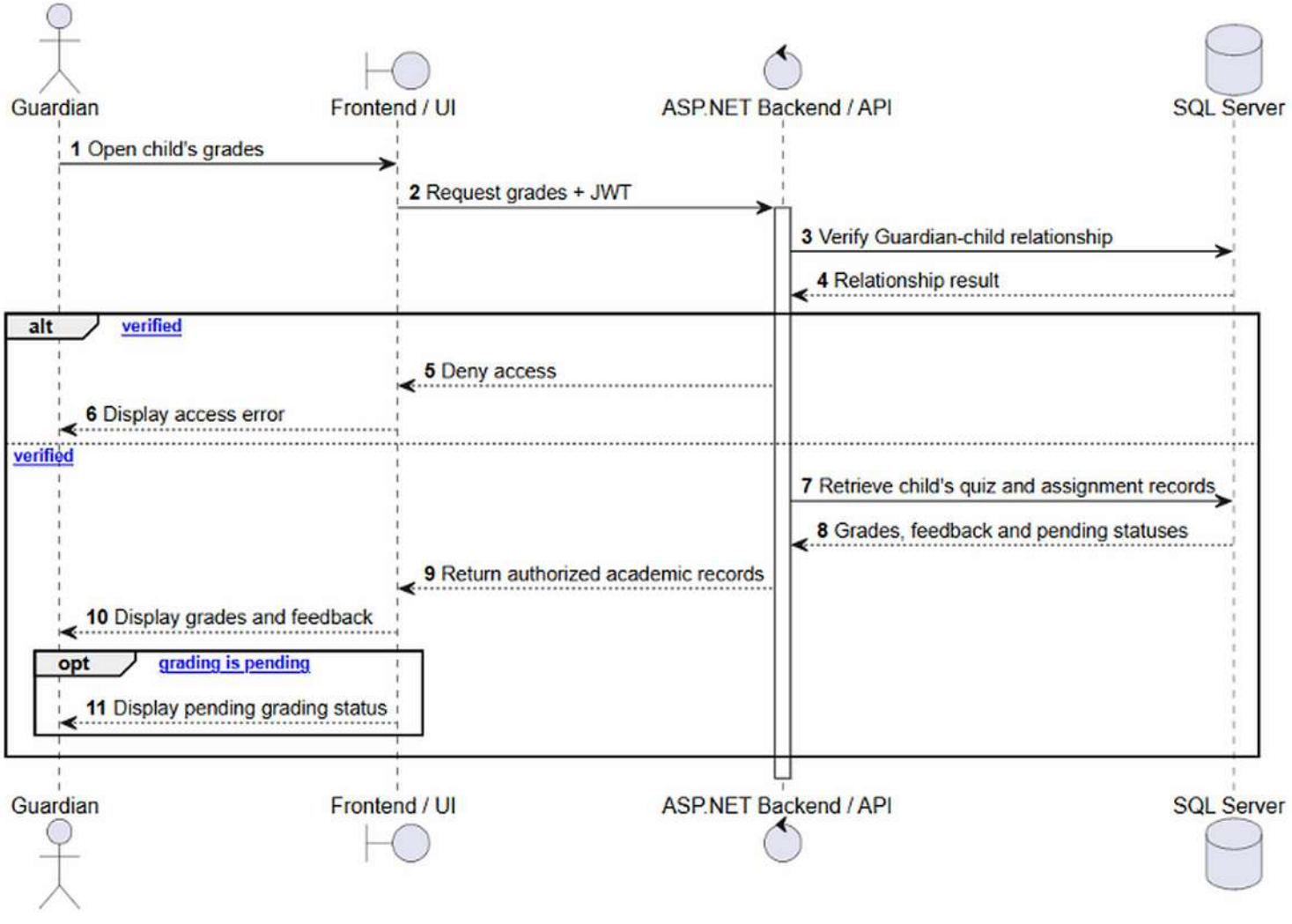


SD-GUA-01: View Linked Child's Courses




SD-GUA-02: View Linked Child's Progress
Guardian
Frontend / UI
ASP.NET Backend / API
SQL Server
1 Select linked child
2 Request progress + JWT
3 Verify Guardian-child relationship
4 Relationship result
alt
unverified
5 Deny access
6 Display access error
verified
7 Retrieve child's progress records
8 Authorized learning records
9 Apply approved progress calculation
10 Return progress summary
11 Display learning progress
Guardian
Frontend / UI
ASP.NET Backend / API
SQL Server

SD-GUA-03: View Child's Grades



SD-ADM-01: Review and Decide on Paid Course

Administrator
Frontend / UI
ASP.NET Backend / API
Authentication / JWT
SQL Server

1 Open pending paid courses
2 Request pending queue + JWT
3 Validate Administrator role
4 Authorization result

alt not authorized
5 Reject request
6 Display access error

authorized
7 Retrieve pending course and review data
8 Course details and status
9 Return review information
10 Display course structure, assessments and price
11 Record review decision
12 Submit decision
13 Verify readiness and acknowledgement
14 Validation result

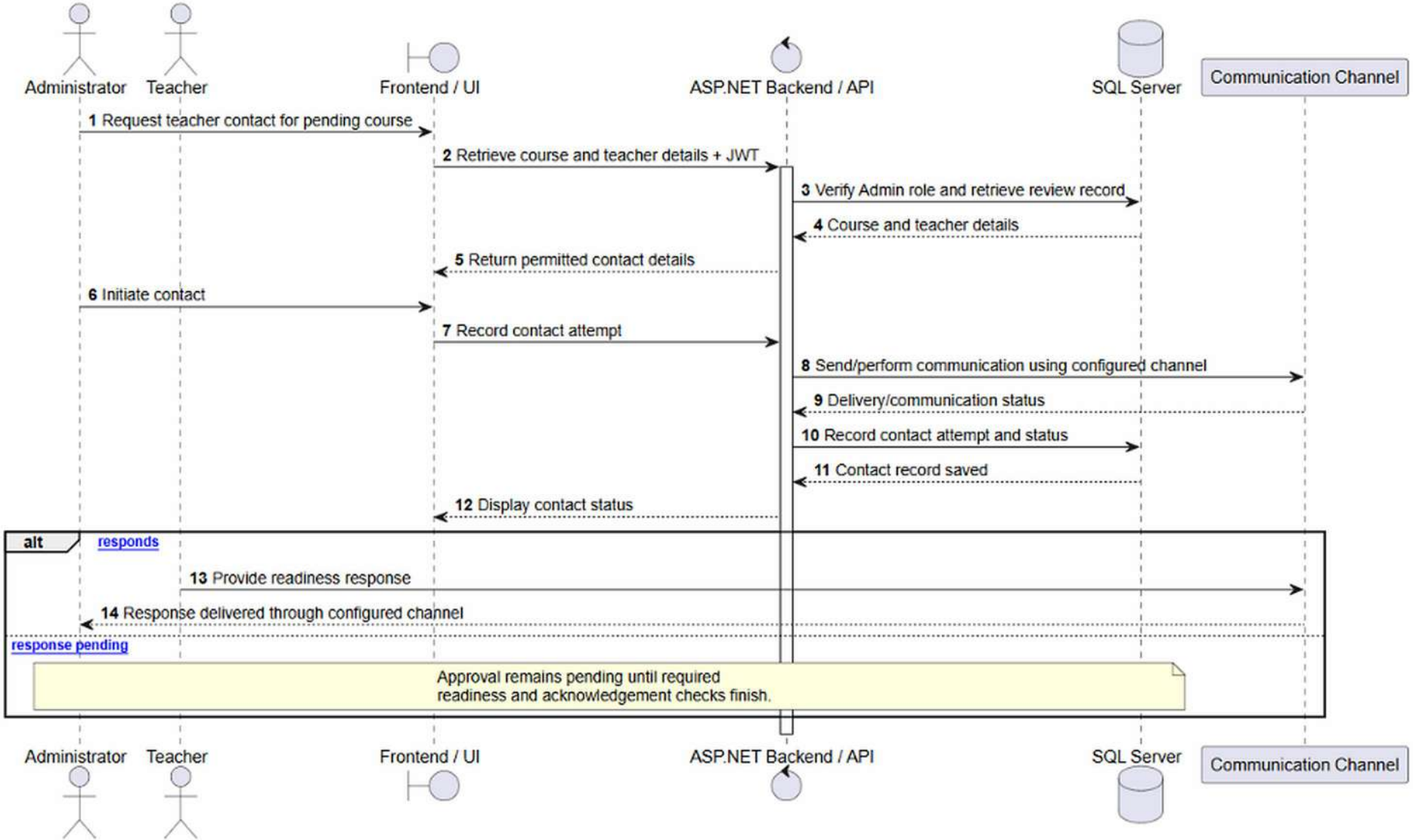
alt first module not prepared or acknowledgement missing
15 Withhold approval and request changes
16 Display readiness issue

rejected or changes requested
17 Record rejection or request for changes
18 Decision recorded
19 Confirm decision
20 Display decision

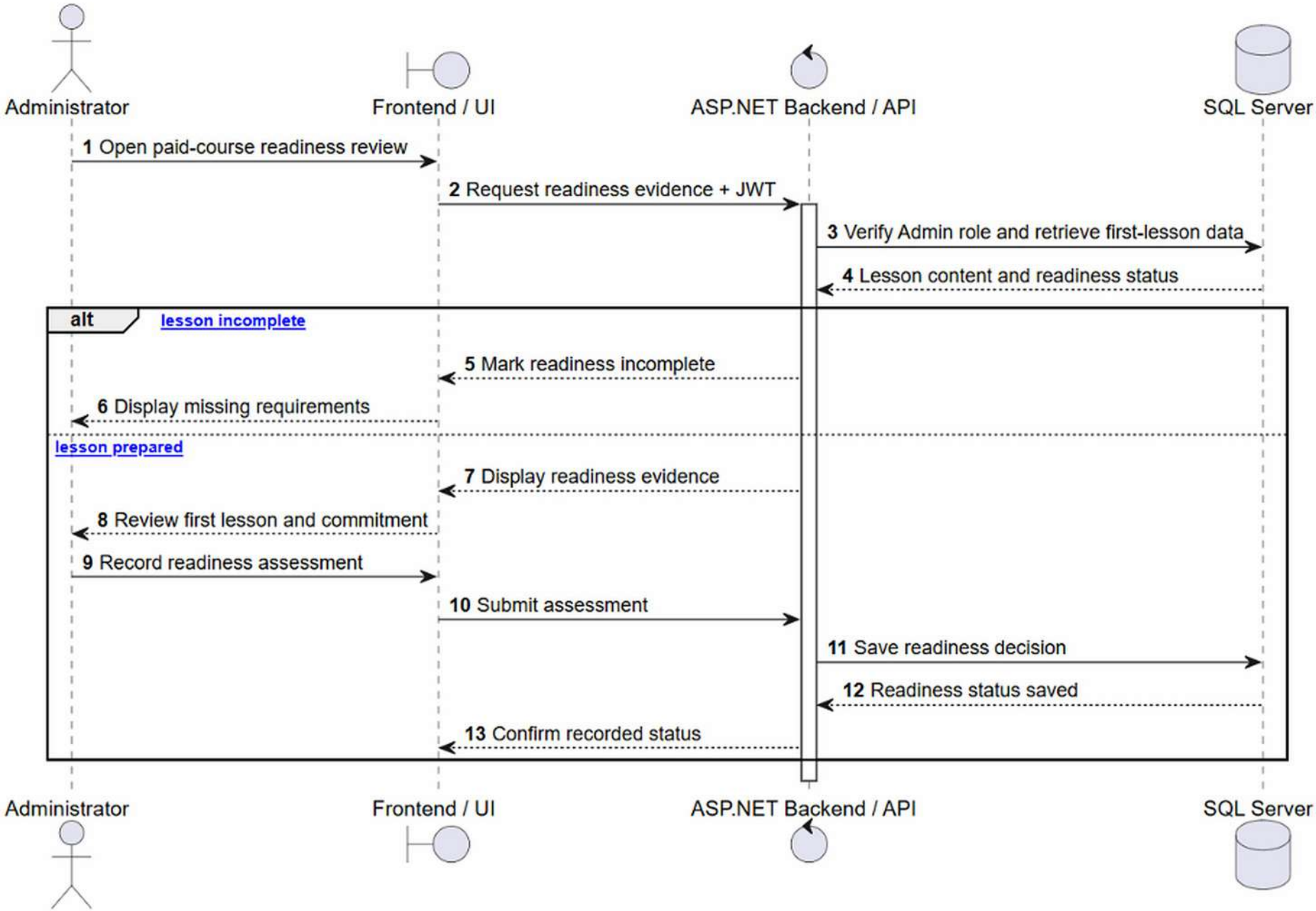
approval conditions satisfied
21 Set course status to Approved
22 Approval recorded
23 Confirm approval
24 Display approval confirmation

Administrator
Frontend / UI
ASP.NET Backend / API
Authentication / JWT
SQL Server

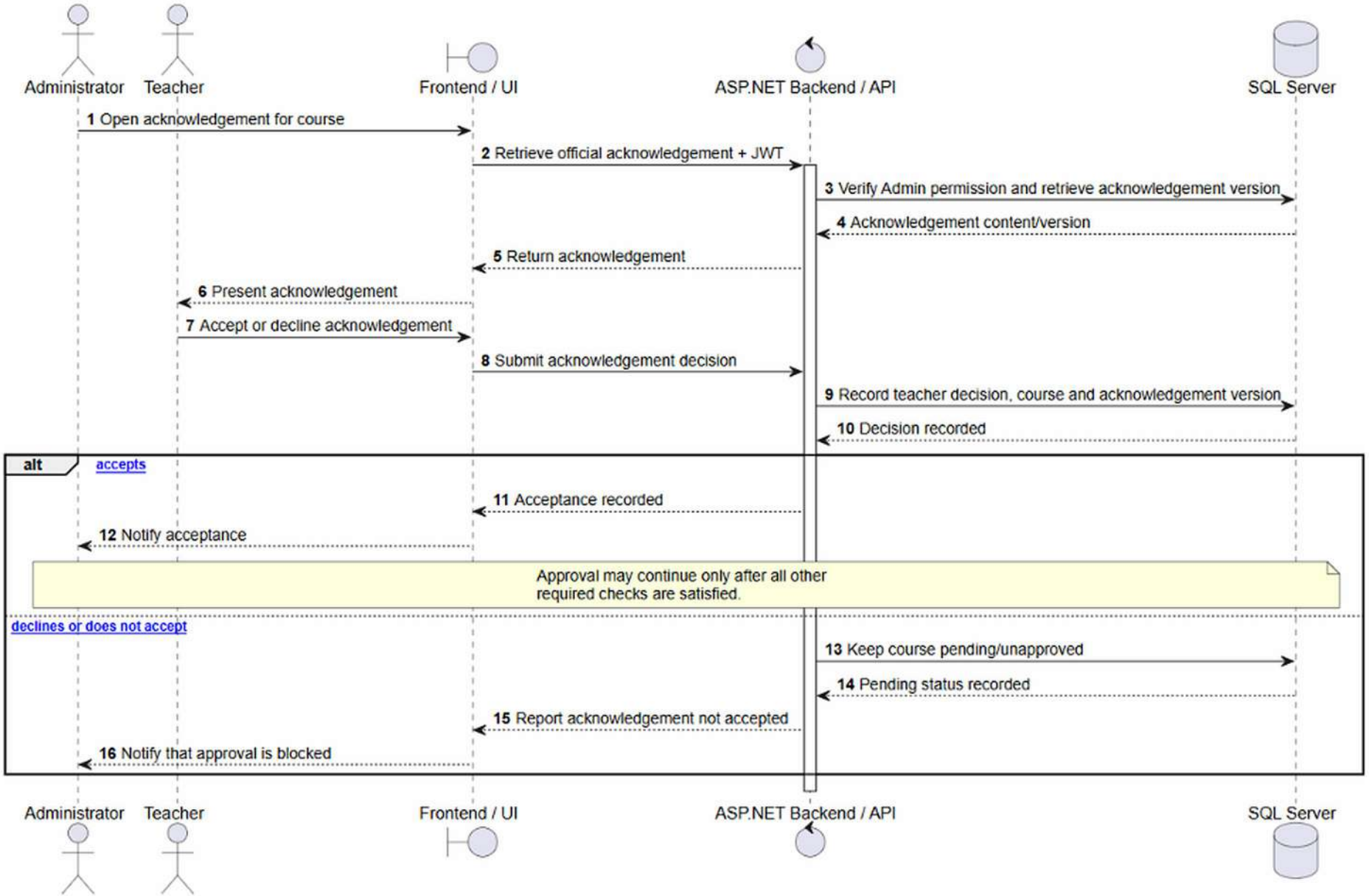
SD-ADM-02: Contact Teacher During Approval



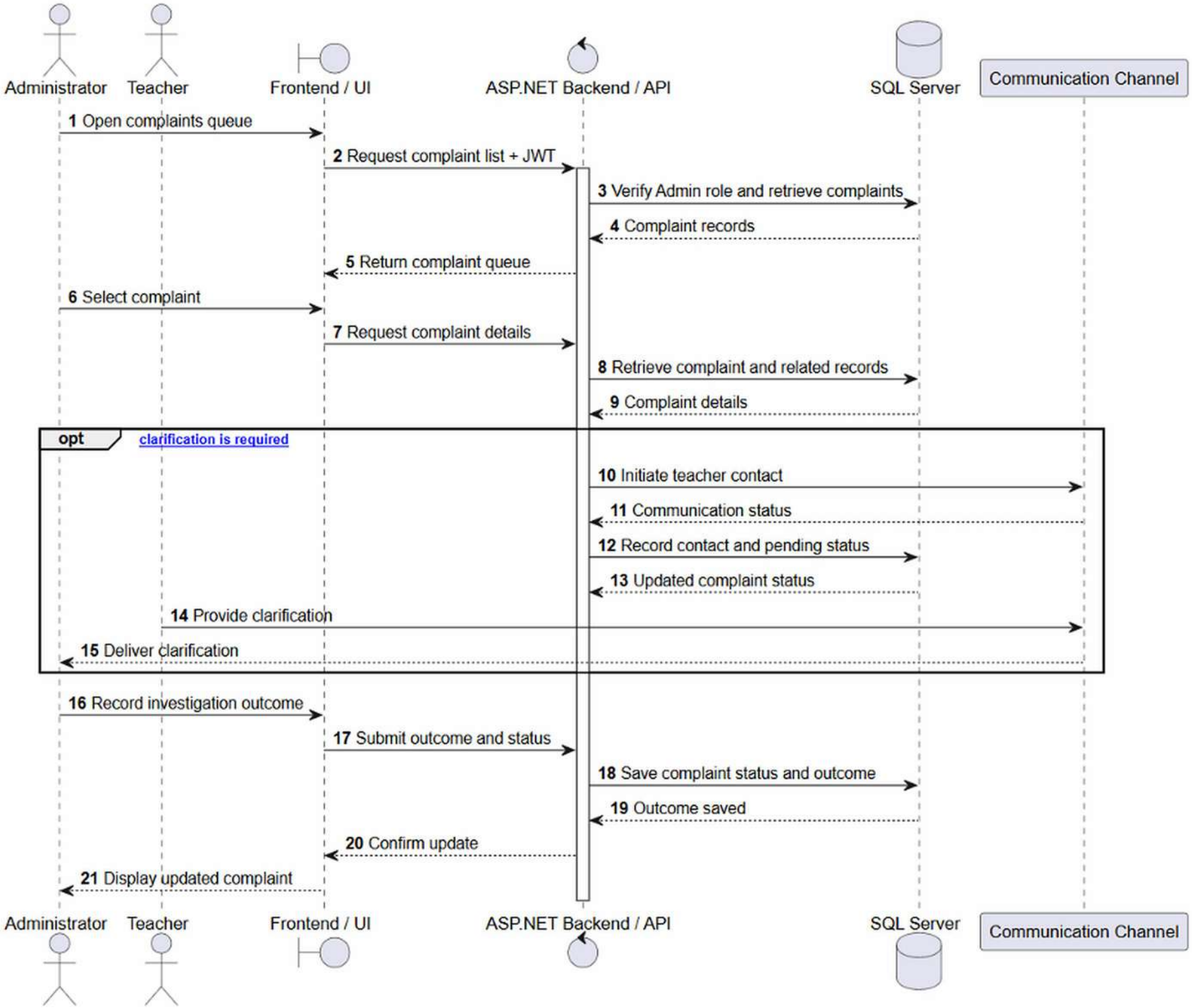
SD-ADM-03: Verify Teacher Commitment and First Lesson



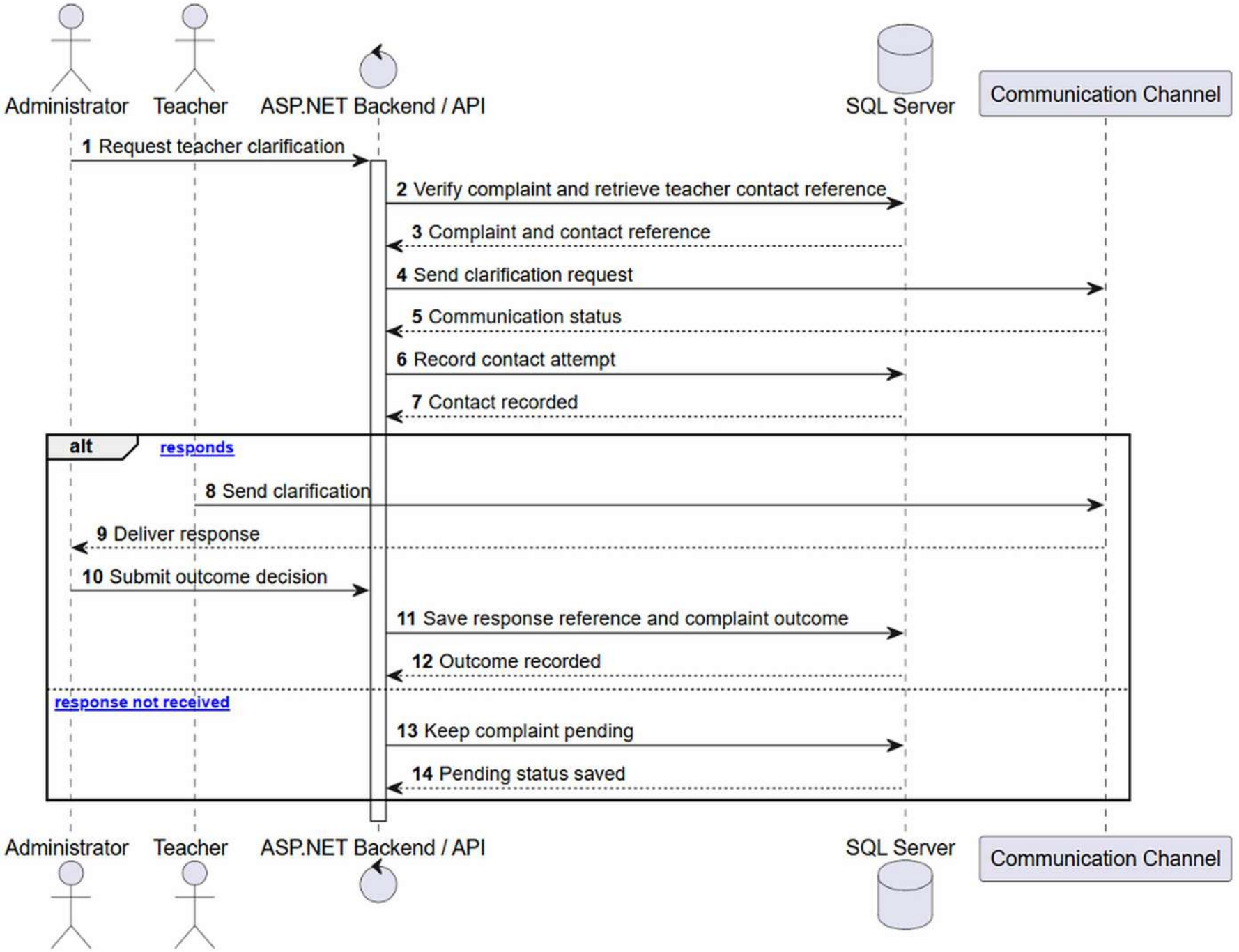
SD-ADM-04: Official Teacher Acknowledgement



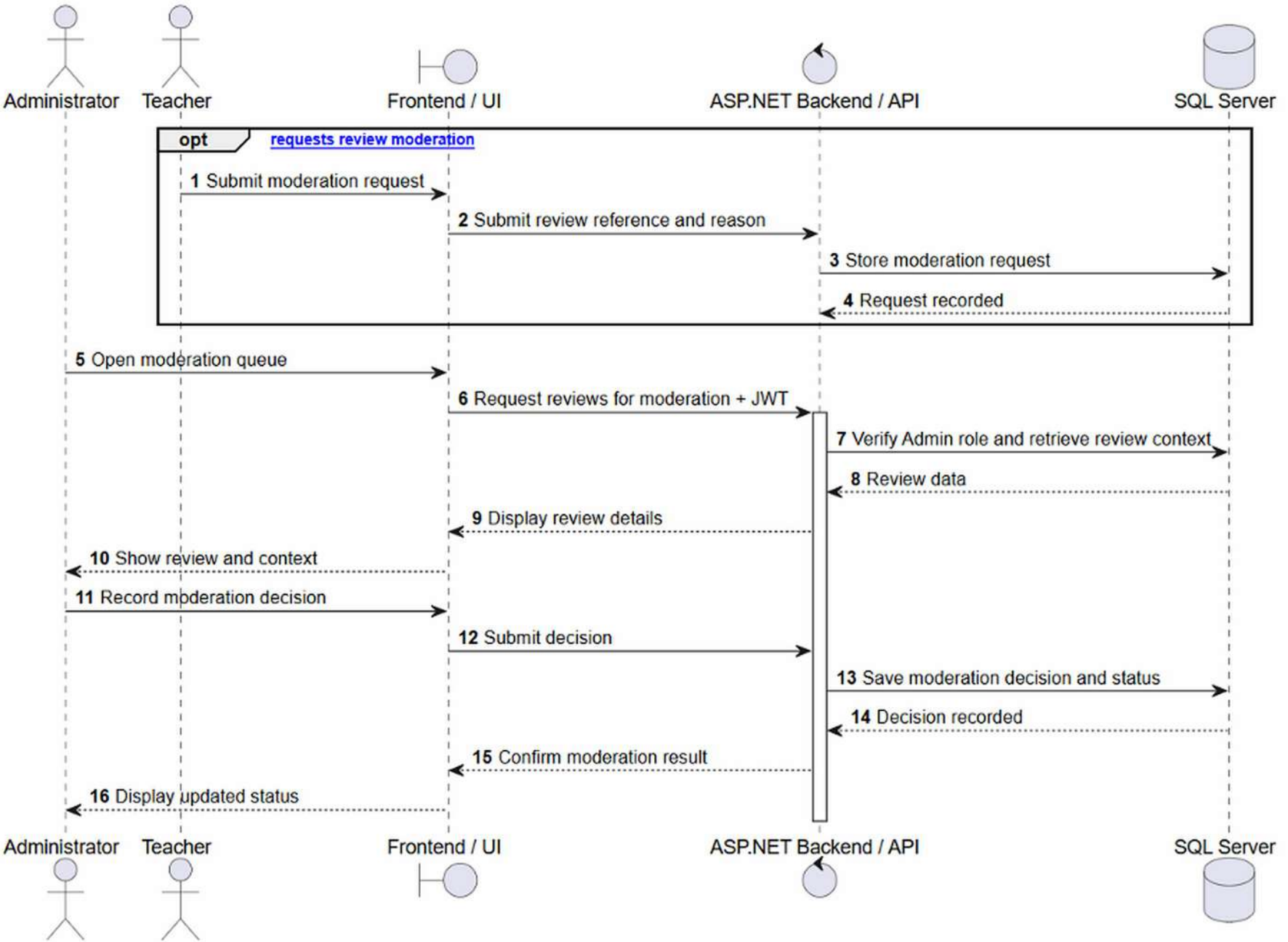
SD-ADM-05: Handle Student Complaint

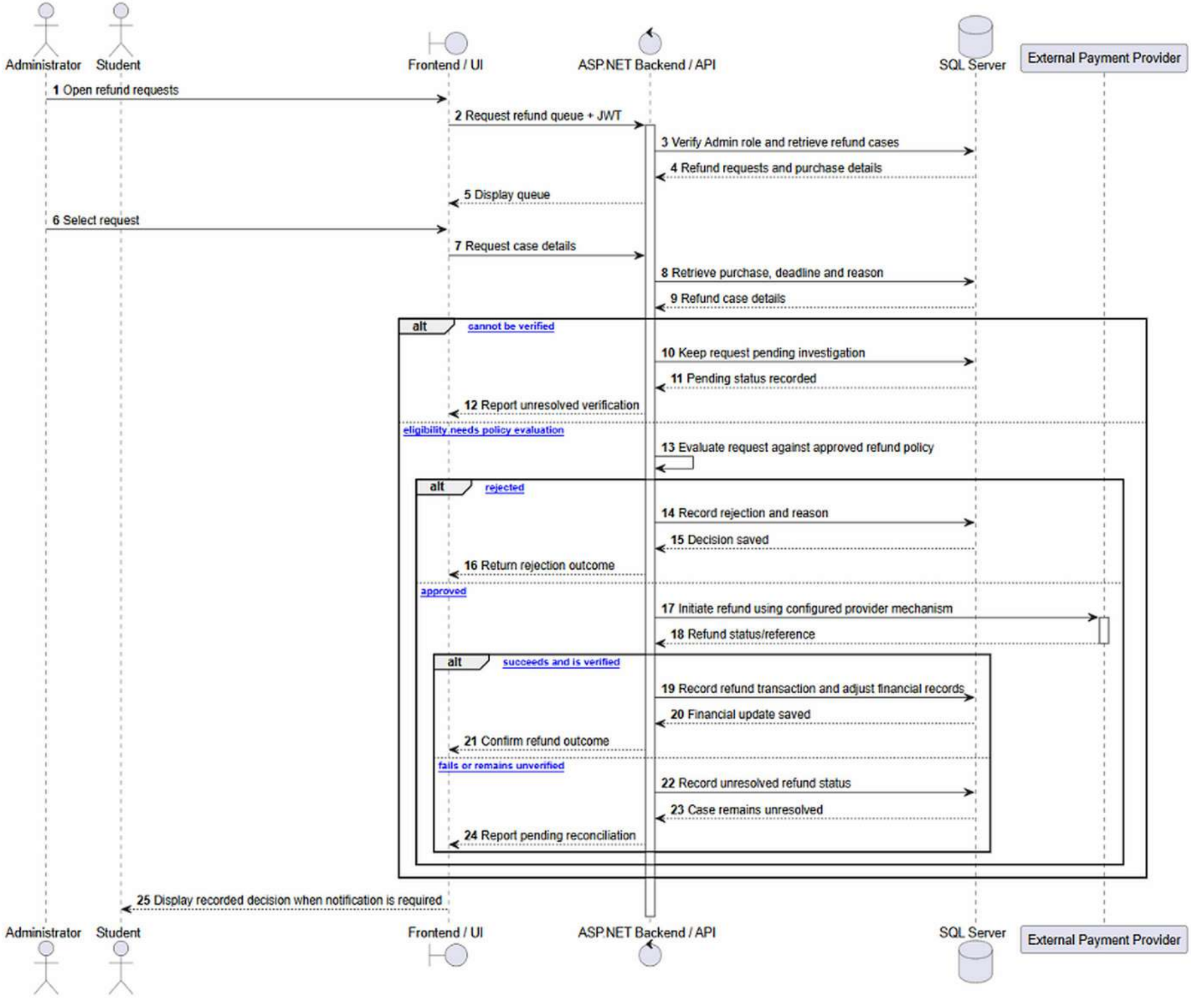


SD-ADM-06: Teacher Clarification and Complaint Outcome

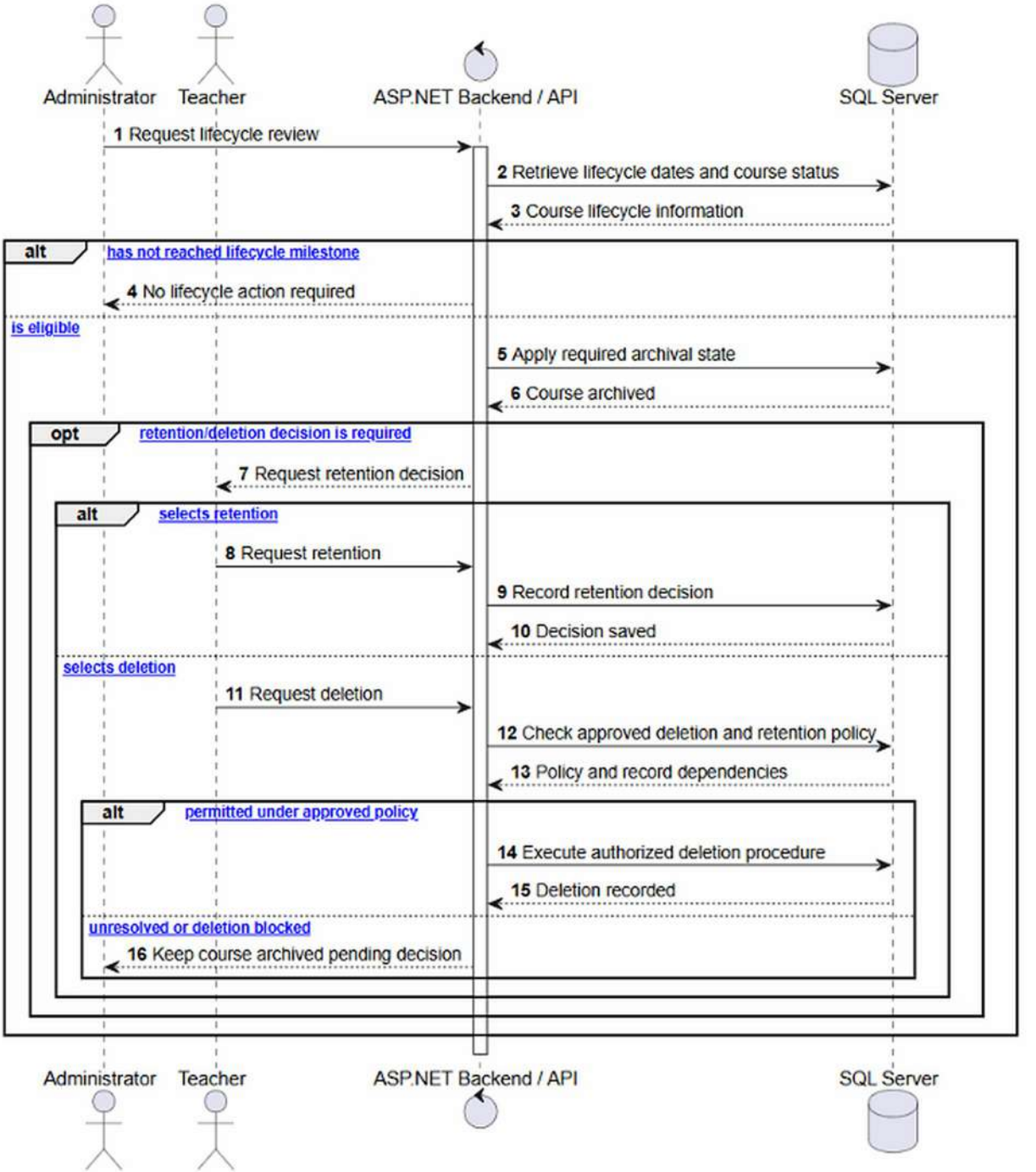


SD-ADM-07: Moderate Course Review

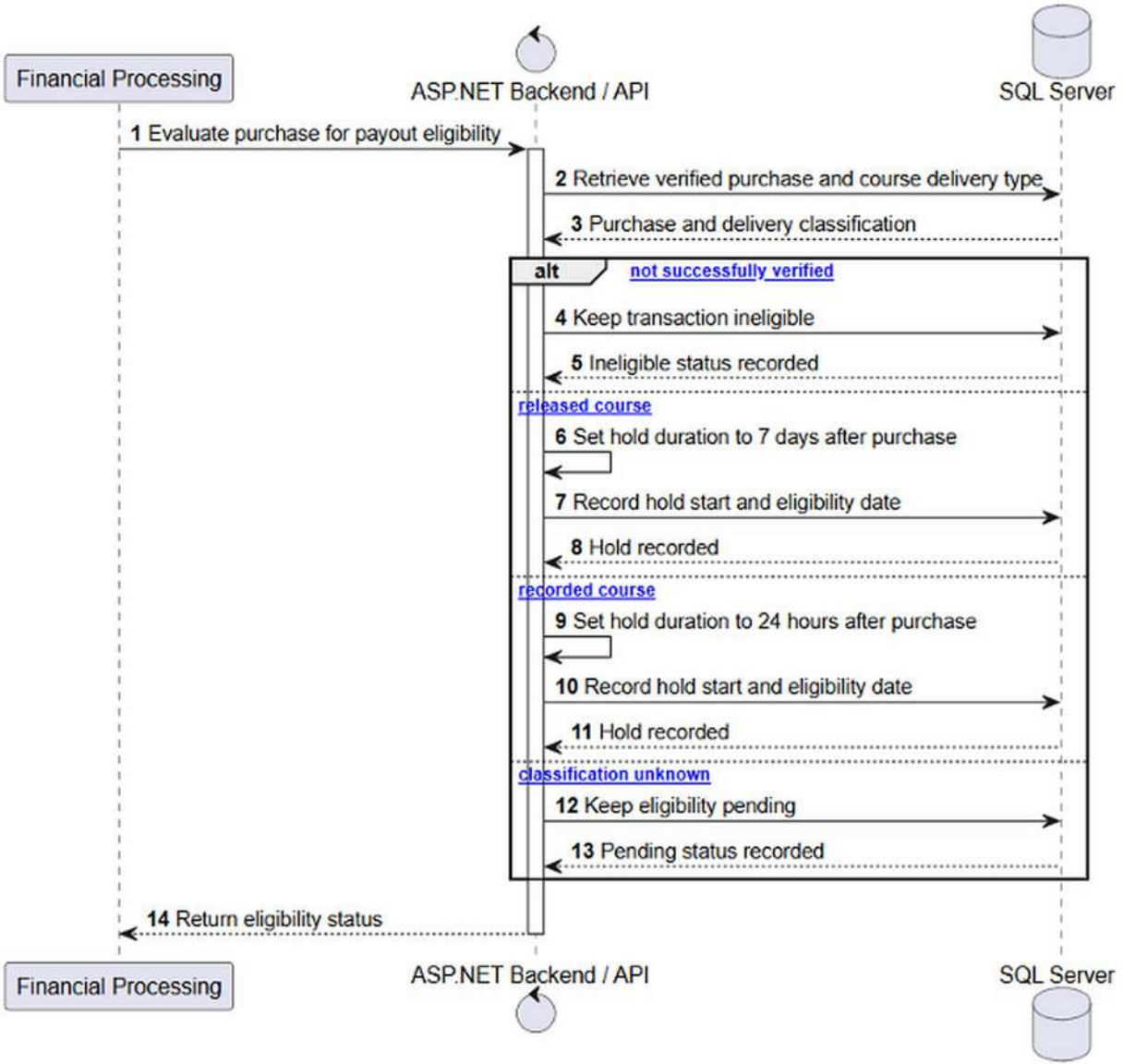



SD-ADM-08: Review and Handle Refund Request
Administrator
Student
Frontend / UI
ASP.NET Backend / API
SQL Server
External Payment Provider
1 Open refund requests
2 Request refund queue + JWT
3 Verify Admin role and retrieve refund cases
4 Refund requests and purchase details
5 Display queue
6 Select request
7 Request case details
8 Retrieve purchase, deadline and reason
9 Refund case details
alt
cannot be verified
10 Keep request pending investigation
11 Pending status recorded
12 Report unresolved verification
eligibility needs policy evaluation
13 Evaluate request against approved refund policy
alt
rejected
14 Record rejection and reason
15 Decision saved
16 Return rejection outcome
approved
17 Initiate refund using configured provider mechanism
18 Refund status/reference
alt
succeeds and is verified
19 Record refund transaction and adjust financial records
20 Financial update saved
21 Confirm refund outcome
fails or remains unverified
22 Record unresolved refund status
23 Case remains unresolved
24 Report pending reconciliation
25 Display recorded decision when notification is required
Administrator
Student
Frontend / UI
ASP.NET Backend / API
SQL Server
External Payment Provider

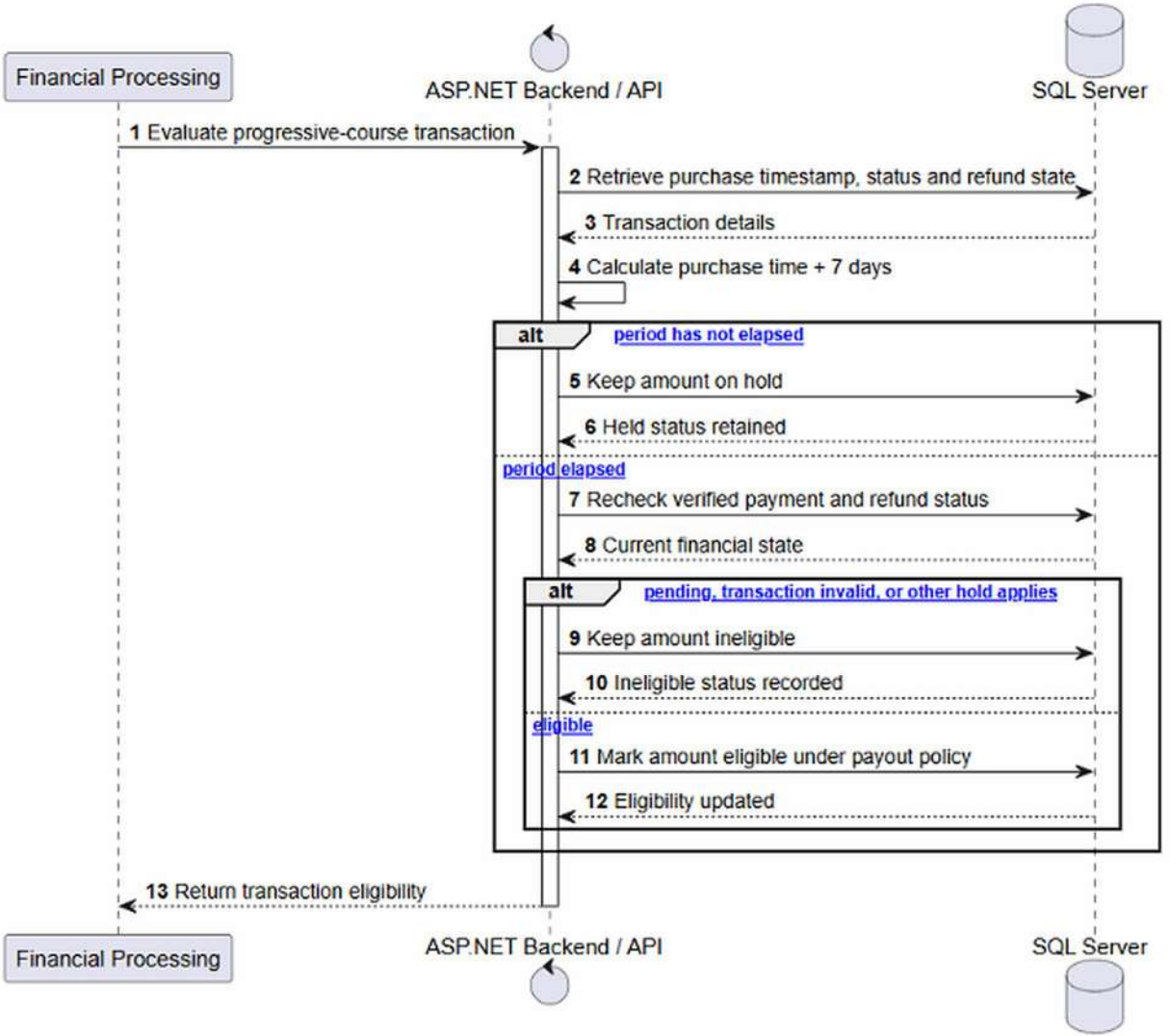
SD-ADM-09: Course Lifecycle and End-of-Life



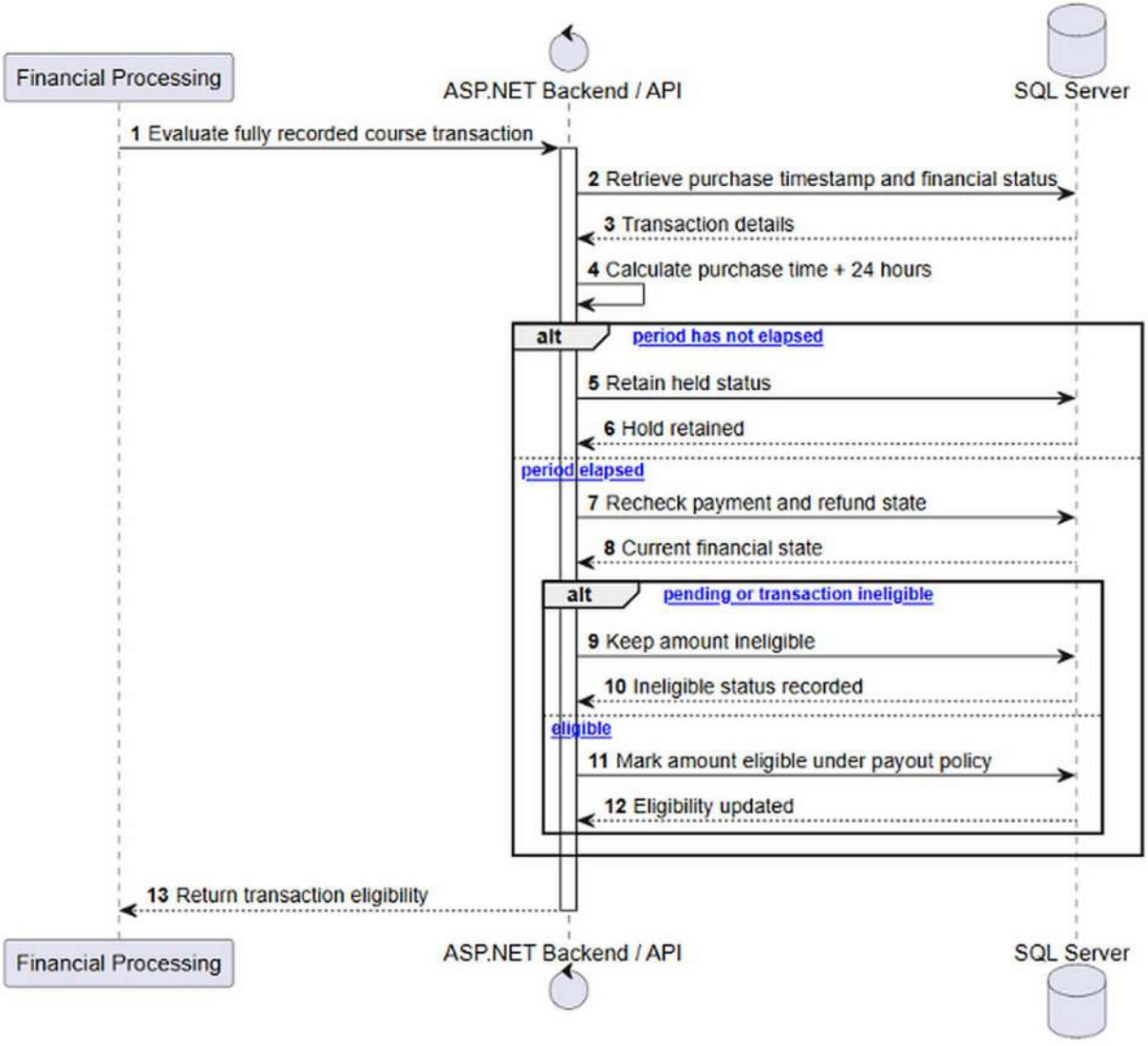
SD-FIN-01: Select Refund and Payout Hold Policy



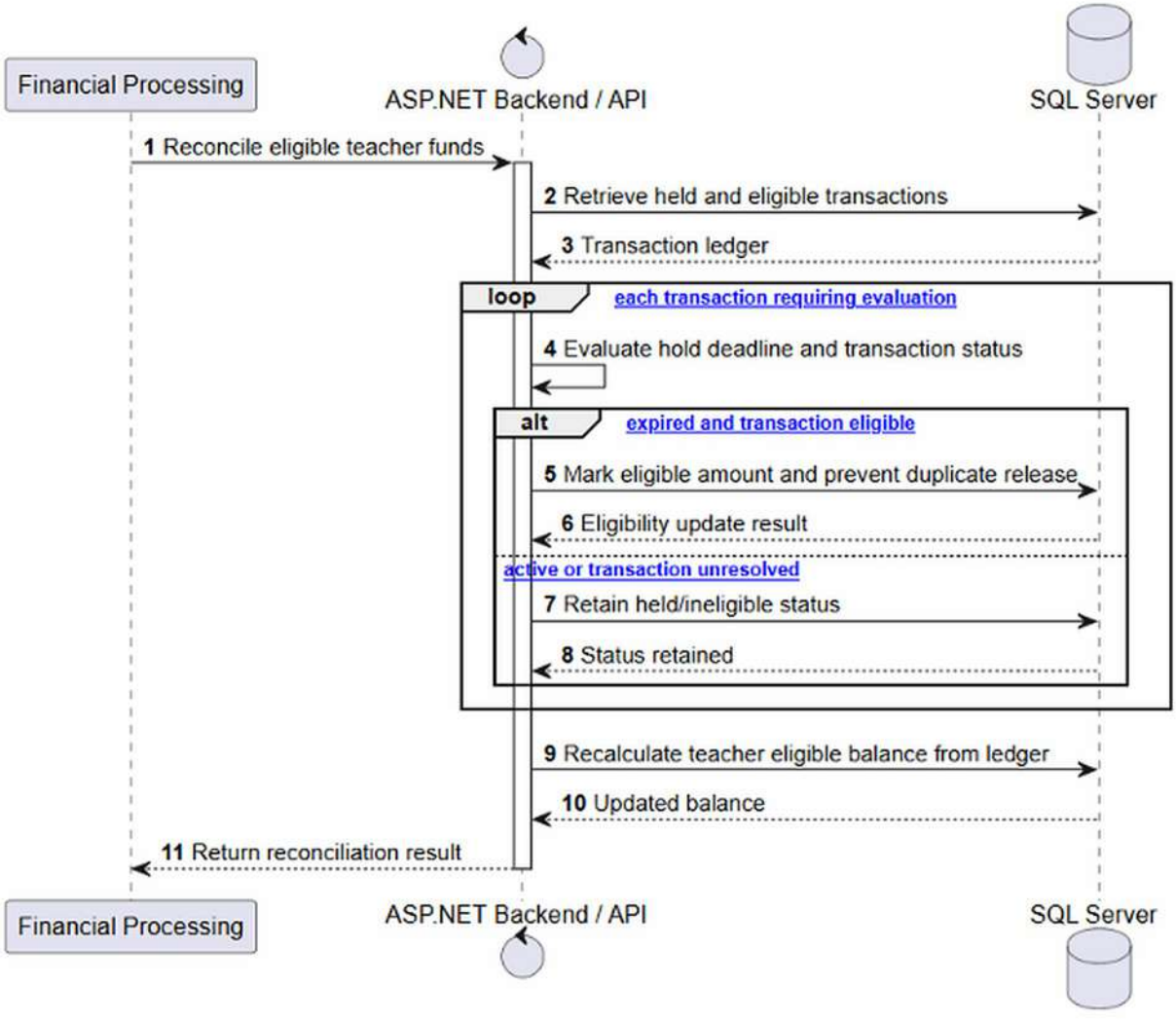
SD-FIN-02: Seven-Day Hold for Progressive Courses

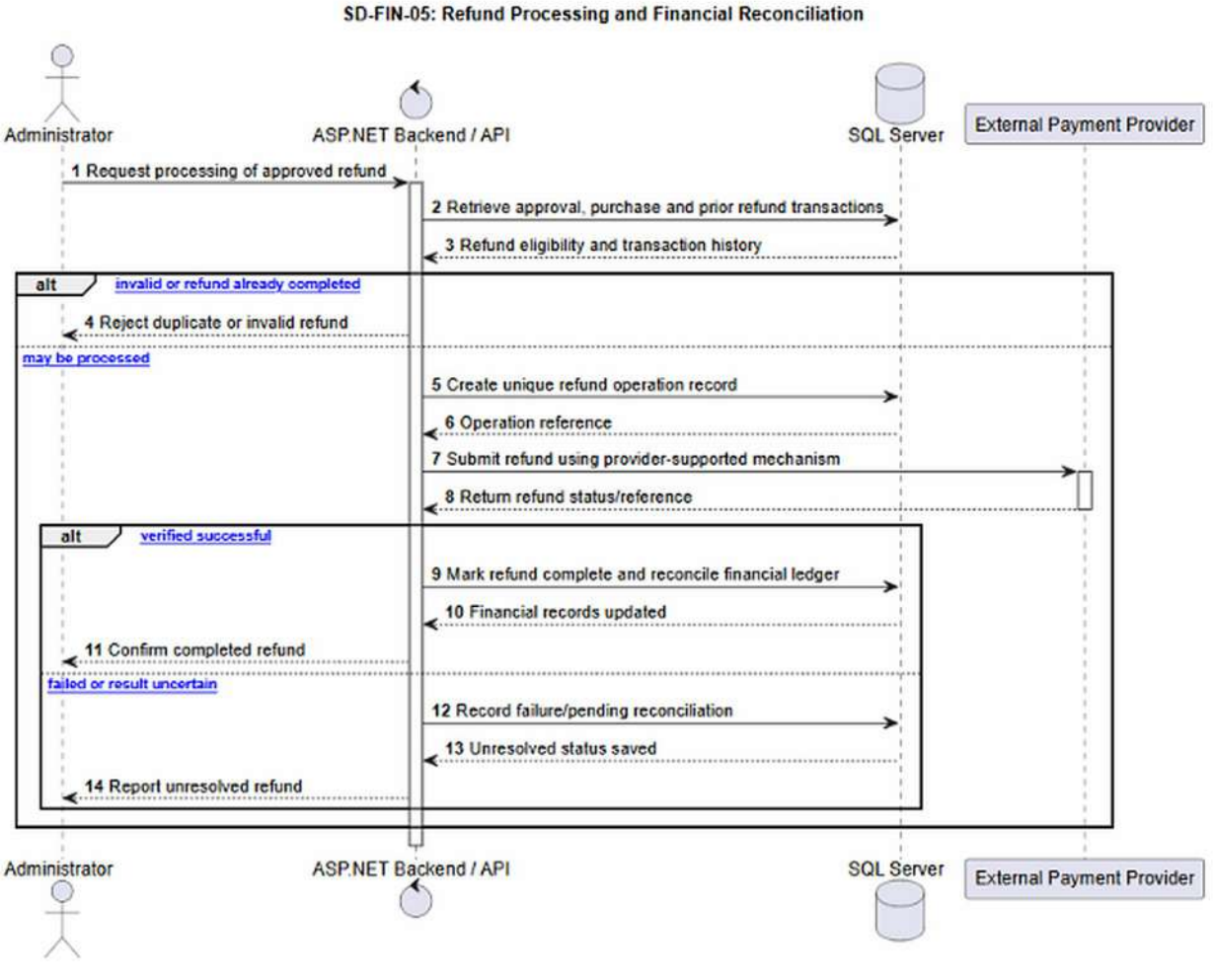


SD-FIN-03: 24-Hour Hold for Fully Recorded Courses

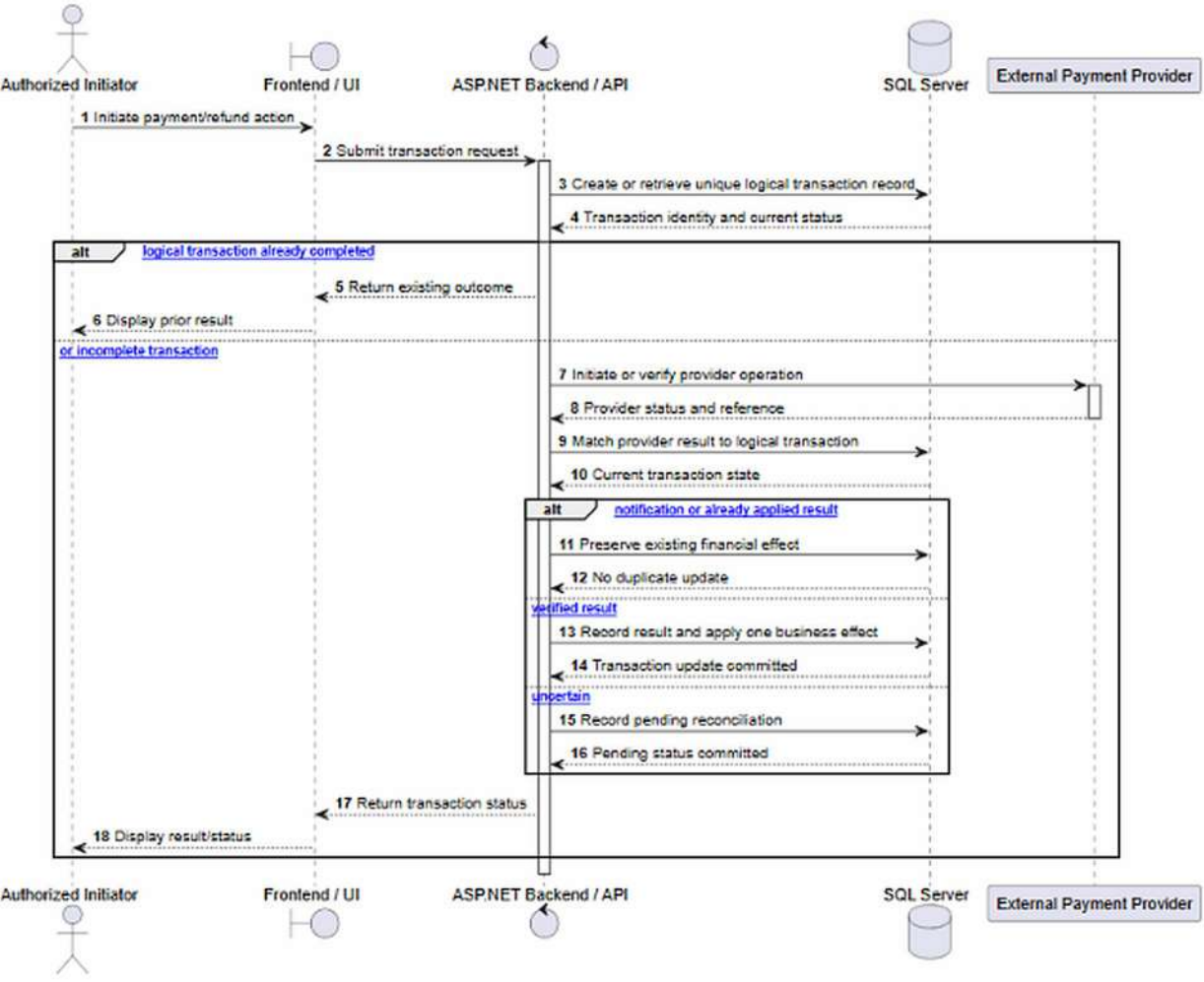


SD-FIN-04: Update Teacher Eligible Balance

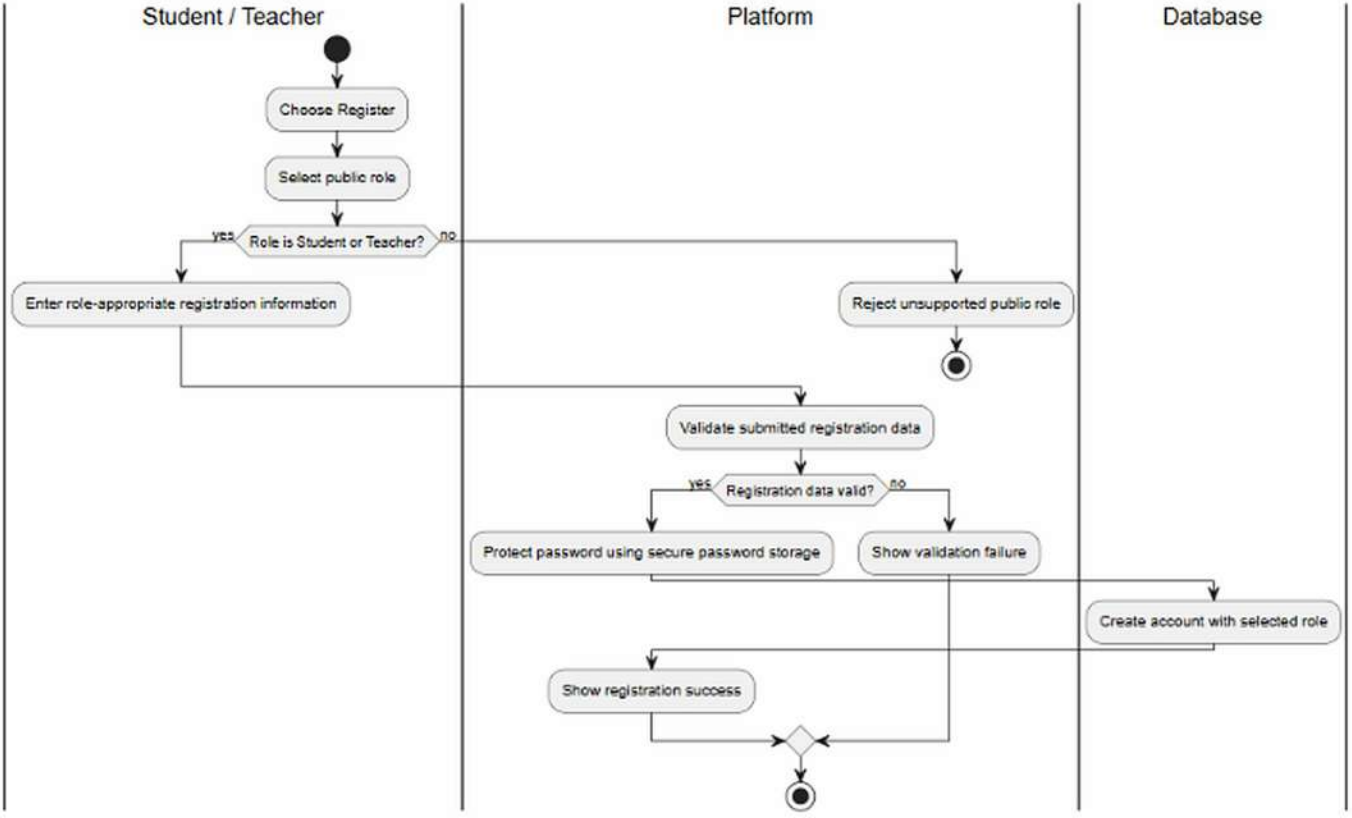



SD-FIN-05: Refund Processing and Financial Reconciliation
Administrator
ASP.NET Backend / API
SQL Server
External Payment Provider
1 Request processing of approved refund
2 Retrieve approval, purchase and prior refund transactions
3 Refund eligibility and transaction history
alt
invalid or refund already completed
4 Reject duplicate or invalid refund
may be processed
5 Create unique refund operation record
6 Operation reference
7 Submit refund using provider-supported mechanism
8 Return refund status/reference
alt
verified successful
9 Mark refund complete and reconcile financial ledger
10 Financial records updated
11 Confirm completed refund
failed or result uncertain
12 Record failure/pending reconciliation
13 Unresolved status saved
14 Report unresolved refund
Administrator
ASP.NET Backend / API
SQL Server
External Payment Provider

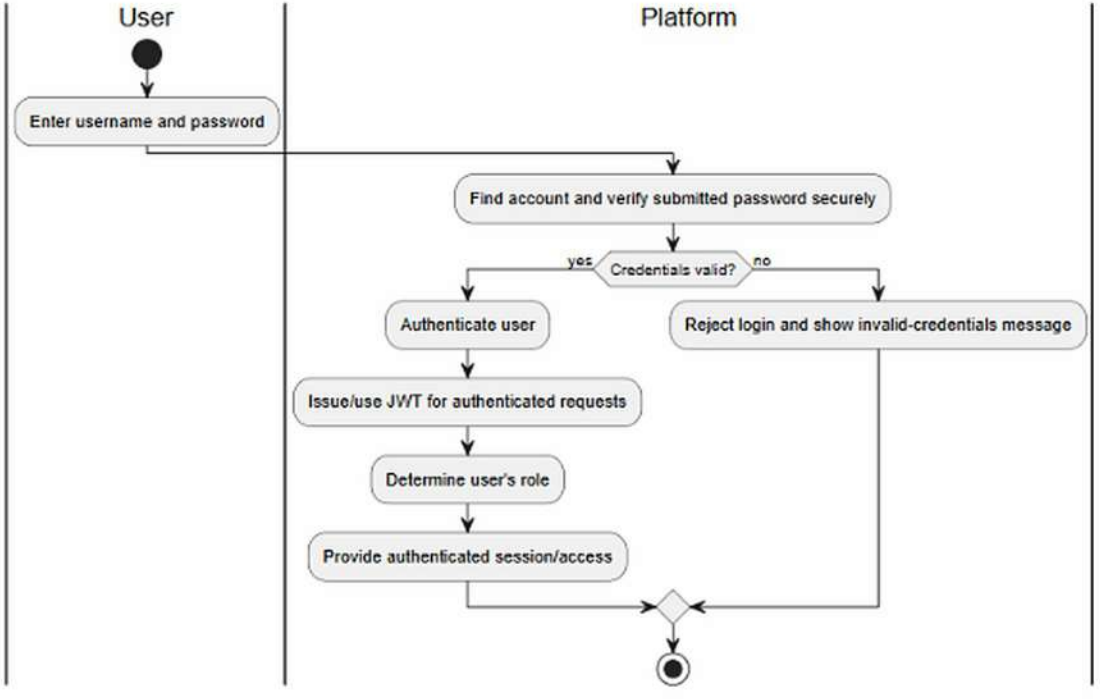
SD-FIN-06: Transaction Traceability and Duplicate Prevention



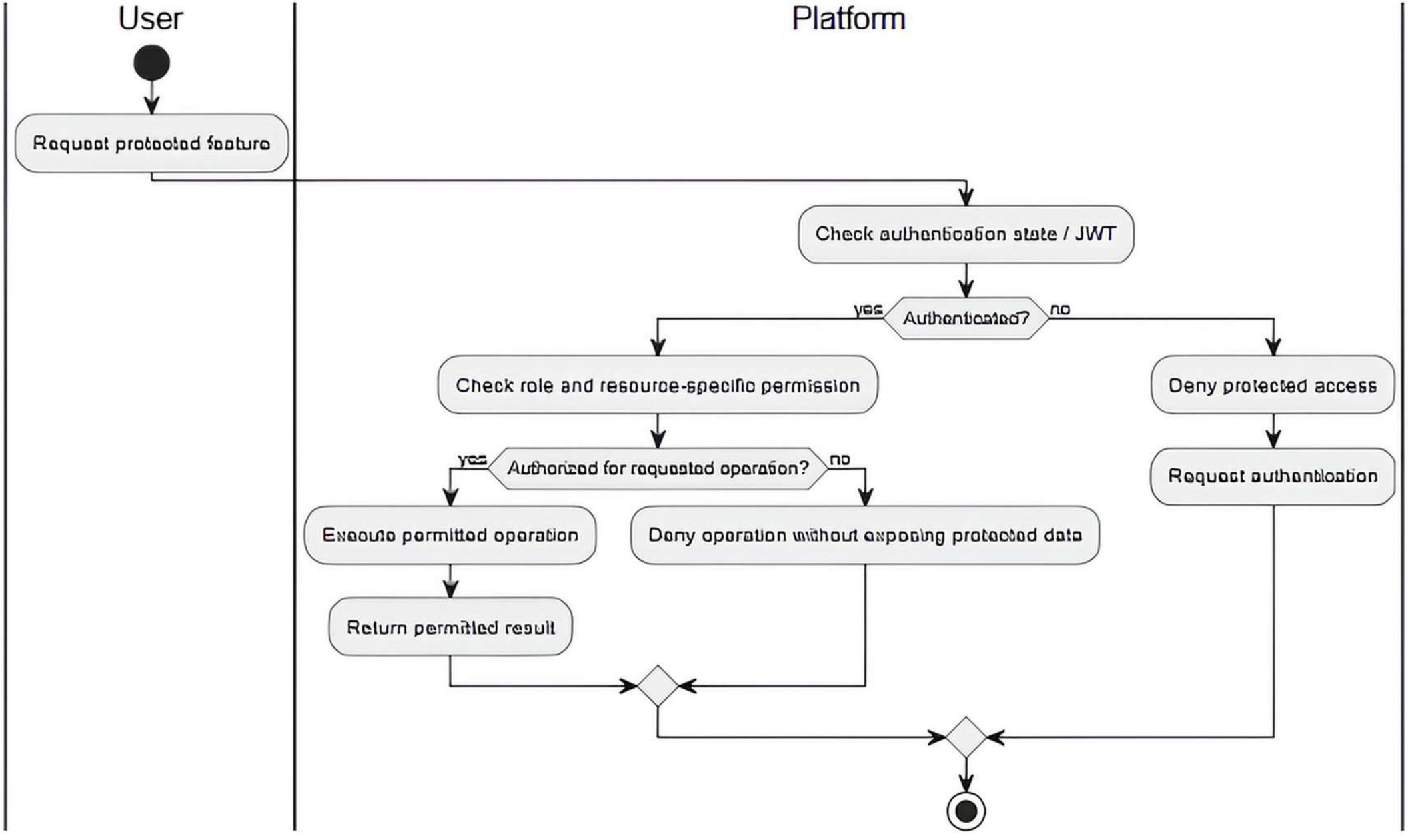
AD-01 — User Registration



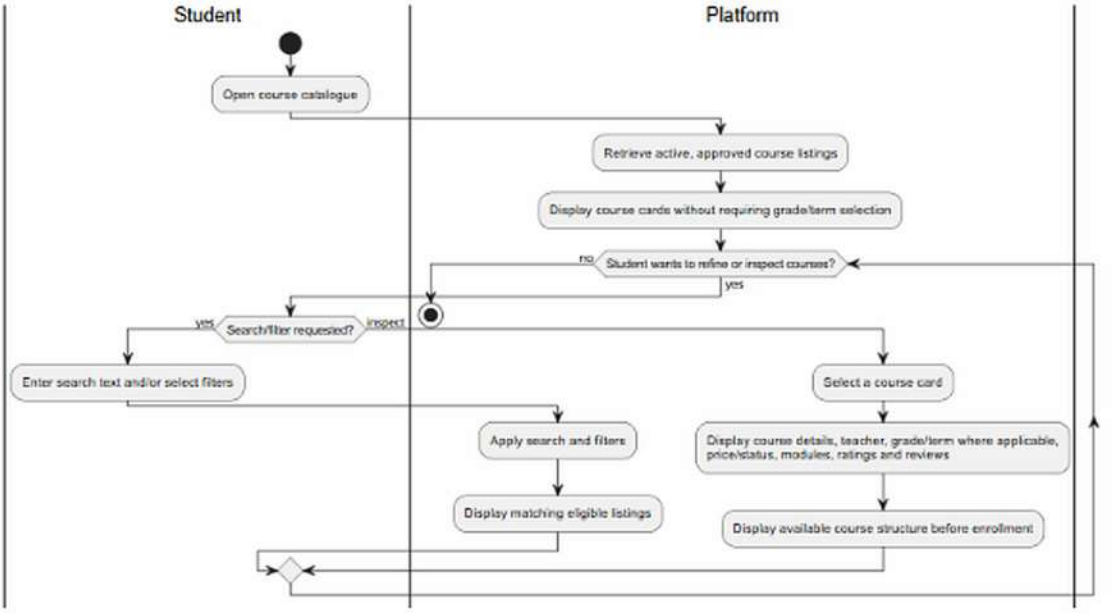
AD-02 — User Login and Authentication

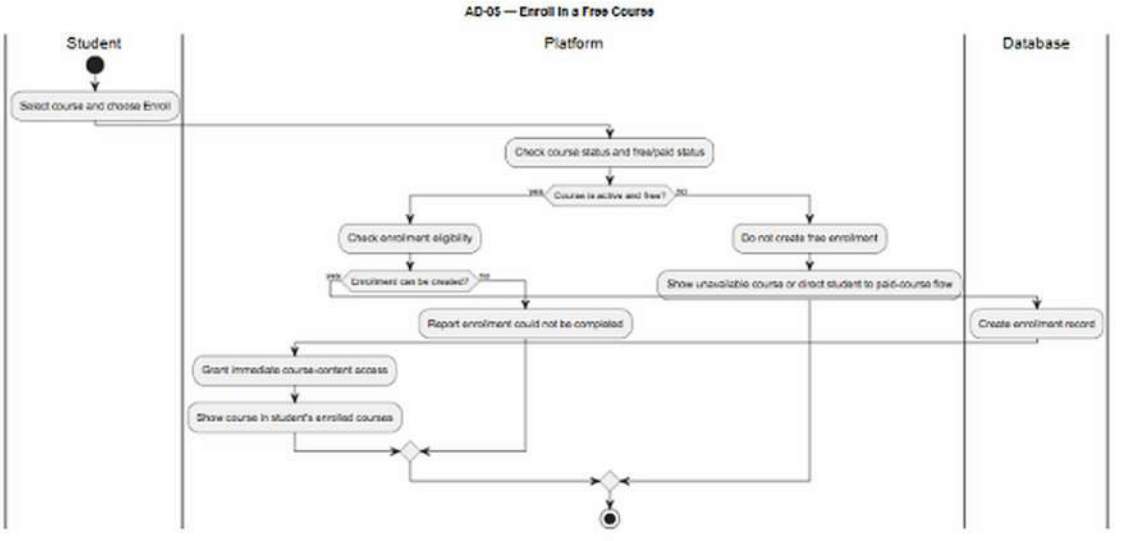


AD-03 — Role-Based Authorization and Protected Access

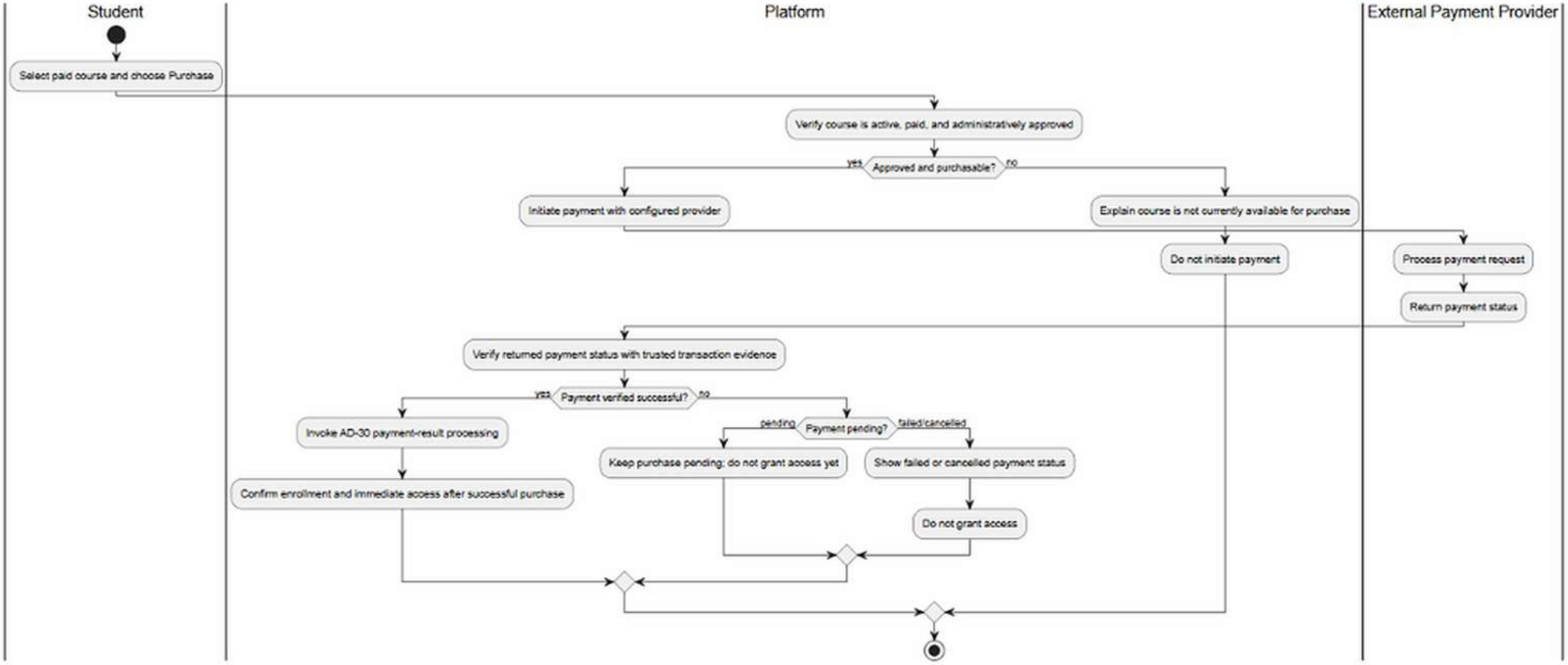


AD-04 — Browse, Search, Filter, and View Courses

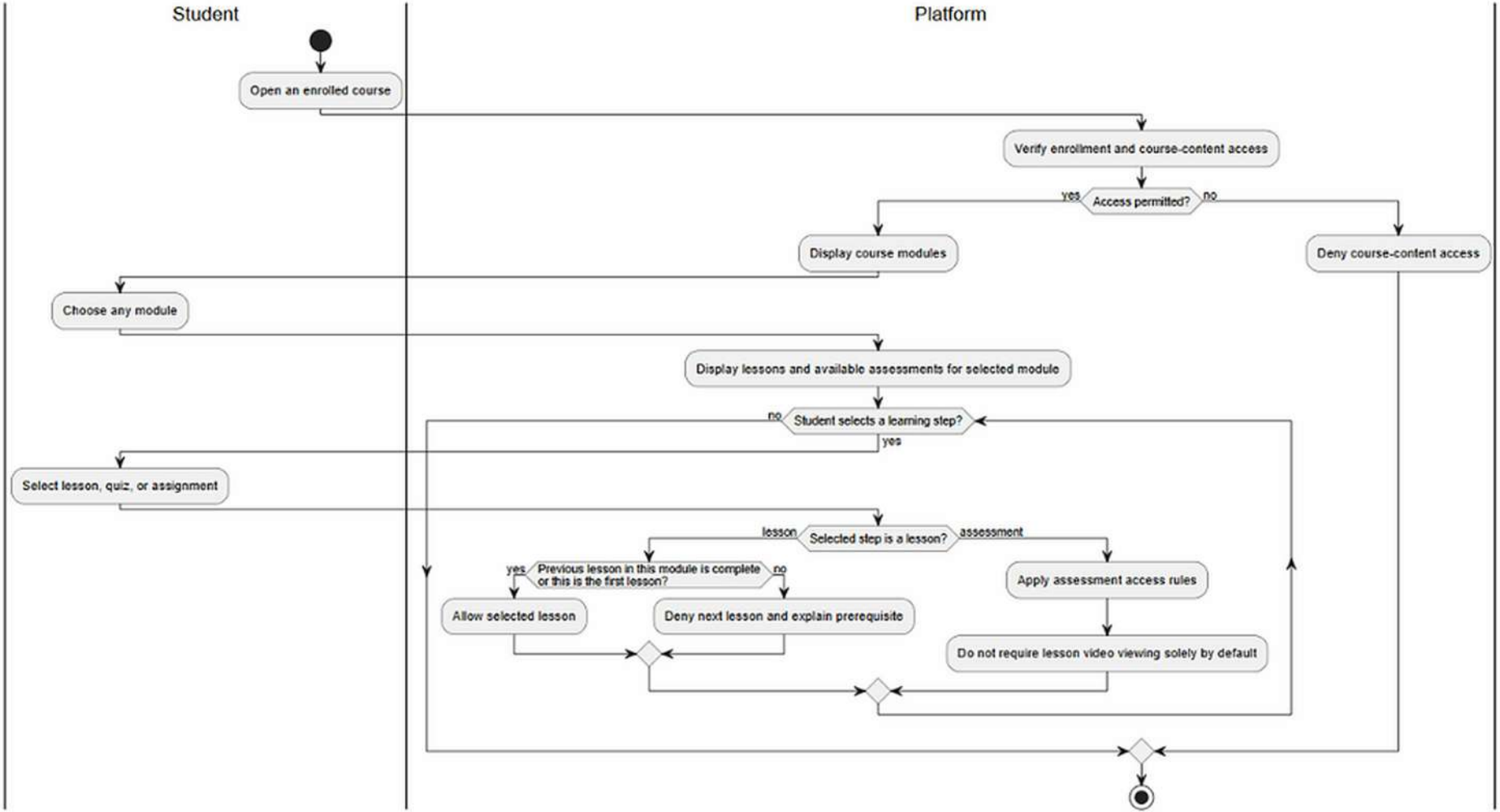



AD-05 — Enroll in a Free Course
Student
Platform
Database
Select course and choose Enroll
Check course status and free/paid status
yes
Course is active and free?
no
Check enrollment eligibility
Do not create free enrollment
yes
Enrollment can be created?
no
Show unavailable course or direct student to paid-course flow
Report enrollment could not be completed
Create enrollment record
Grant immediate course-content access
Show course in student's enrolled courses

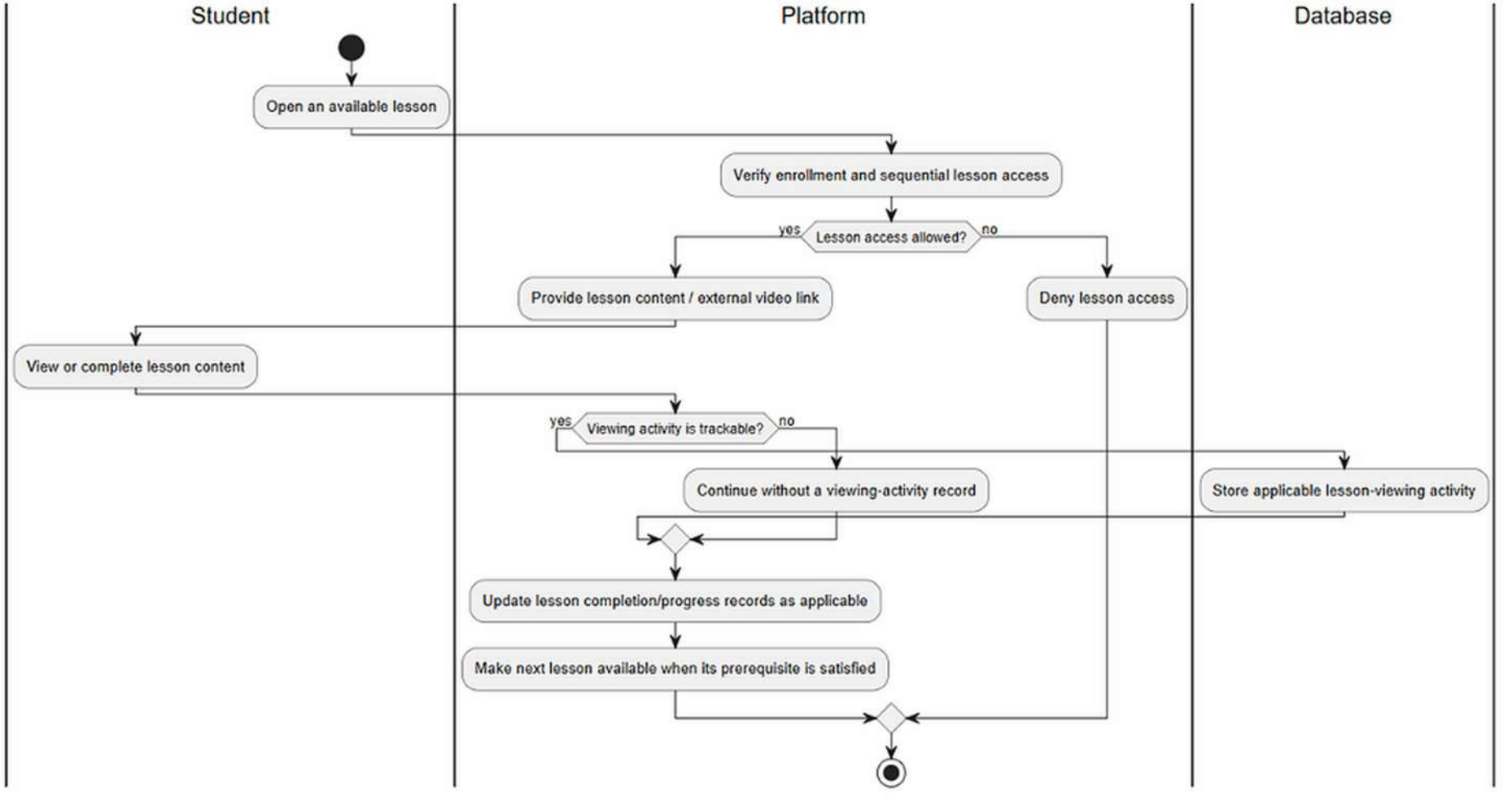
AD-06 — Purchase and Enroll in a Paid Course



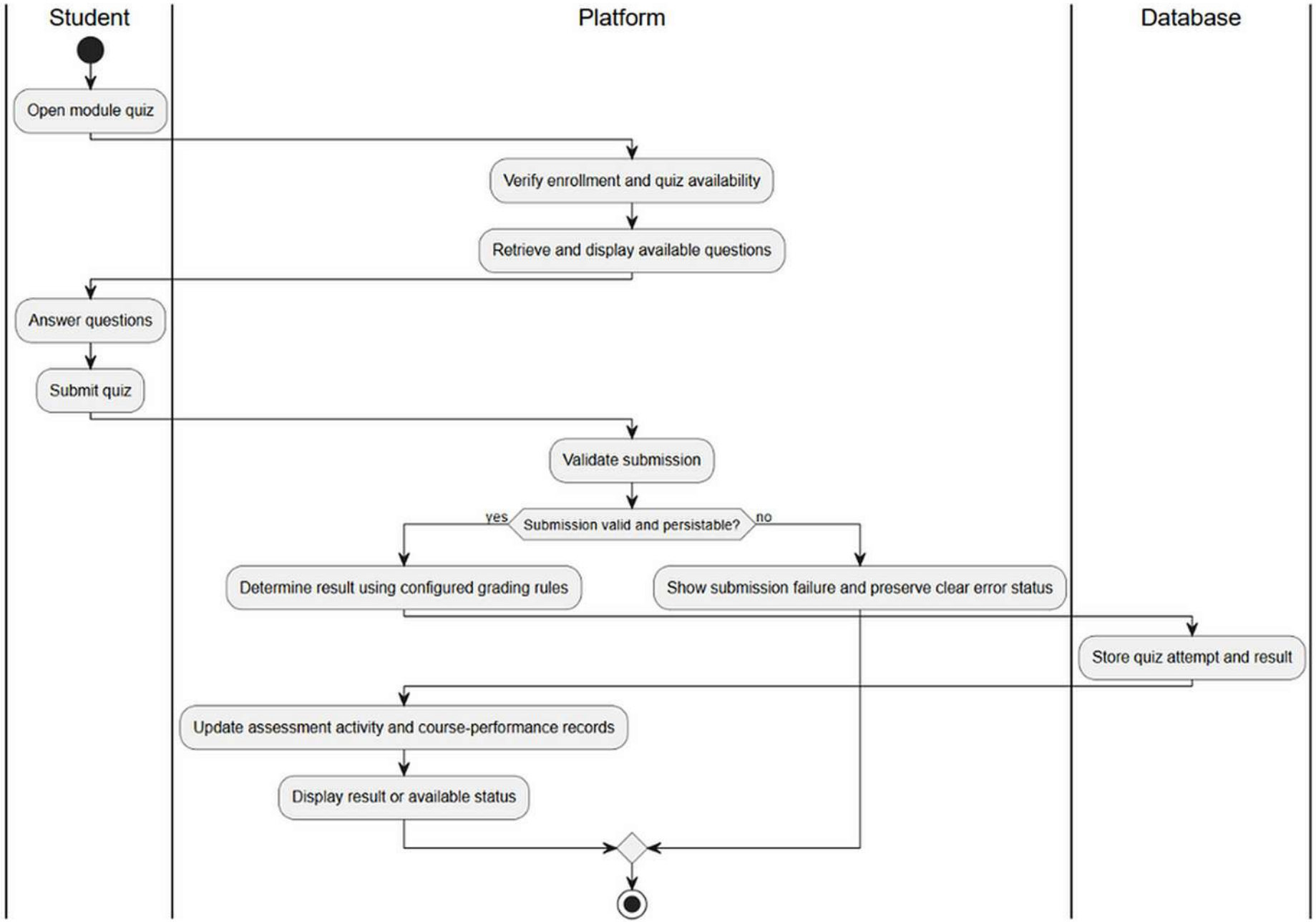
AD-07 — Access Modules and Sequential Lessons



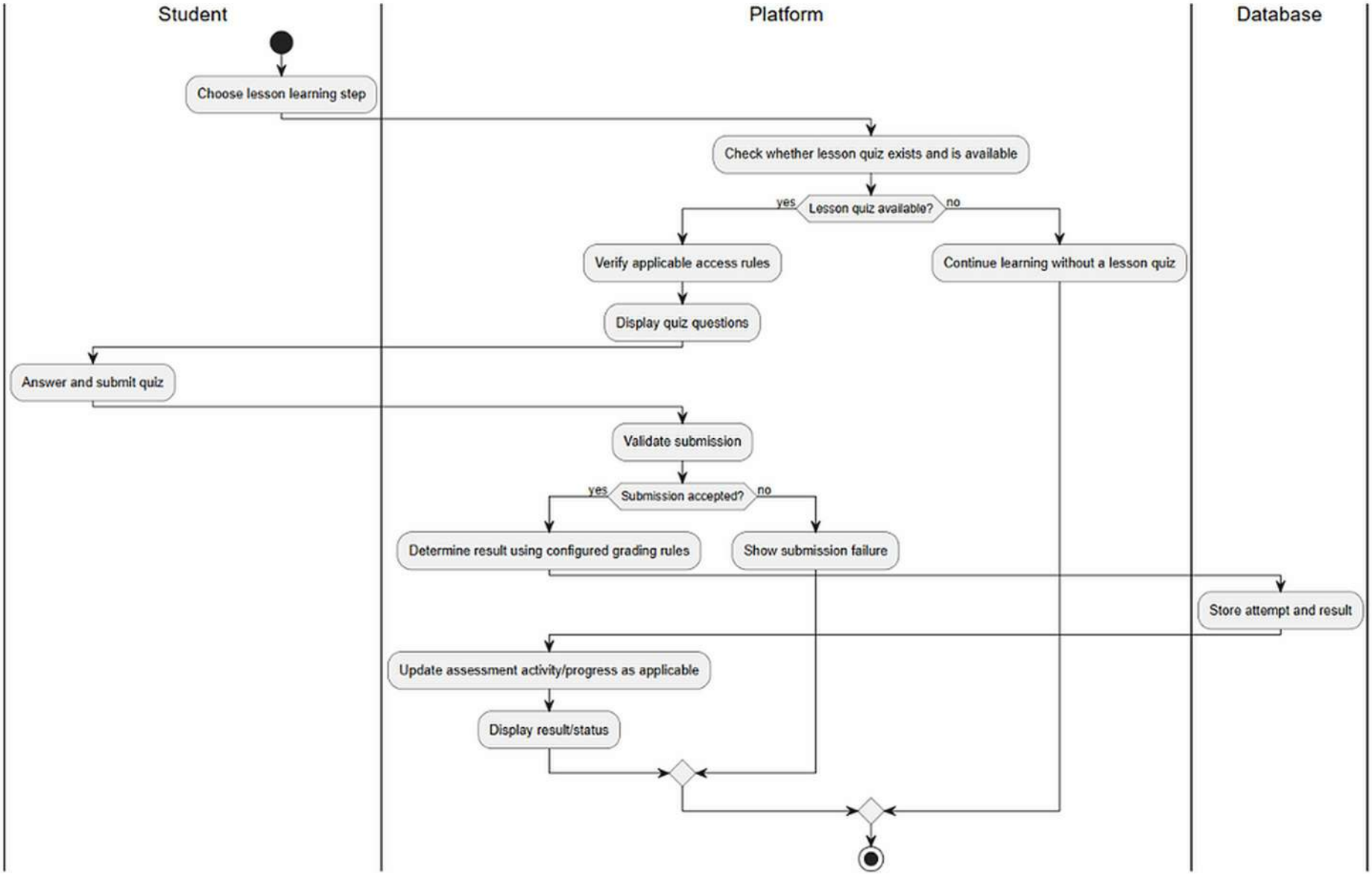
AD-08 — Complete Lesson and Track Progress



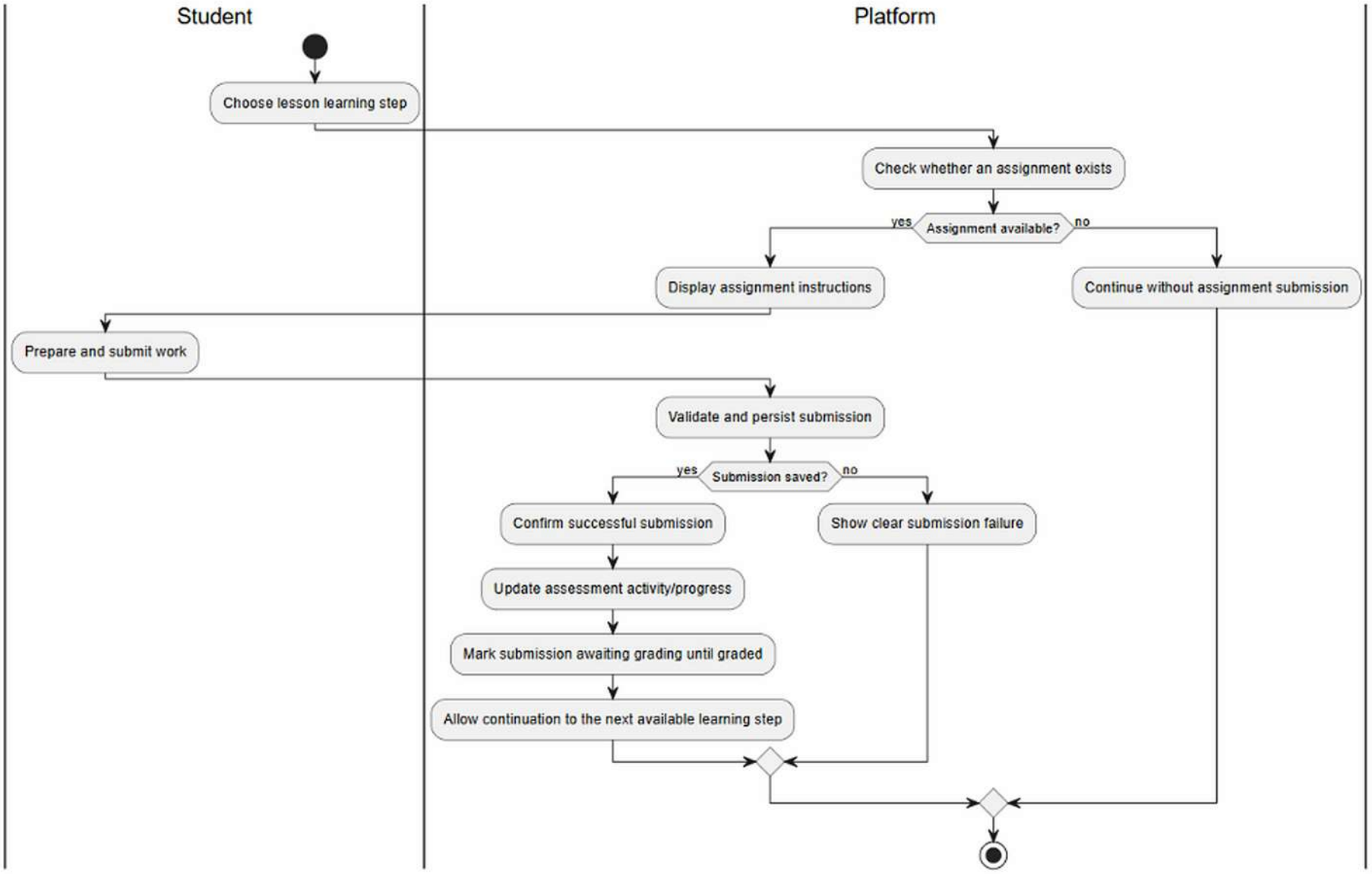
AD-09 — Attempt Mandatory Module Quiz



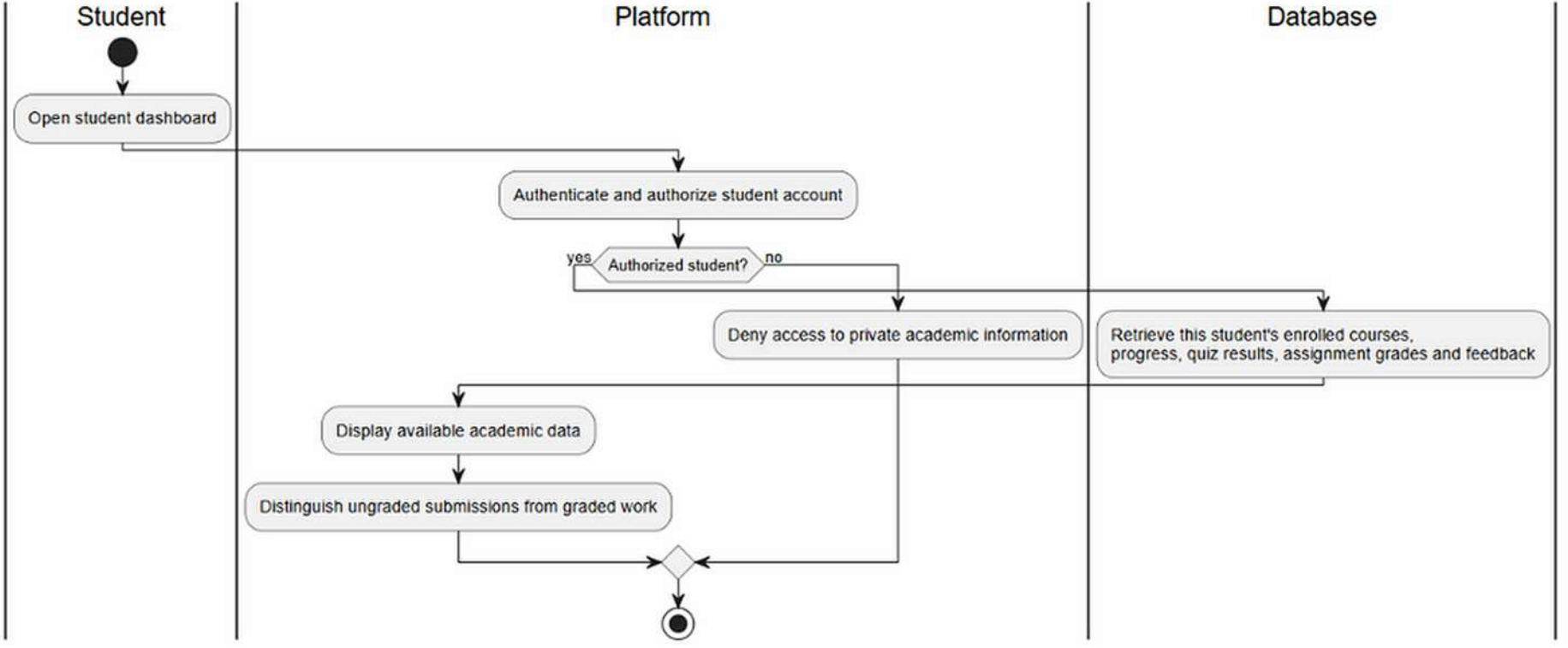
AD-10 — Attempt Optional Lesson Quiz



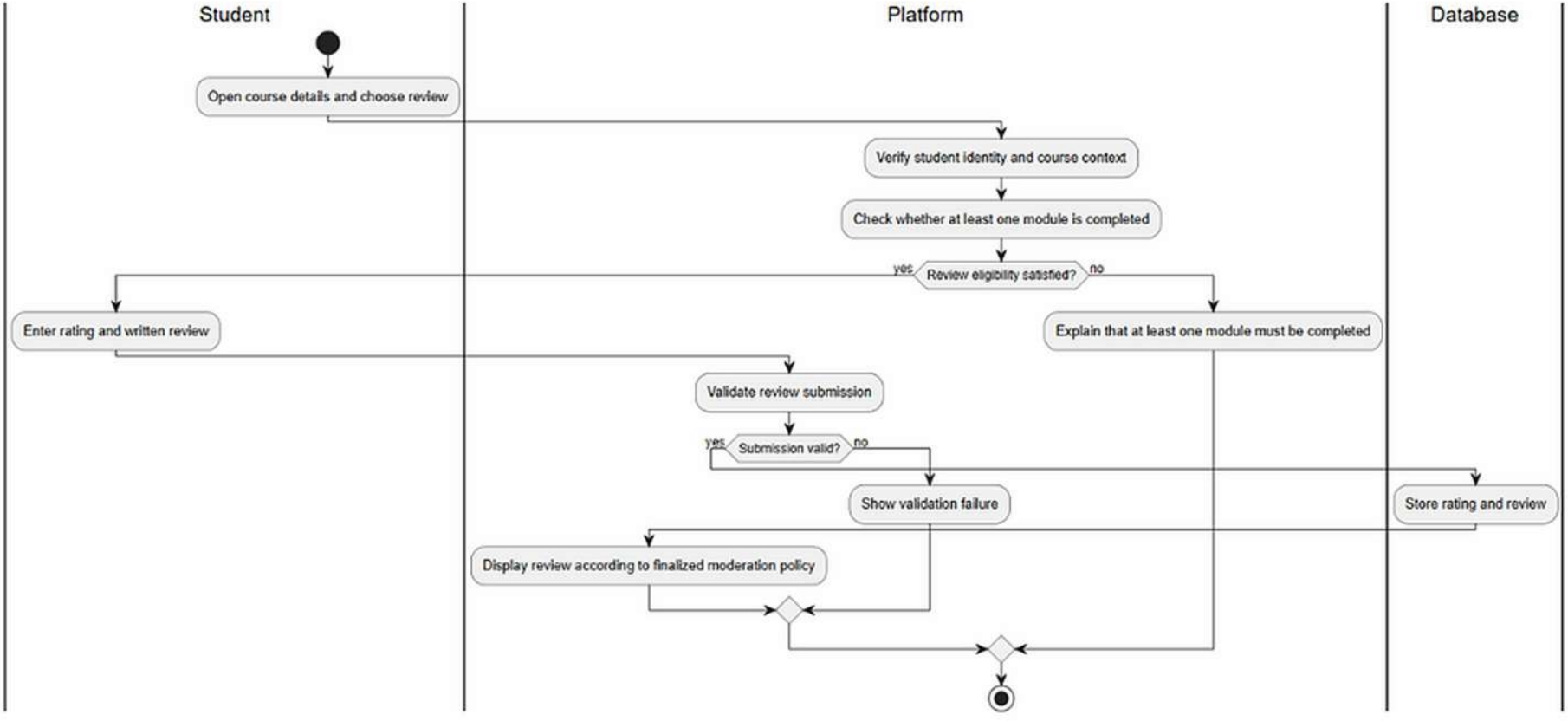
AD-11 — Submit a Lesson Assignment



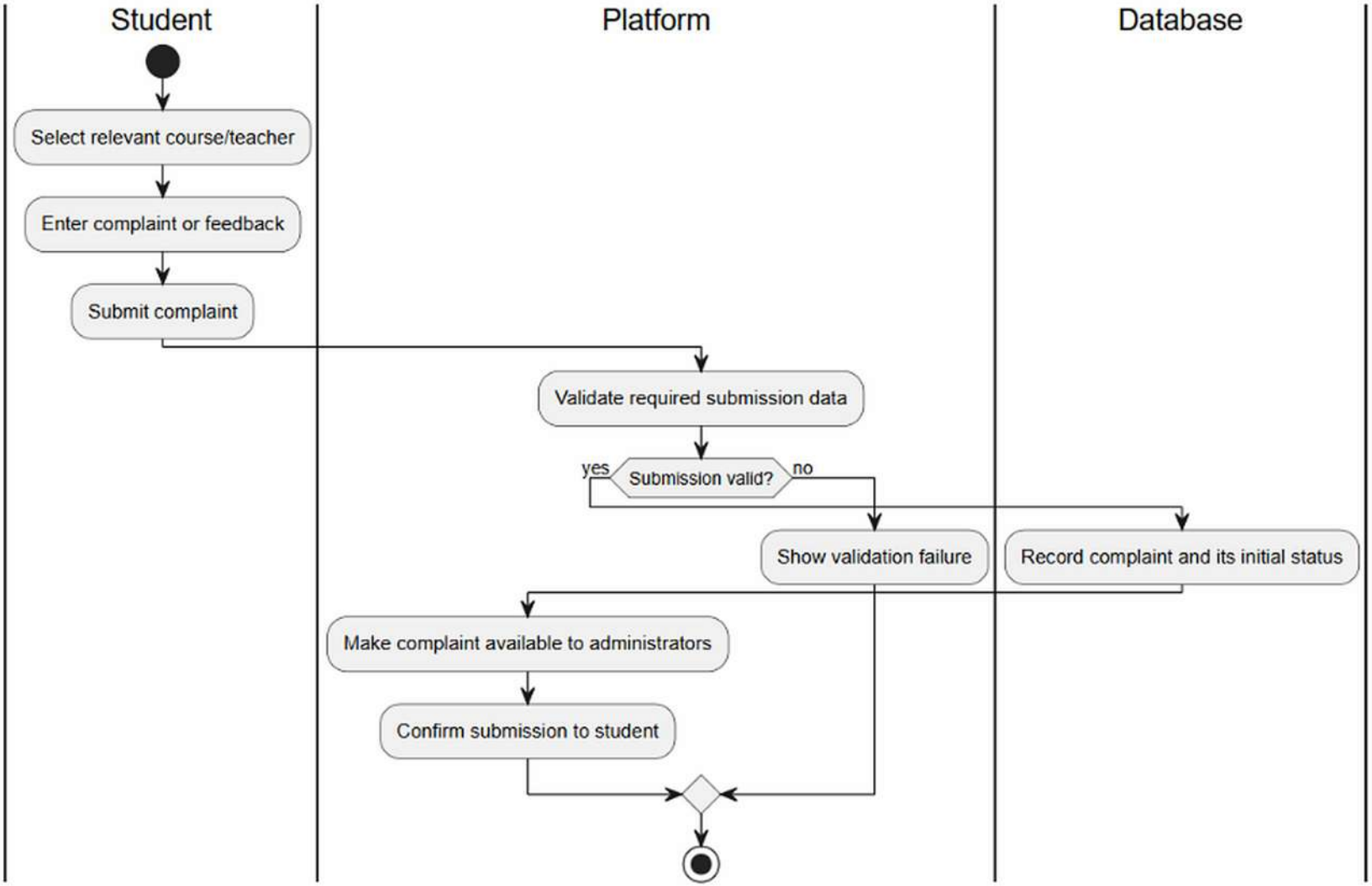
AD-12 — View Student Grades, Feedback, and Progress



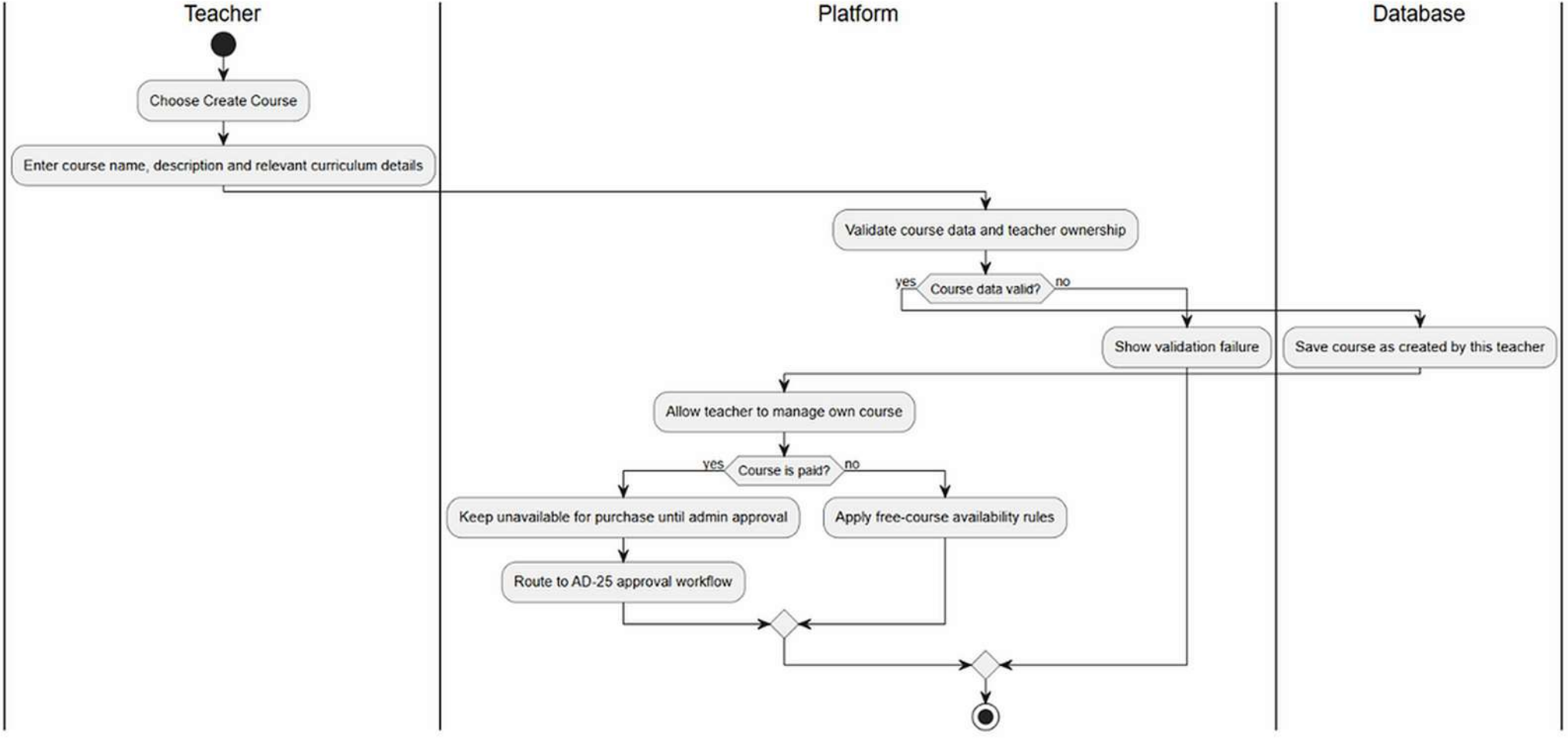
AD-13 — Submit a Course Review and Rating



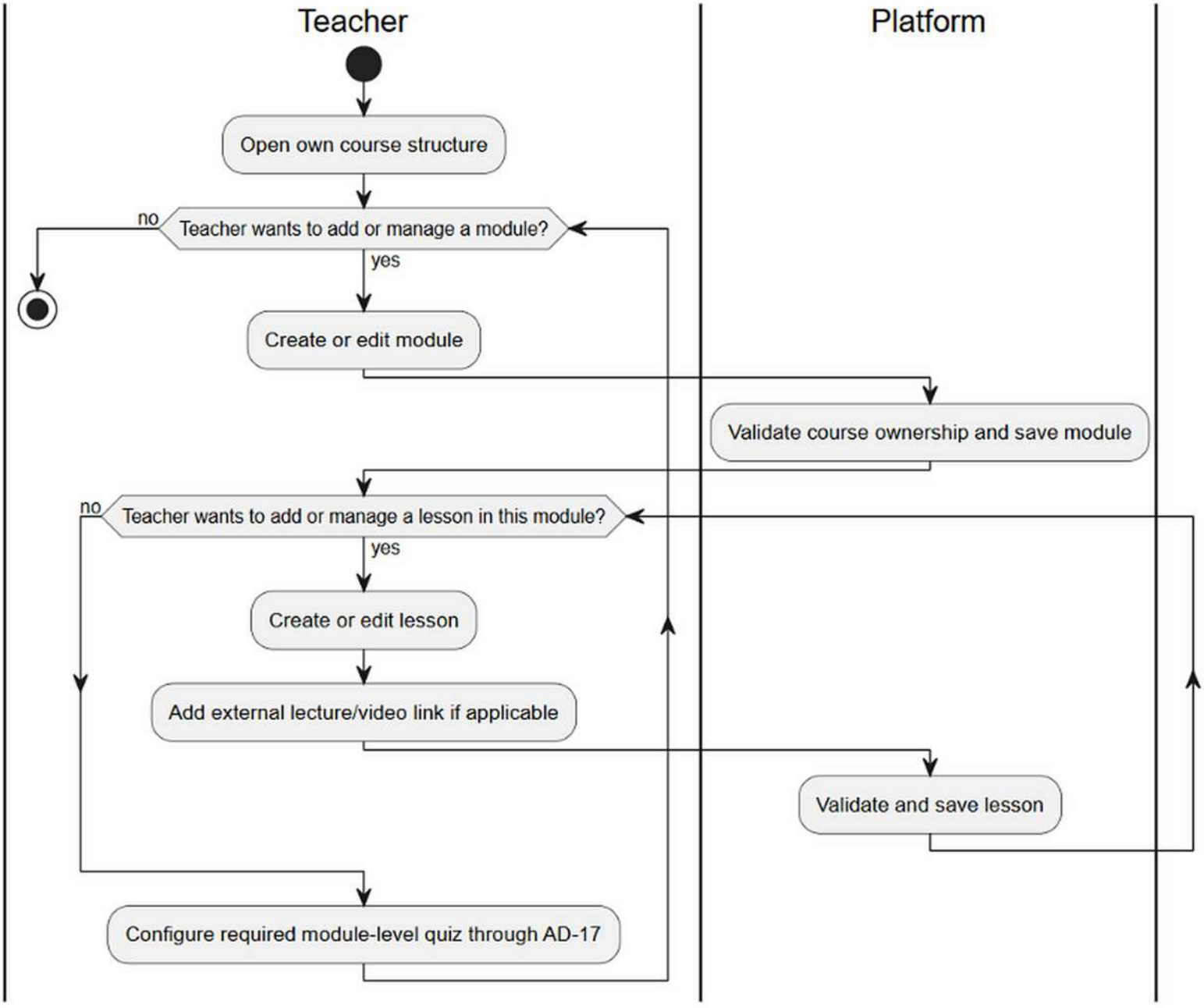
AD-14 — Submit a Complaint or Feedback



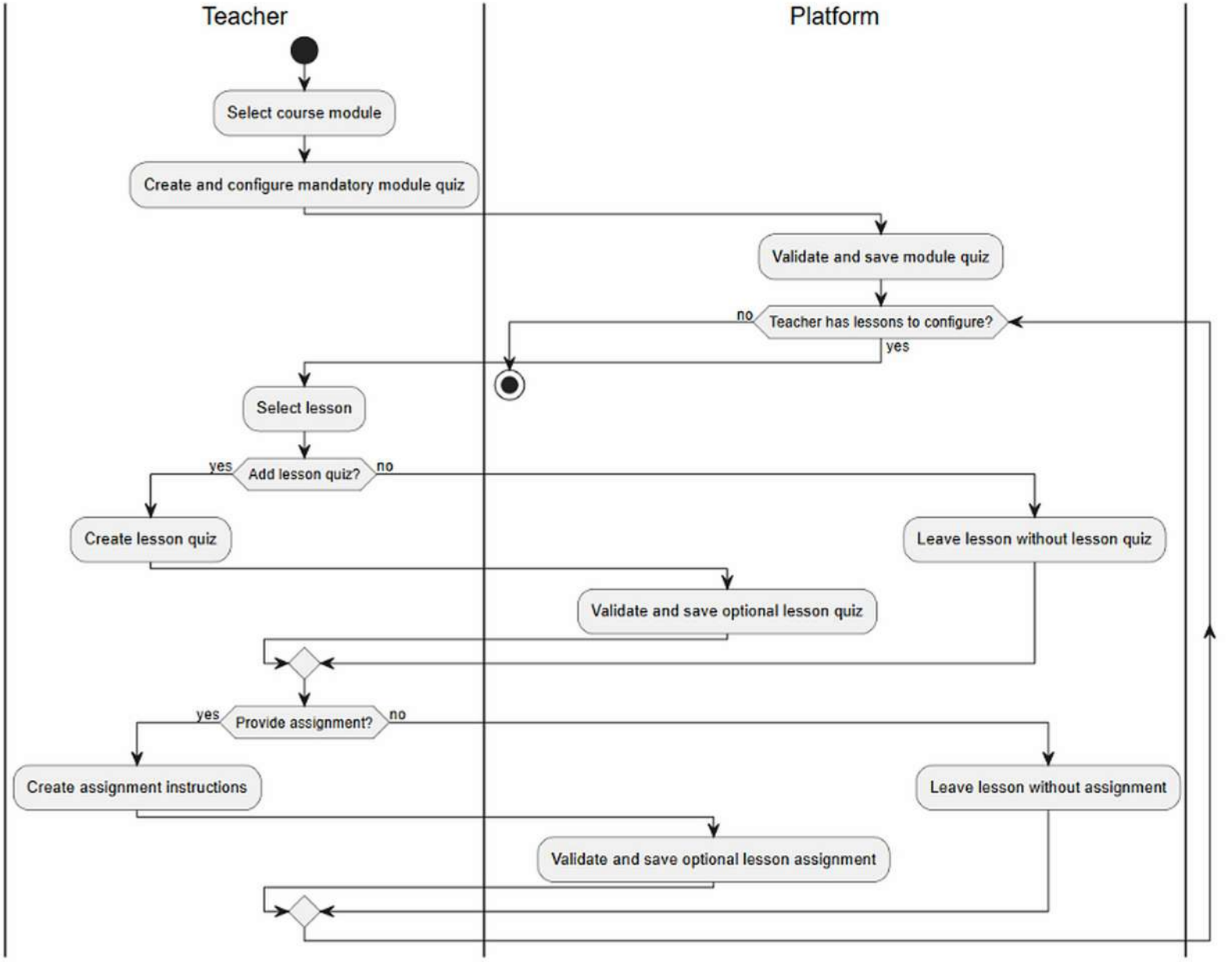
AD-15 — Create and Manage a Course



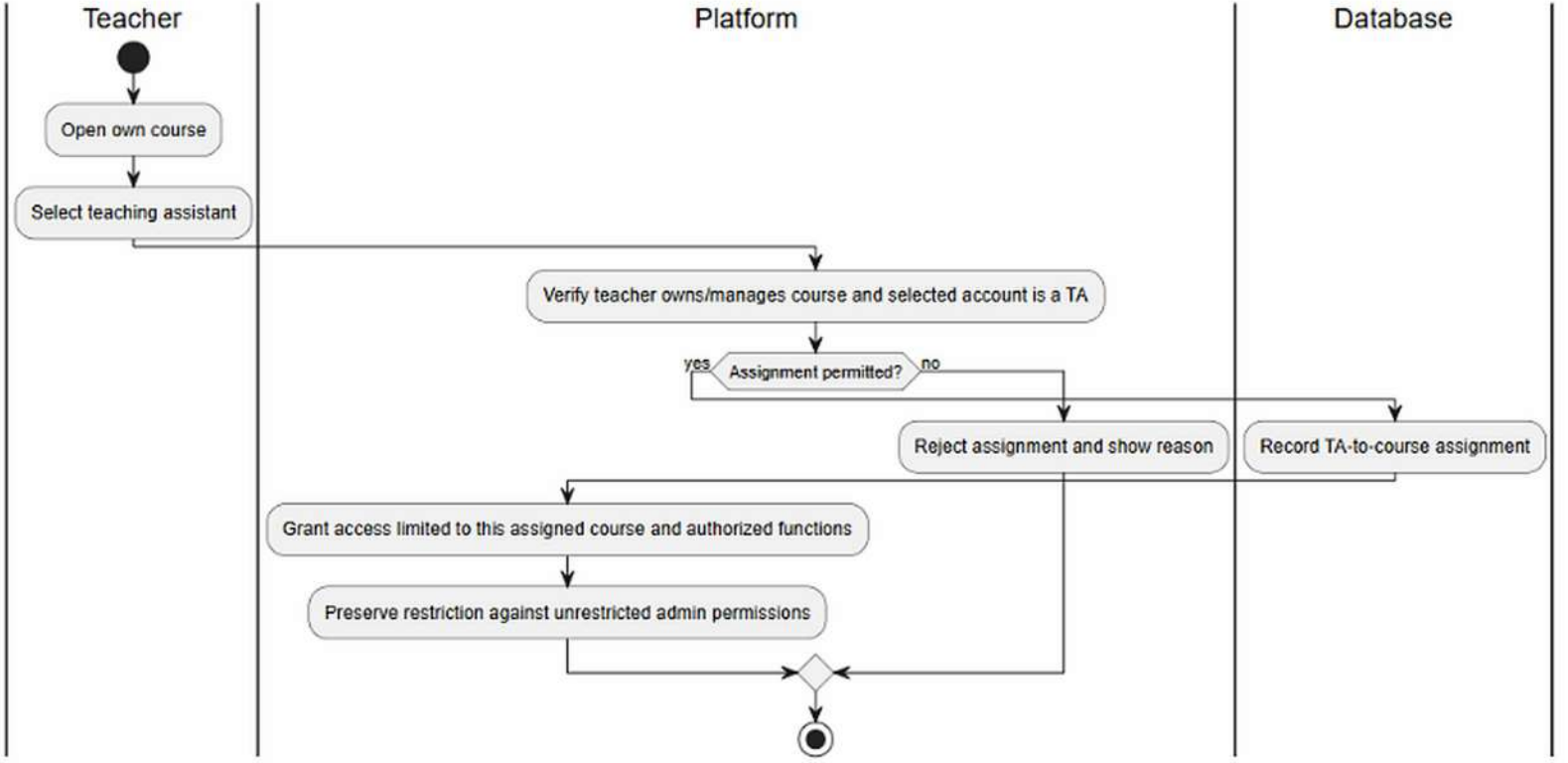
AD-16 — Create and Manage Modules and Lessons



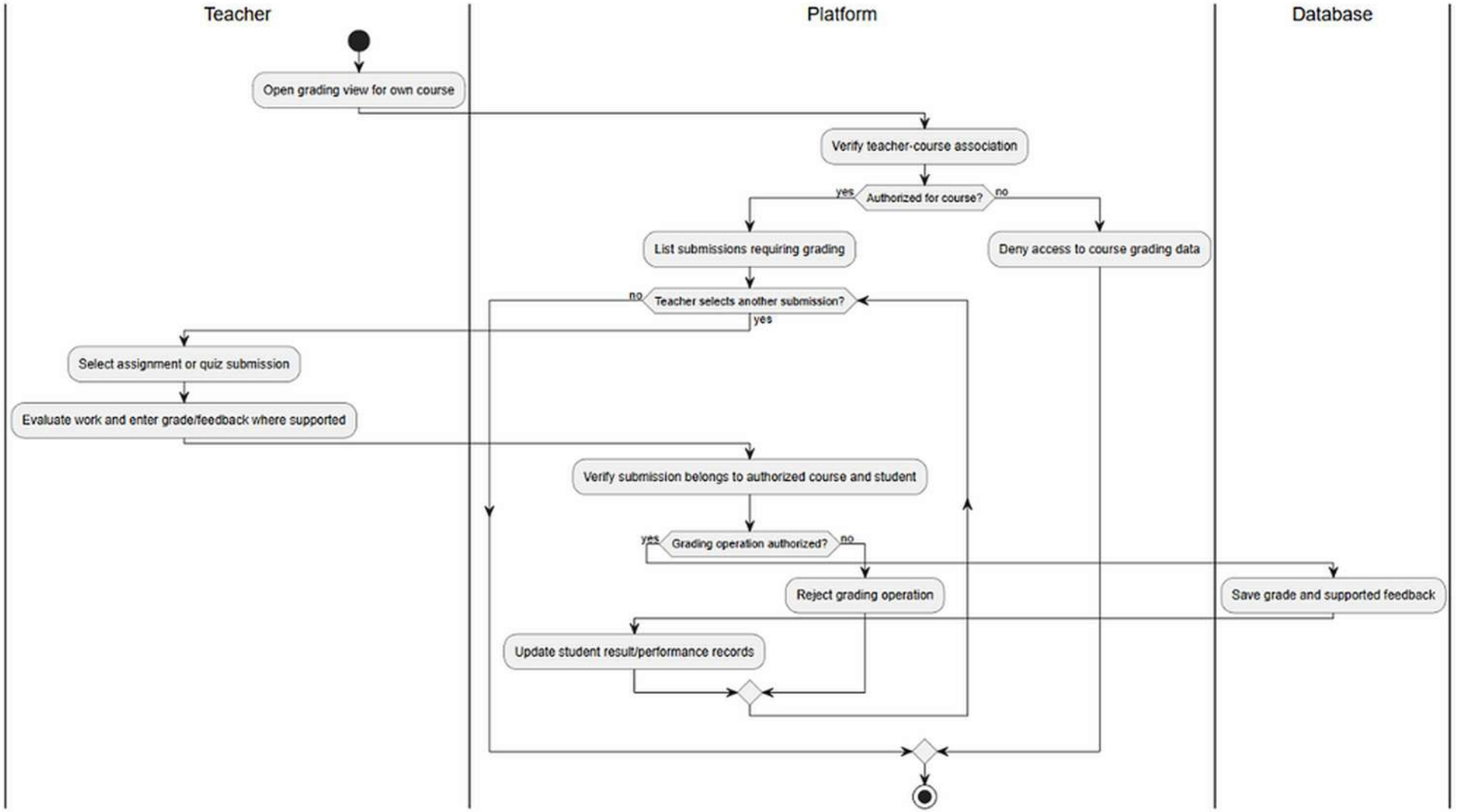
AD-17 — Create and Configure Quizzes and Assignments



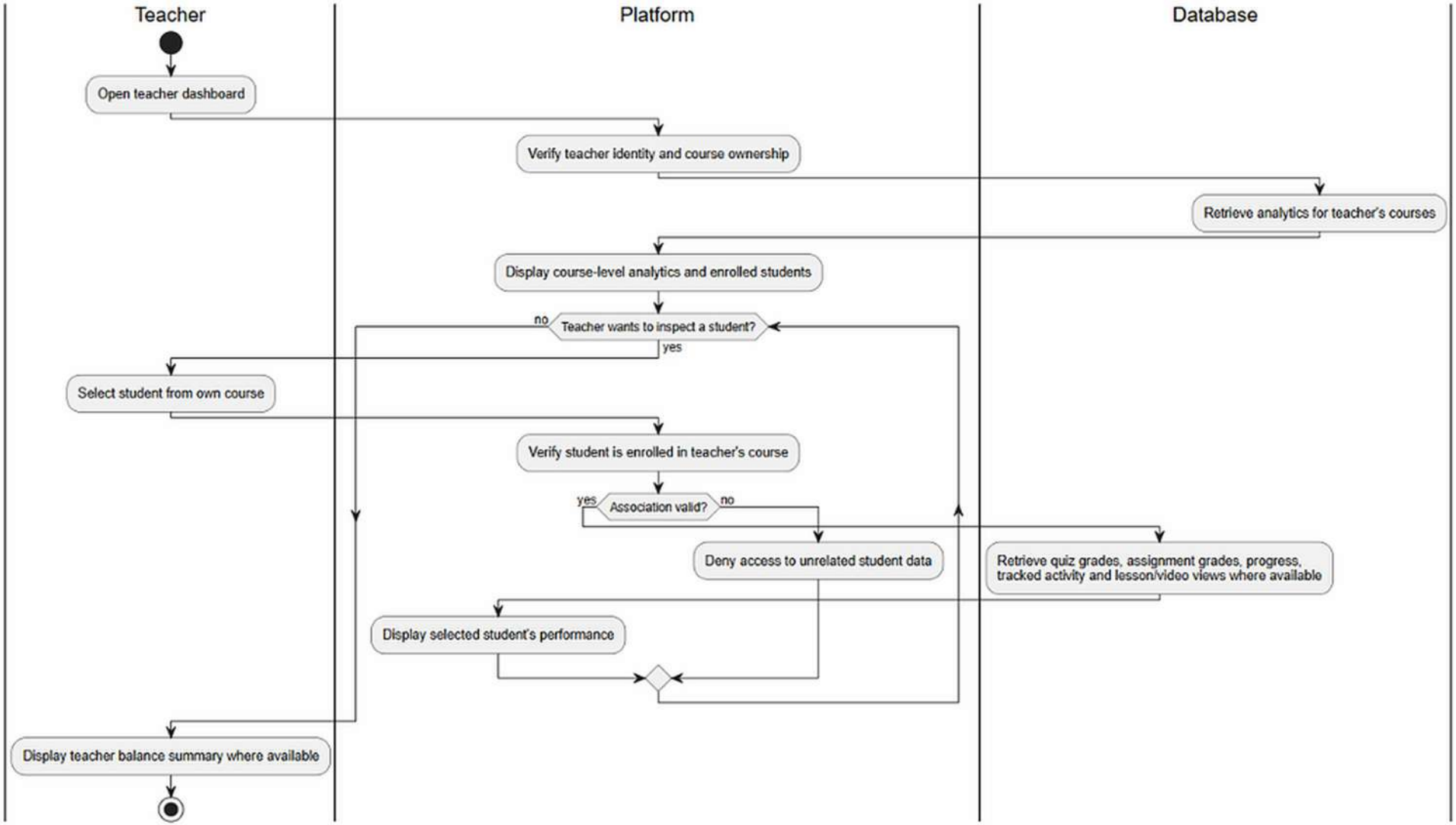
AD-18 — Assign a Teaching Assistant to a Course



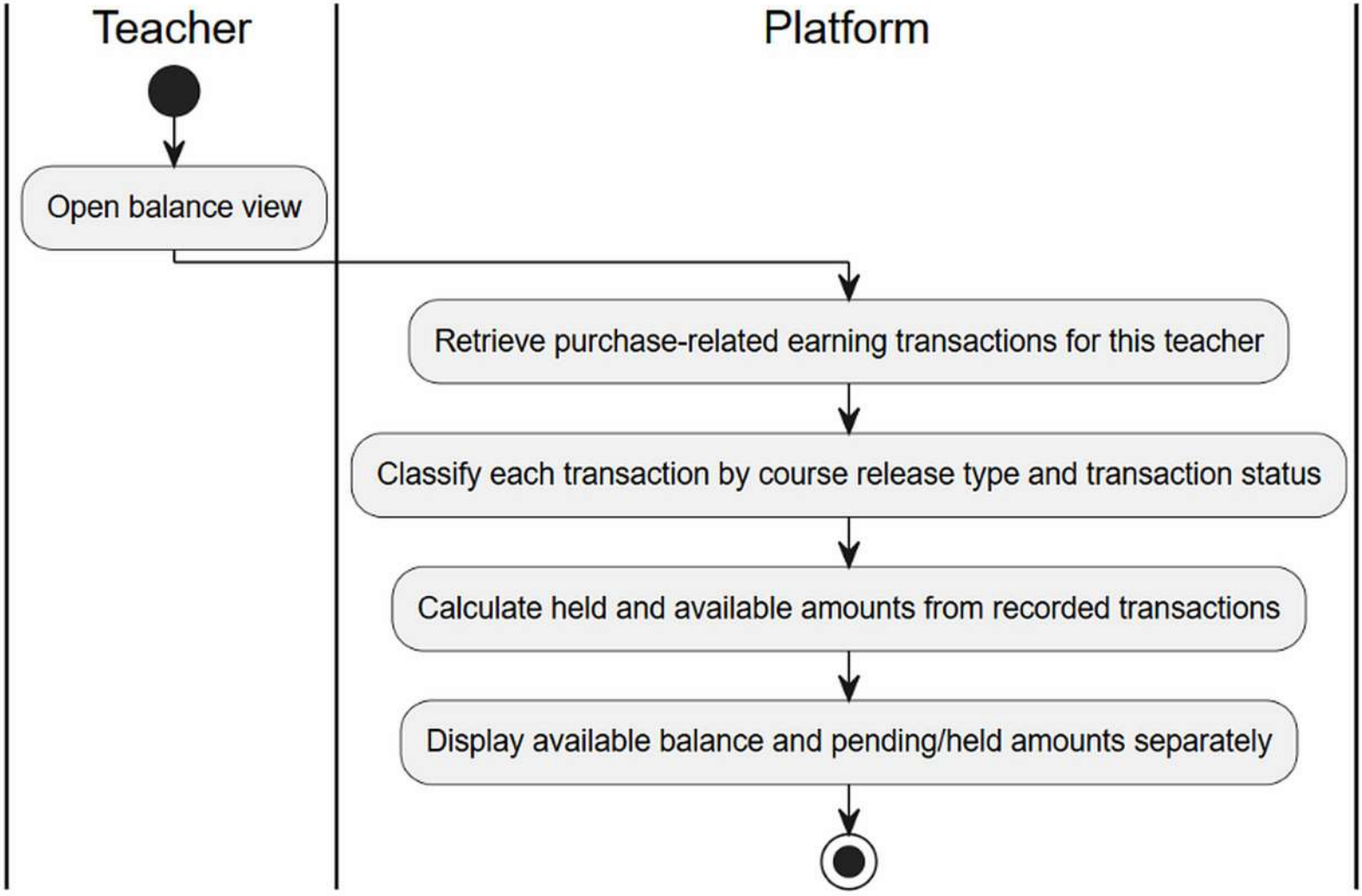
AD-19 — Grade Assignments and Authorized Quizzes



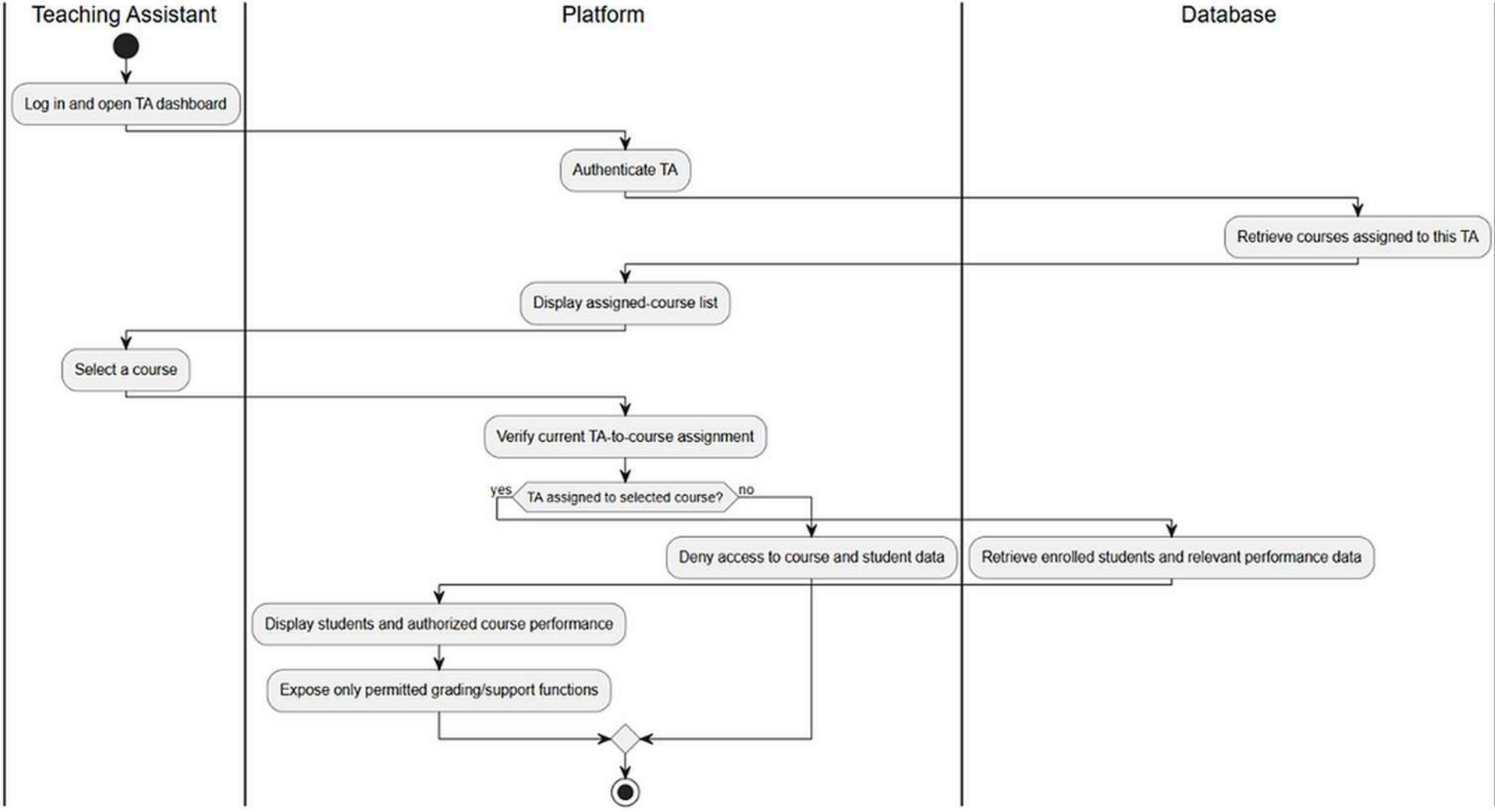
AD-20 — Teacher Dashboard and Student Performance



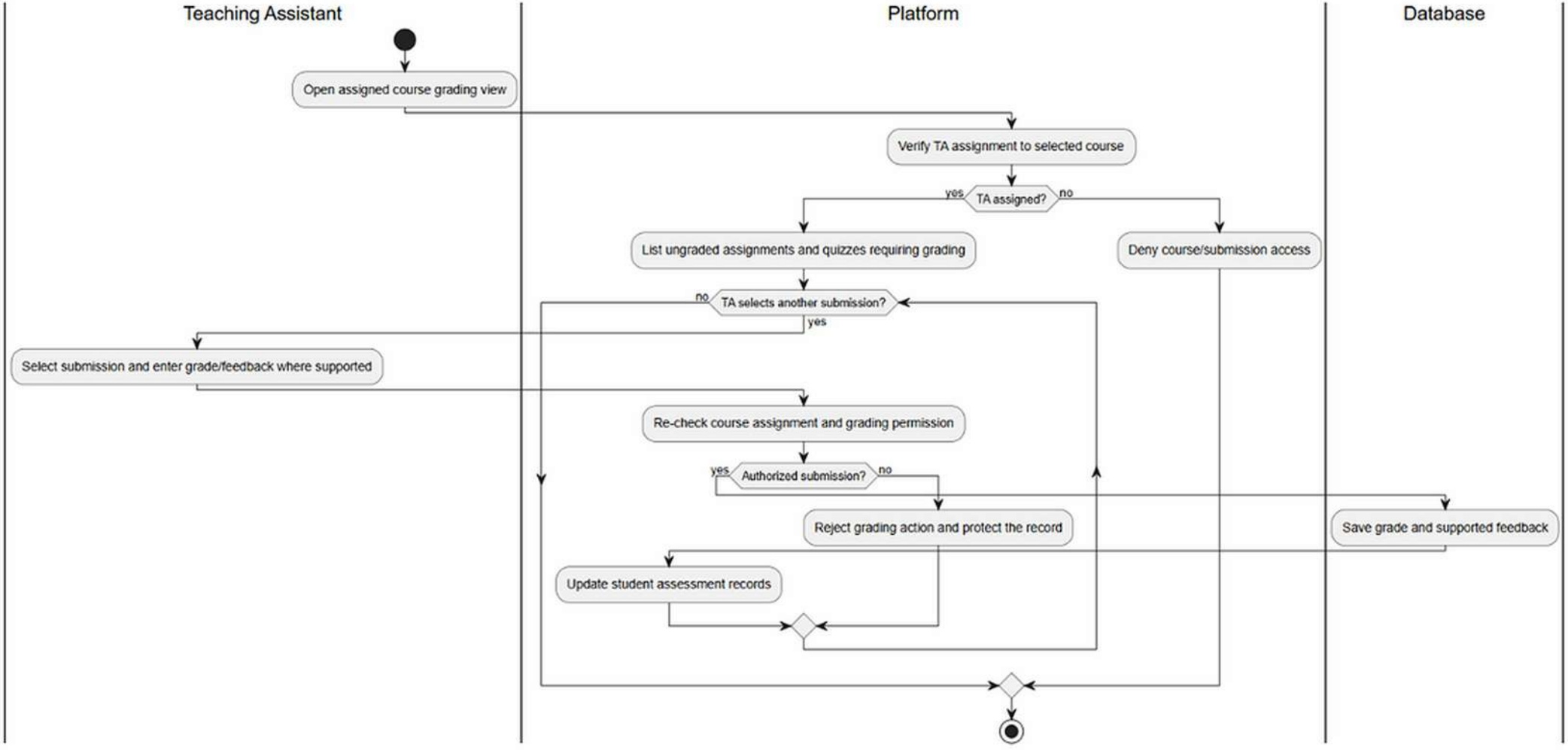
AD-21 — Teacher Balance and Eligible Earnings



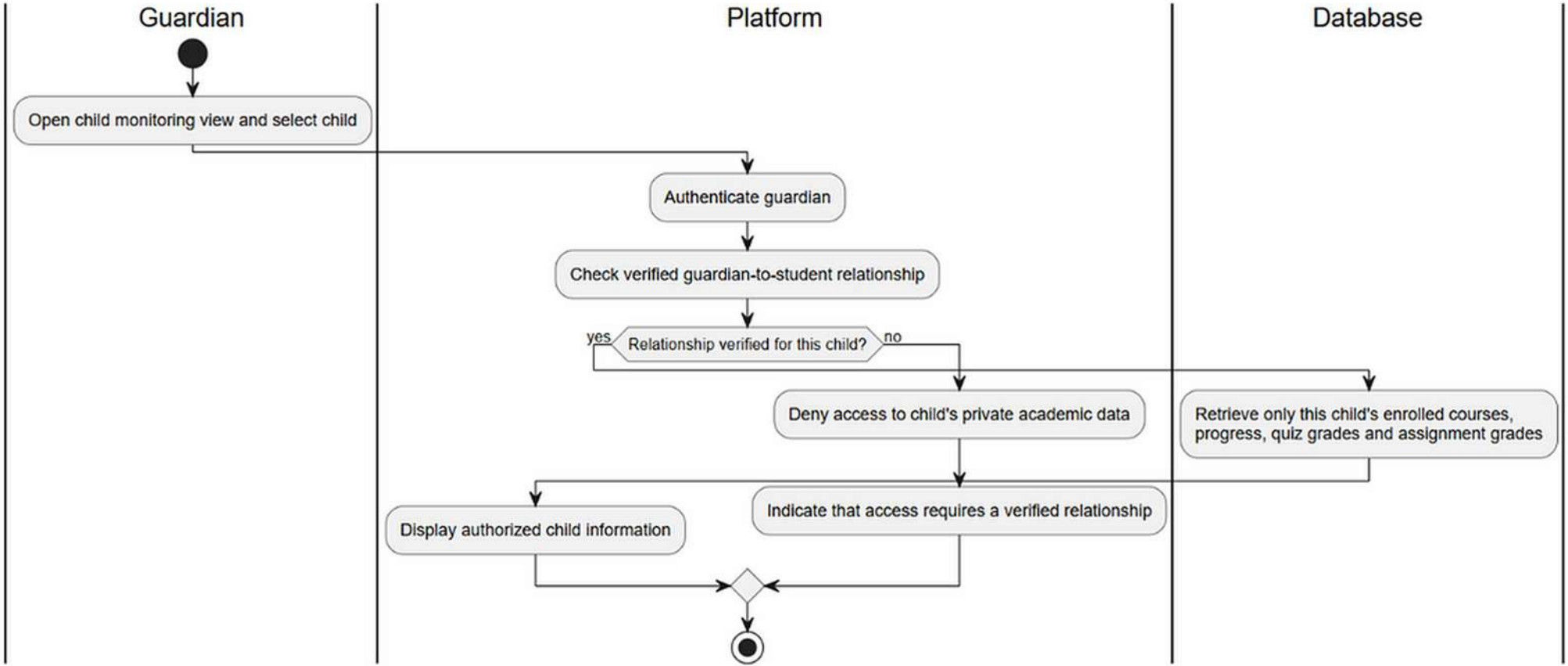
AD-22 — TA Access to Assigned Courses and Students



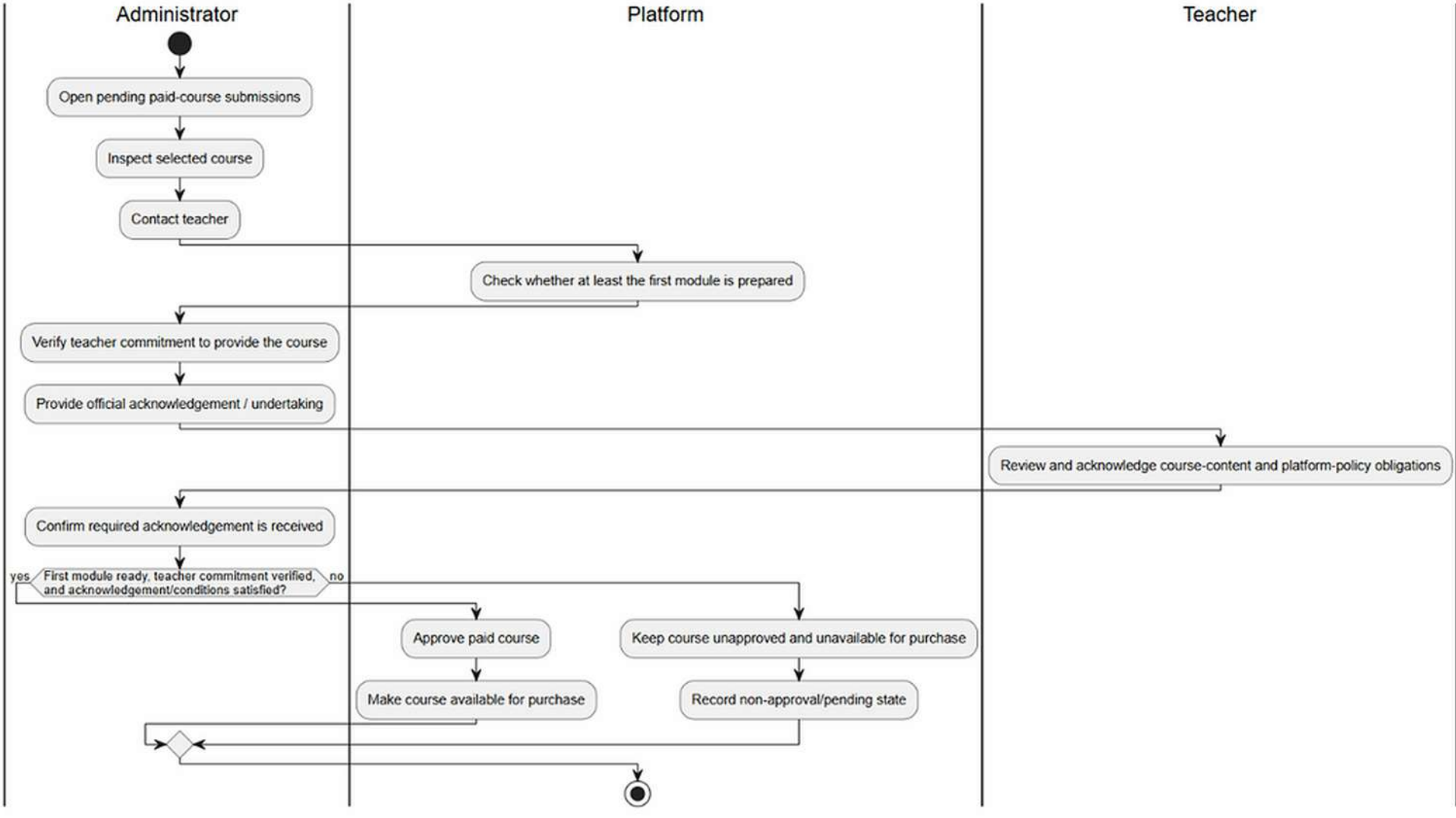
AD-23 — Grade Assigned-Course Submissions



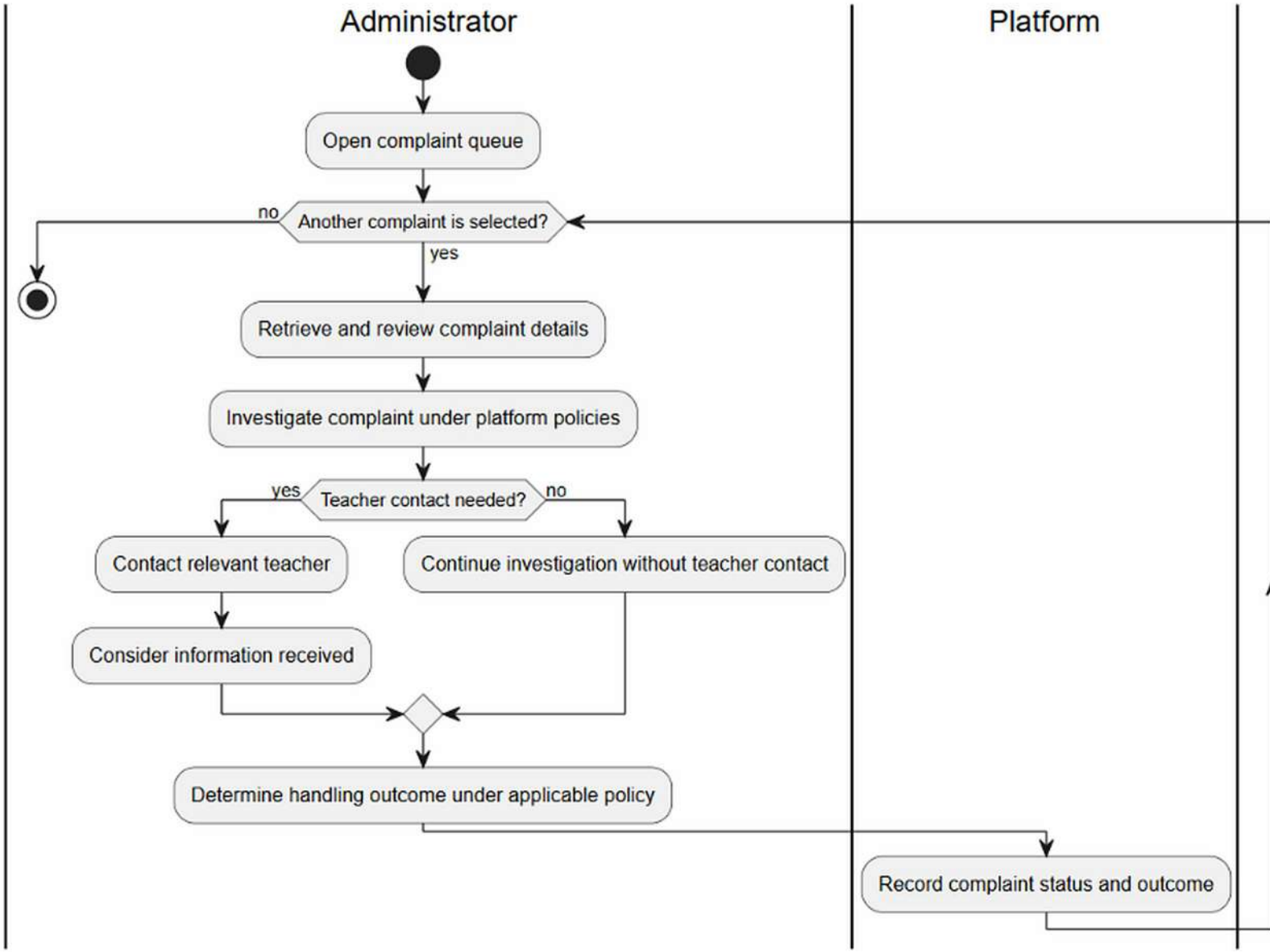
AD-24 — Guardian Access and Child Monitoring



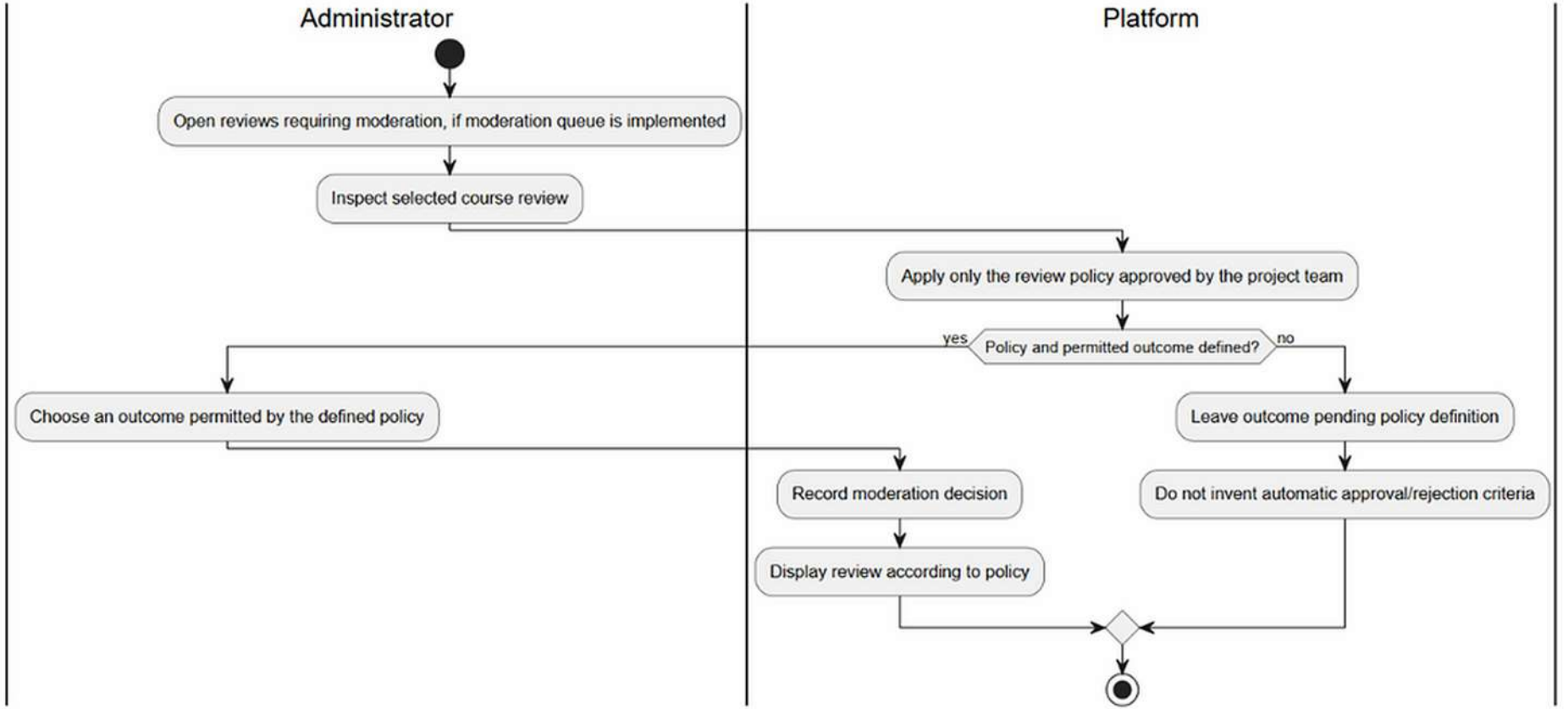
AD-25 — Review and Approve a Paid Course



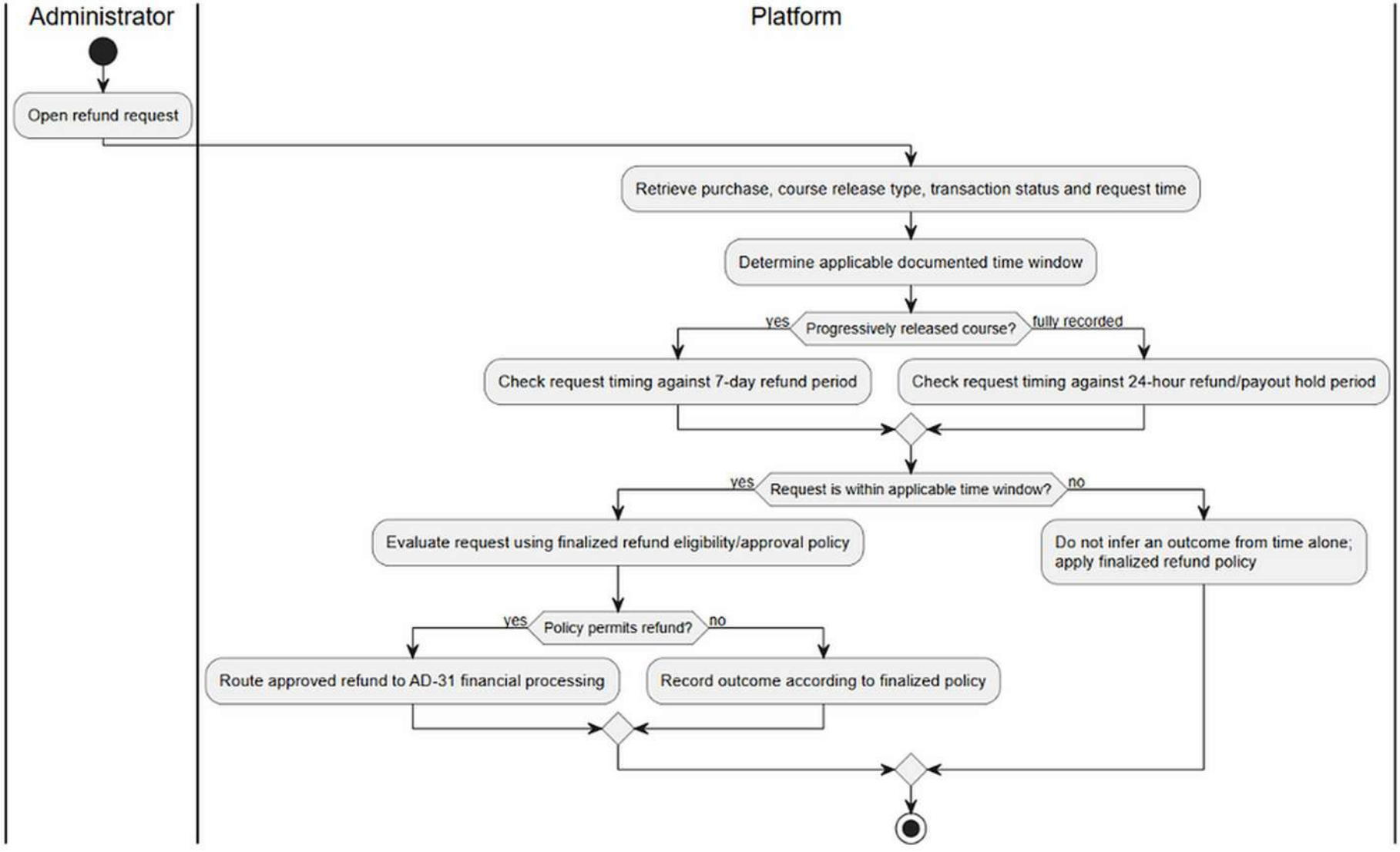
AD-26 — Handle Student Complaints



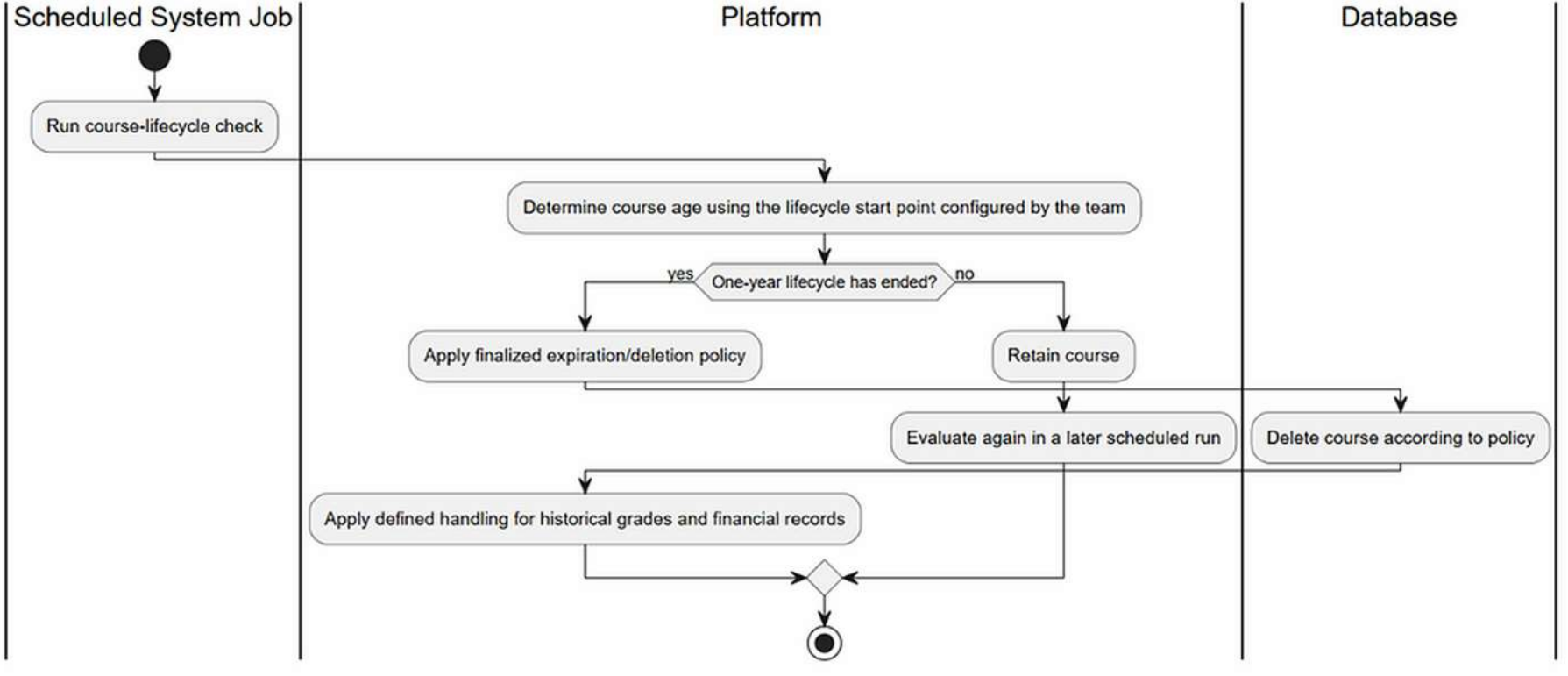
AD-27 — Moderate Course Reviews



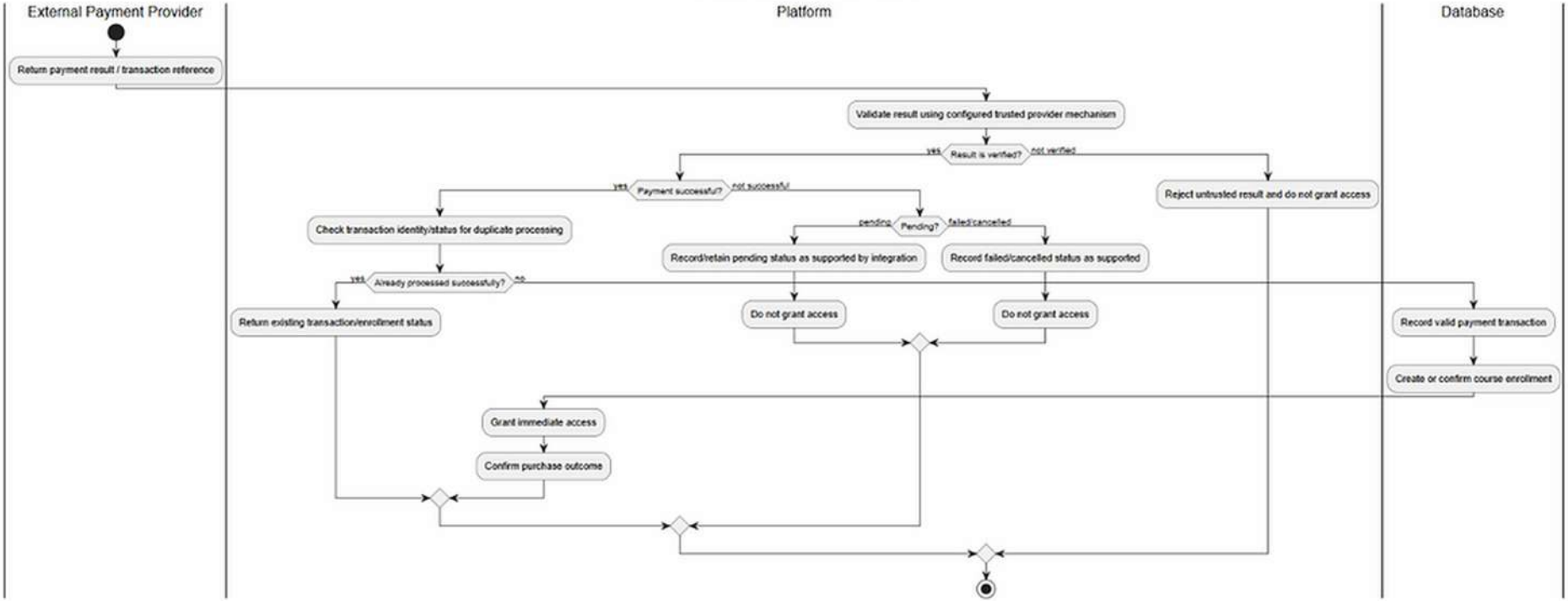
AD-28 — Handle Refund Requests

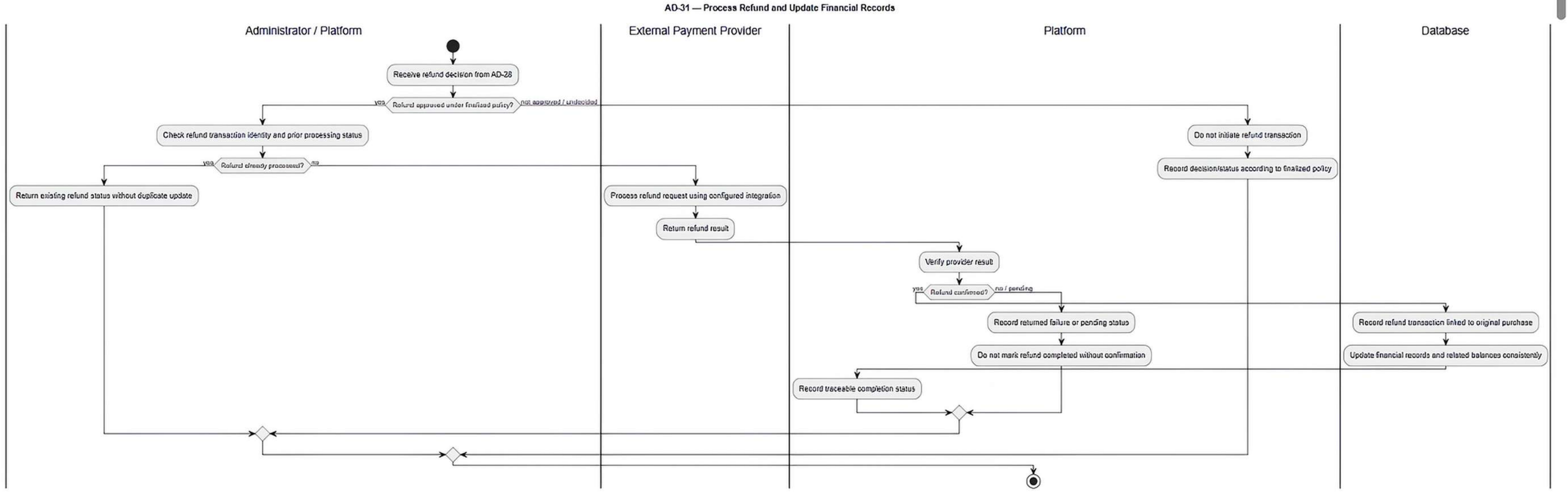


AD-29 — Course Lifecycle Expiration

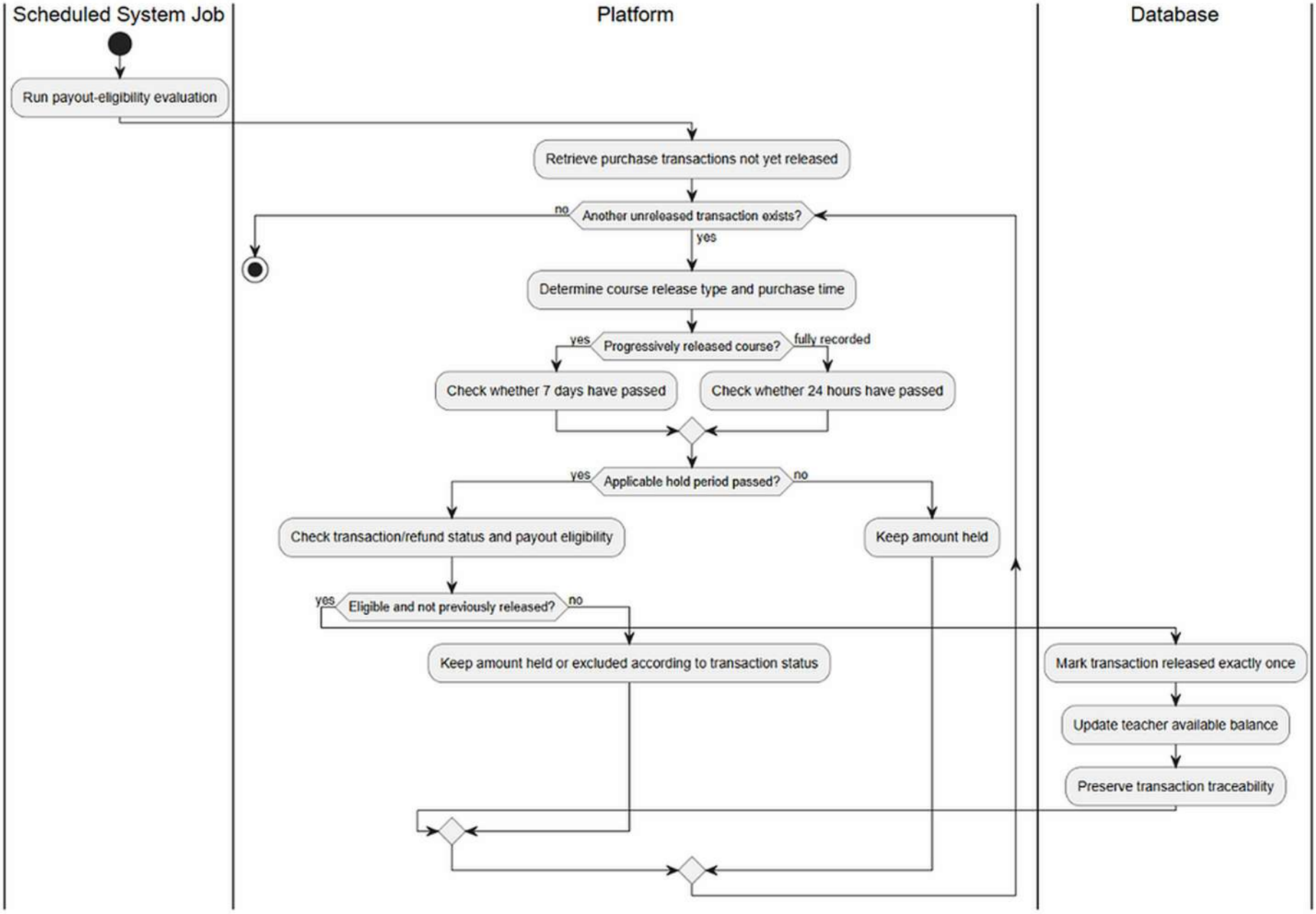


AD-30 — Process Payment Result and Enrollment

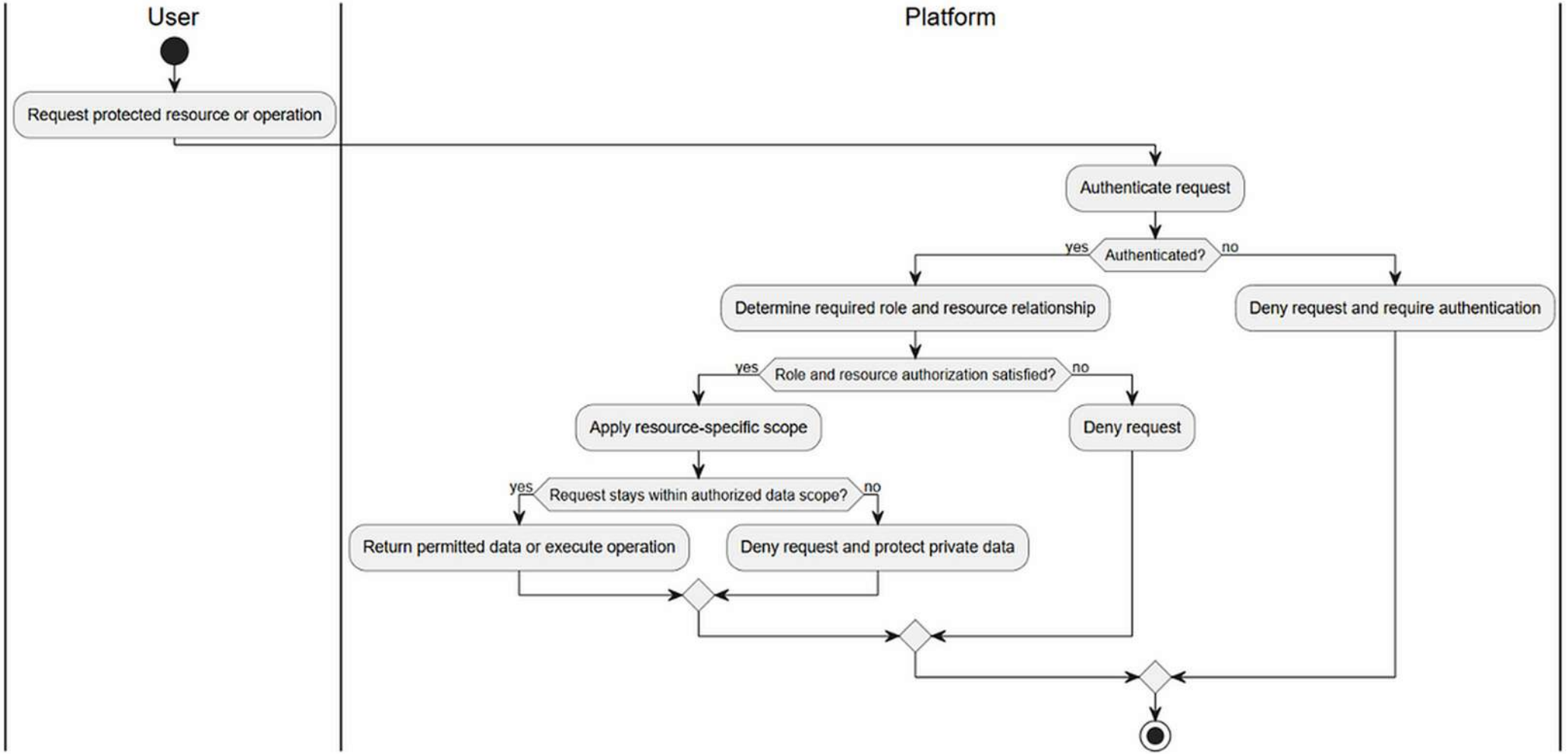



AD-31 — Process Refund and Update Financial Records
Administrator / Platform
External Payment Provider
Platform
Database
Receive refund decision from AD-28
Refund approved under finalized policy?
yes
not approved / undecided
Check refund transaction identity and prior processing status
Refund already processed?
yes
no
Return existing refund status without duplicate update
Process refund request using configured integration
Return refund result
Verify provider result
Refund confirmed?
yes
no / pending
Record returned failure or pending status
Do not mark refund completed without confirmation
Record traceable completion status
Do not initiate refund transaction
Record decision/status according to finalized policy
Record refund transaction linked to original purchase
Update financial records and related balances consistently

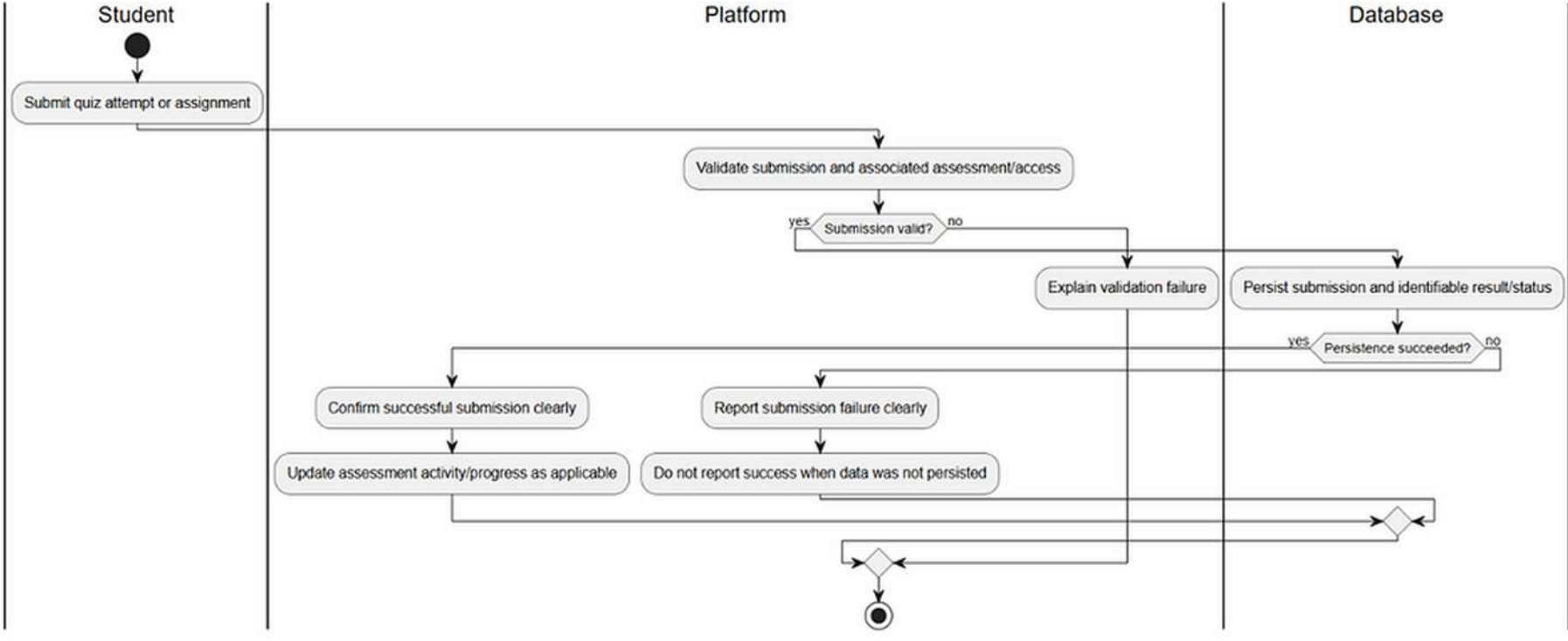
AD-32 — Scheduled Payout Eligibility and Teacher Balance Update



AD-33 — Cross-Cutting Unauthorized Access and Data Protection



AD-34 — Reliable Assessment Submission



AD-35 — Financial Transaction Integrity

